Paper version 3.25 (2026 September 9). Sample rescoped from 20 pc to the full current search volume (40 pc). Live snapshot 2026 September 9, 15:58 UT: 114 of 208 planned target/bands (55%) across 86 of the 168 census stars, 448 spectral windows. The candidate criterion is now the symmetric single-position star-versus-control test; see the version-control changelog in the Data Availability section.
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A SEARCH FOR TIME-VARIABLE AND FREQUENCY-DRIFTING NARROWBAND TECHNOSIGNATURES TOWARD ALMA-OBSERVED STARS WITHIN 40 PC OF THE SUN: SURVEY DESIGN AND FIRST RESULTS

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Version September 9, 2026

ABSTRACT

We define a census of 168 stars within 40 pc of the Sun having usable public ALMA coverage and search the archival data for narrowband emission from time-variable and/or sufficiently frequency-drifting carriers. The per-channel temporal-median subtraction that rejects stationary ground-based interference also suppresses phase-steady carriers, which are therefore outside this paper's scope entirely. This release is a live snapshot of an ongoing archival campaign: at 2026 September 9 it has processed 114 of 208 planned target/bands across 86 of the 168 census stars and 448 spectral windows, spanning 1.3–38.8 pc. No signal matching the searched class was detected above the nominal instrumental threshold during the analysed archival intervals. This version adopts the fully symmetric single-position star-versus-control statistic as the operative candidate criterion, replacing the region-maximum statistic of earlier versions, which compared a maximum over the source region against single-position controls and was therefore biased towards detection. Four target/bands are flagged: two towards β Pictoris and one towards HD 48370 are circumstellar or foreground CO, and one towards CP-72 2713 is a marginal threshold crossing consistent with the 0.87 flags expected by chance across 448 windows. The AU Mic crossing of earlier versions fails the symmetric test and is no longer counted.

Nominal 5σ thresholds span 2.0×10^{12}–2.8×10^{17} W in EIRP (median 1.0×10^{15} W); because most windows are channelised at 15.625 MHz, these are power thresholds for unresolved narrowband emission, not Hz-resolution sensitivity. A 1200-trial injection campaign on real data recovers the drifting searched class at ∼42 per cent at threshold and ∼83 per cent at twice threshold power in fine-channel windows; coarse-channel windows recover nothing within the measured amplitude range. Zero credible detections bound the conditional occurrence of transmitters in the searched class: at EIRP $\gtrsim 10^{16}$ W the 95-per-cent upper limit is 6.9 per cent under unit recovery, ∼16 per cent under a uniform measured completeness, and 45 per cent under the fully frequency-integrated per-system completeness this release can calibrate (24 of 42 systems).

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

SCOPE OF THIS VERSION, AND WHAT IT DOES NOT YET COVER

This version begins the rescope of the survey from the 20 pc sample of versions ≤3.24 to the full volume now being searched, $d < 40$ pc. The rescope is *partial*, and because a half-rescoped paper is easy to misread, the state of every part is set out here explicitly.

Rescoped in this version: the census and sample definition (§3), the headline counts in the abstract and version line, the candidate criterion (§5.4), the candidate set and its dispositions, and the survey-level false-alarm accounting (Appendix J). These are stated for the 40 pc sample at the snapshot epoch in the version line: 114 of 208 planned target/bands across 86 of 168 census stars, 448 spectral windows, 101 execution blocks, 1.3–38.8 pc.

Not yet rescoped, and still reporting the ≤20 pc release of version 3.24 wherever they appear: the detailed results subsections that quote 60 of 148 target/bands, 46 stars, 225 windows and 57 execution blocks; the unique post-veto bandwidth (82.3 GHz); the data-quality exclusion accounting; every figure; the per-target appendix table; and the occurrence limits of §6, whose completeness calibration exists only for that subsample. Those quantities are internally consistent with each other and with version 3.24, and each remains a correct statement about the ≤20 pc release; they are simply not yet statements about the 40 pc volume. Extending them requires the per-target table, the frequency-union mask, the figures, and the injection campaign to be regenerated over the wider volume, which is deferred to the next version. Readers should therefore not combine a 40 pc headline count with a 20 pc derived quantity; where the two appear in the same section, the sample is named at the point of use.

Processing is ongoing and the snapshot counts move as it proceeds; the version line always carries the epoch to which they refer.

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1. INTRODUCTION

A technological civilisation, whether deliberately signalling or simply going about its own affairs, may leave a detectable imprint on the electromagnetic spectrum: a *technosignature* (Sheikh 2020; Vidal et al. 2026). Six decades of radio searches have grown into surveys of many thousands of stars, yet, as Wright et al. (2018) quantified, the combined effort has sampled only an infinitesimal fraction of the multi-dimensional space of transmitter frequency, bandwidth, power, and duty cycle. Three gaps persist: the millimetre and submillimetre regime above $\sim$50 GHz is essentially unsearched; most surveys target interest-selected classes of star rather than the local stellar population; and few studies have systematically reprocessed *existing* interferometric archives.

This paper addresses all three. We construct a census of every star within 20 pc of the Sun that has public ALMA archival coverage, irrespective of spectral type, disc-hosting status, or known planets, and mine that archive for narrowband technosignature candidates and anomalous millimetre continuum, with corrections for stellar proper motion and for the Doppler drift rates plausible for a transmitter on an orbiting platform. The search class is defined by the data processing itself: the per-channel temporal-median subtraction that rejects stationary ground-based interference also suppresses phase-steady carriers, so this paper constrains *time-variable and/or sufficiently frequency-drifting narrowband emission* only (§4.5 measures both sides of that definition). Four archival-specific analyses supplement the classical search: closure-phase vetting, a continuum anomaly screen, a chirp and periodicity re-analysis, and a cross-target frequency-occupancy check.

Section 2 reviews the literature, Section 3 the sample, Section 4 the search methodology, Section 5 the results, Section 6 the constraints and their interpretation; per-target tables are deferred to online-only Appendix I. We publish at 41 per cent completion because the citable content is methodological – the census definition, the empirically calibrated false-alarm machinery, and the measured completeness – together with a first set of per-target limits, every number tied to a frozen pipeline and a named data snapshot that later work will supersede row by row without invalidating this one.

2. BACKGROUND

Six decades of searches divide into wide-field commensal surveys and targeted programmes on preselected stars. The modern generation is dominated by Breakthrough Listen, whose GBT, Parkes, and MeerKAT programmes have published multi-thousand-star limits from 1.1 to 8 GHz (Isaacson et al. 2017; Enriquez et al. 2017; Price et al. 2020; Czech et al. 2026), with parallel efforts on FAST, LOFAR, NenuFAR, the Sardinia Radio Telescope, and commensal VLA, MeerKAT, and MWA back-ends (Zhang et al. 2020; Johnson et al. 2023; Valente et al. 2025; Tremblay et al. 2024); machine-learning rejection of RFI-dominated hit lists has become central (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026) – a capability the classical threshold search we adopt lacks, as §6 returns to. Two structural points from this history shape the present survey. First, Margot et al. (2023) showed that a non-detection bounds the transmitter fraction only if the sample is representative of the underlying population – a condition interest-selected samples violate. Second, essentially all of this activity lies below $\sim$10 GHz: above it, the millimetre and submillimetre regime has been searched only once before, also with ALMA (Mason et al. 2024); sixteen programmes of historical detail are tabulated in Appendix E.

Mason et al. (2024) searched archival Band 3 windows toward 28 "stellar bycatch" stars and placed $\mathrm{EIRP_{min}} > 7 \times 10^{17}$ W for the nearest, identifying the specific challenges of high-frequency work: drift rates scale with observing frequency, smearing erodes sensitivity unless a drift search is run, and the molecular line forest confuses. Our earlier pilot (White 2026; under review at the time of submission – the citation is provisional and will be updated on acceptance) extended that methodology to TRAPPIST-1 across Bands 3, 6, and 7 ($\mathrm{EIRP_{min}} \sim 5$–$10 \times 10^{13}$ W, no detection) and built the ranked 100-star list from which this survey grew. The incremental content of this submission is the census selection (§3), the 512-position control-ring false-alarm calibration, closure-phase vetting, the spectral-type-normalised continuum screen, the chirp and periodicity extension, and the cross-target frequency-occupancy check. No execution block is analysed in both papers: the pilot's 100-star search products are superseded, not re-used, here, and every window analysed in this paper was re-reduced and re-searched from the raw archive data with the pipeline of §4. [calibrator-only usage – and state it here; otherwise delete this bracket and the sentence stands as the complete answer.]

Figure 1 places this release's nominal EIRP thresholds among the published searches, under two conventions. A quoted EIRP threshold is defined at the producing survey's channel width, and those widths span nearly six orders of magnitude between Hz-scale backends and ALMA's native correlator channels: panel (a), which divides each limit by its native channel width to give $\mathrm{W\,Hz^{-1}}$, is the like-for-like spectral comparison (the primary panel), and panel (b) the total-power comparison – our limits are numerically comparable to the deepest L-band limits in panel (b) only because the nearest targets lie an order of magnitude closer, despite channels some 10^6 times coarser. Both panels provide context, not a sensitivity ranking. Our linewidth convention is set out in §4.

A non-detection must be translated into a quantitative exclusion statement. Wright et al. (2018) formalised the "Cosmic Haystack" search volume; Margot et al. (2023) the conversion of a non-detection rate into an upper bound on the transmitter-hosting fraction, requiring sample representativeness; and Sheikh et al. (2025a) inverted the question to ask where Earth's own technosignatures would be detectable – a calibration point any survey's sensitivity can be read against.

The case for a census over interest-selected lists follows directly. Most surveys select by proximity, spectral type, planets, or Earth Transit Zone membership – reasonable per unit time, but an obstacle to prevalence inference (Margot

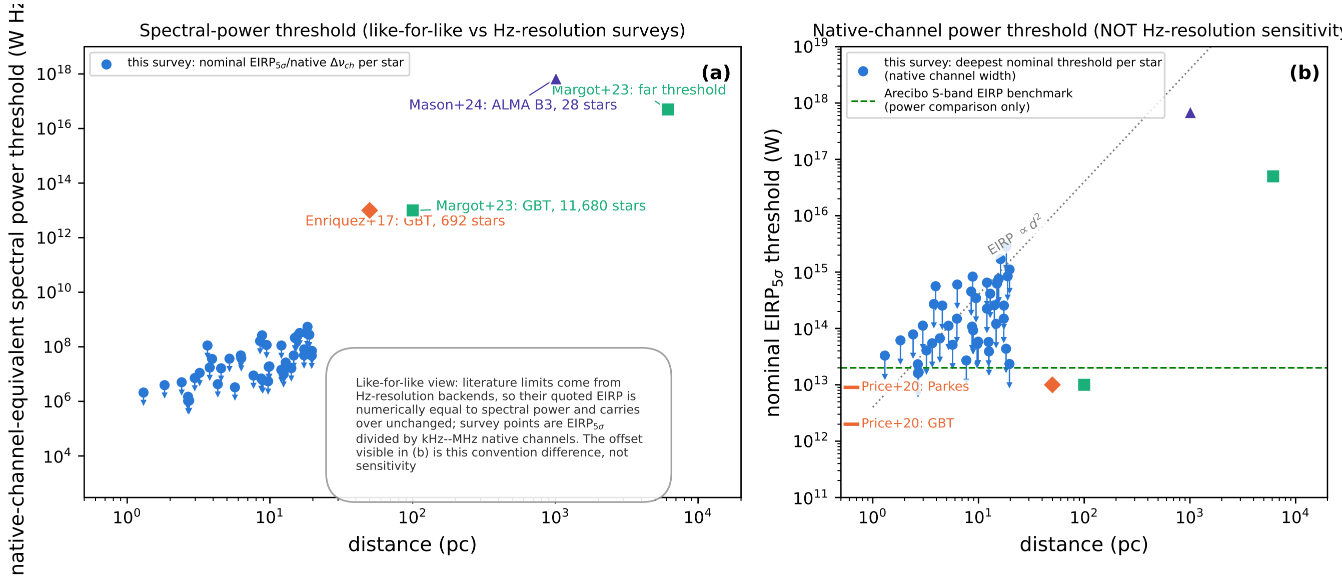

FIG. 1.— Literature context of this release's nominal thresholds. (a) Spectral power threshold (W Hz^{-1}), the primary and like-for-like comparison: the same literature points carry over unchanged (one-Hz channels make their quoted EIRP and spectral power identical), while our kHz–MHz channels separate the two by up to six orders of magnitude – this survey (blue; nominal 5σ thresholds under the boxed reading rule of §4) against Enriquez et al. (2017) (orange diamonds), Margot et al. (2023) (aqua squares), Mason et al. (2024) (violet triangles). (b) Total transmitter power required for detection at each survey's native spectral resolution – not narrowband spectral sensitivity – with the deepest per-target limits of Price et al. (2020) as left-edge ticks; the grey dotted line marks EIRP $\propto d^2$, the green dashed line an Arecibo-like planetary radar (2×10^{13} W). Conventions differ among surveys (thresholds, drift coverage, confidence); context, not a ranking.

et al. 2023). The ALMA-specific form is disc selection: ALMA-detected discs are orders of magnitude brighter than the Solar System's zodiacal cloud would appear at interstellar distances, so disc-selected samples skew young and dynamically active. This survey therefore identifies every star within 20 pc in the usable field of at least one public ALMA observation (§3) – a bycatch census of the archive within a fixed volume. The census removes technosignature-specific selection, not the astrophysical selection embodied in the archive itself; our 120 stars are ∼5 per cent of the known stellar population within 20 pc, and that residual coverage selection is characterised as a completeness function (§3, Table 7) rather than left implicit.

The coming decade will widen the search: the SKA (Siemion et al. 2015; Tremblay et al. 2026) and ngVLA (Ng et al. 2022) bring wide bandwidth and commensal architectures below ∼50 GHz, but neither offers ALMA's combination of resolution and collecting area above it in the near term, so a systematic mm/submm census of the kind begun here remains an ALMA-specific measurement for some years.

3. SAMPLE SELECTION

Our sample comprises every star within 40 pc of the Sun (Gaia DR3 parallaxes, all distances direct inversions $d = 1/\pi$; every entry has $\pi \geq 25$ mas with $\sigma_\pi/\pi \leq 0.8$ per cent) with at least one public ALMA observation, of any band, science category, or proposal type, in which the star's proper-motion-corrected position at the epoch of observation falls within the field of view of at least one execution block – not necessarily at the pointing centre, since mosaics and wide pointings serendipitously cover more than one catalogued star. No disc, planet, or spectral-type weighting is applied; those properties are descriptive metadata, not selection criteria. This identifies 168 unique target stars, processed nearest-first – a deliberately sensitivity-ranked order, not an unbiased draw, and the prevalence statements of §6 are conditional on it.

The composition follows directly from this archival selection: just under two thirds of the 120 stars are M dwarfs (76 stars, 63 per cent), followed by 12 K, 11 G, and 8 F dwarfs, a single A-type star (β Pic), and six white dwarfs; six faint companions lack tabulated types. Fifteen of the 46 processed stars are confirmed exoplanet hosts – a direct imprint of the observing-history bias – and well-studied debris-disc hosts (β Pic, AU Mic, ϵ Eri, τ Cet) contribute continua that include circumstellar dust, a distinction the continuum lane's spectral-type-normalised screen (§4) keeps from masquerading as an anomaly.

What the sample does and does not cover should be explicit: the 120 stars are every star within 20 pc with public ALMA coverage, a complete census conditional on archive coverage – complete with respect to the intersection of the 20-parsec reference catalogue and the public-archive snapshot (2026 September 8) that defines this release, and conditional on that intersection only – *not* a complete census of every known star within 20 pc. Against the reference Gaia DR3 catalogue of the solar neighbourhood (2357 stars within 20 pc; Figure 13, online-only appendix), the 120 stars are ∼5%, and that reference list is itself incomplete at both ends, so 5% is a denominator convention: across an independent SIMBAD TAP denominator bracket the ALMA-covered fraction is 4.4–5.1%, and small under any convention is the only use we put it to. How the covered stars differ from the census is quantified in Table 15 (online-only appendix) so the selection function is inspectable rather than asserted: a factor ∼5 enrichment within

5 pc (nearby famous stars attract ALMA time, falling to parity by 10–20 pc), mild M-dwarf depletion with 1.8–1.9× F/G enrichment, planet-hosting in 24 of 120 entries, and 113 systems in 120 entries; age and disc-hosting flags are not uniformly available for the census and are documented qualitatively, with a machine-readable selection-function table part of the release.

ALMA coverage itself is strongly astrophysically selected – archival coverage inherits the proposers' interests – so any prevalence statement this survey eventually makes is a statement about the ALMA-observed subset of the solar neighbourhood, not the neighbourhood as a whole; we state this plainly rather than letting "complete census of the ALMA-observed stars" be read as "complete census of the solar neighbourhood". The stellar counts quoted throughout are catalogue-entry counts: the 120 entries comprise 113 distinct physical systems (seven designation-linked component pairs),[4] and because the 44 narrowband-covered entries include the spatially unresolved UV Ceti and AT Mic pairs, the 60 target/bands, 225 windows, and 57 execution blocks derive from 42 independent systems – the paired rows are bookkeeping, not independent trials.

The operative meaning of "usable public ALMA coverage" is frozen as a five-criterion decision tree, enforced for this release and not changed subsequently without an erratum: (i) at least one public execution block (EB) with QA2-calibrated visibilities covering the star's epoch-propagated position in the primary beam, with no minimum integration, sensitivity, antenna count, mosaic-tile count, or array configuration imposed beyond QA2 state itself; (ii) propagated position within one primary-beam FWHM of the field centre (the largest accepted offset is $\sim$0.7 FWHMs, Sirius B); (iii) each spectral window admitted individually, on the pipeline's quality vetting alone; (iv) no programme-category restriction – science, calibration, and observatory projects enter on identical terms; and (v) a fixed archive snapshot date (2026 September 8), re-evaluated only at release boundaries. No star or window of this release was excluded by any criterion not listed here; none is excluded for short integration (the shortest retained are 1.0 and 1.5 min), which keeps the boundary reproducible at the price of a few shallow limits.

Whether each star was the pointed target or a serendipitous field source is quantified from ALMA project metadata: 43 of the 44 narrowband-covered stars are the pointed science target in at least one of their EBs, and of the 57 EBs, 55 point at a census star, one is another programme's science field, and one is a calibrator-only execution – leaving Wolf 219 the sole genuinely serendipitous star. The residual selection function of an archival census is the union of individual ALMA target lists: the contrast with interest-selected technosignature samples is one of degree rather than kind, but a *quantified* degree – a bycatch fraction of 1 star in 44, 1 EB in 57.

4. DATA AND METHODOLOGY

Survey-specific vocabulary – target/band dataset, spectral window, execution block, control ring, candidate, molecular-line exclusion mask – is defined once in the glossary of Appendix B (Table 11) and used consistently thereafter.

The design is summarised in Figure 2: one shared archive-ingestion stage feeds two independent search lanes. For each target we mine the public ALMA science archive for all calibrated (or locally recalibrated from raw data) execution blocks covering the star's Gaia-propagated position, within a bounded per-target processing budget, and search the resulting visibility data in (i) the *primary* lane – a narrowband, Doppler-drift-corrected search for technosignature candidates, following the methodology of Mason et al. (2024) and Enriquez et al. (2017), converting each non-detection into an equivalent isotropic radiated power limit $\mathrm{EIRP}_{5\sigma} = 4\pi d^2 S_{\min}\Delta\nu$, where $S_{\min}$ is five times the per-channel rms of the searched window (primary-beam corrected), $\Delta\nu$ its native channel width, and d the target distance; and (ii) an *ancillary* line-excluded continuum lane (§4.4). Throughout, "5σ" denotes this local instrumental threshold – not a survey-wide false-alarm probability, which is calibrated separately and empirically (§5.4). The chirp/periodicity extension and the molecular-line census are defined in §4.6 and §5.8; every headline limit, candidate disposition, false-alarm calibration, and completeness statement of this paper derives from the primary lane.

Both lanes exclude catalogued molecular transitions, at tolerances appropriate to their statistics. The continuum lane masks the eight common species (CO, ^{13}CO, C^{18}O, HCN, HCO$^+$, CS, CN, SiO) at native resolution before integrating; the narrowband lane applies a $\pm 50\,\mathrm{km\,s^{-1}}$ stellar-frame velocity exclusion mask (frame conventions in §4.1, criterion adopted in §5.4) to any threshold crossing, and the same mask defines the effective searched bandwidth (82.3 of the 87.2 GHz unique frequency union; §5.4). Masked regions reduce the frequency space a candidate can occupy, not the per-channel noise, so no threshold in Table 21 is inflated or deflated by the mask.

All flux densities and EIRP thresholds are Stokes I quantities: Stokes I is total power, so the limits apply to total transmitted power independent of polarisation geometry. What is lost is diagnostic, not sensitivity – polarisation is not used as a discriminator (no Stokes $V/Q/U$ extraction or circularity test is performed), and a polarisation-sensitive test is a natural future discriminator at no sensitivity cost.

How to read every limit in this paper. *Every $\mathrm{EIRP}_{5\sigma}$ value quoted in this paper is a nominal, local instrumental detection threshold: five times the per-channel rms of the searched window, at that window's native channel width, for a transmitter at the star's position matching the searched temporal and spectral morphology – time-variable on track timescales, since phase-steady carriers dwelling in one channel for most of the track are suppressed by the statistic's own continuum subtraction (§4.5). It is not a calibrated completeness limit: the injection campaign measures a recovery of roughly one half at the threshold itself ($\mathrm{EIRP}_{50} = 1.10\,\mathrm{EIRP}_{5\sigma}$ for the searched class), rising to 83 per cent at twice*

[4] Component designations follow SIMBAD; in Table 21 each entry carries its own row and distance, and spatially unresolved pairs (e.g. the UV Ceti entries at 2.7 pc, whose fluxes are blended) are marked in the notes column.

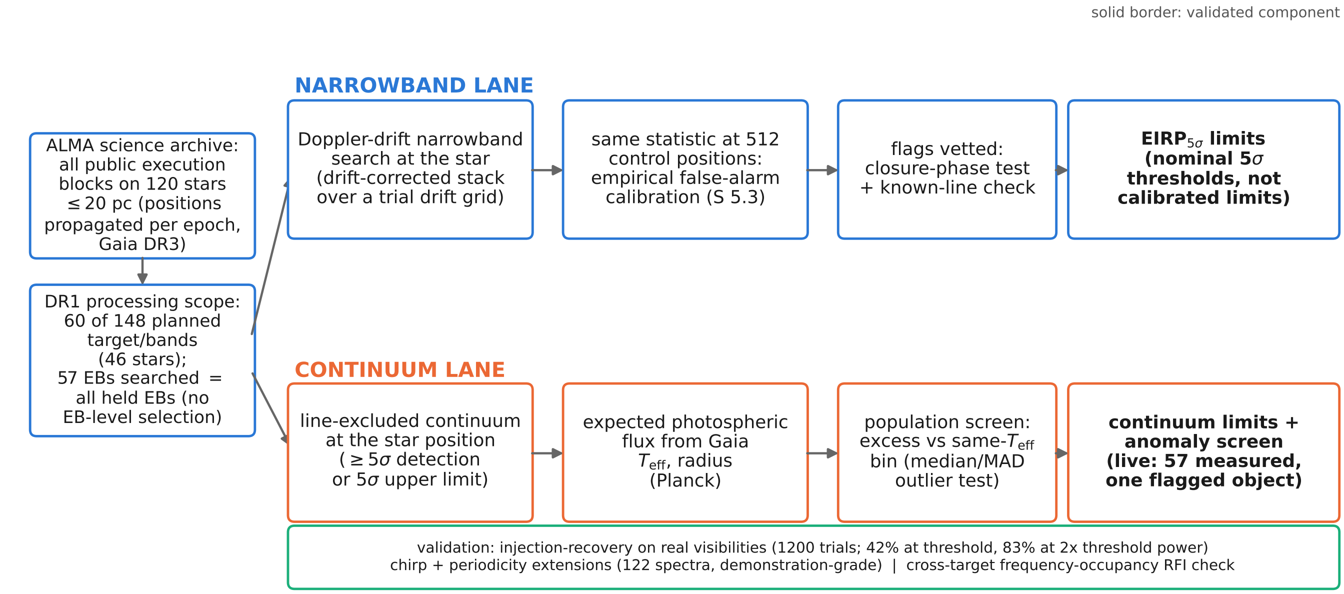

Fig. 2.— Design of the survey. All public execution blocks on the 120 targets are mined from the ALMA science archive, with positions propagated to each epoch from Gaia DR3, and searched along two independent lanes. The narrowband lane runs a Doppler-drift-corrected narrowband search at the star position, calibrates its false-alarm rate empirically against 512 control positions per window, and vets any candidate with a closure-phase point-source test and a known-line check, yielding per-window EIRP$_{5\sigma}$ limits. The continuum lane measures the line-excluded continuum flux at the star position and screens for anomalies against a same-$T_{\rm eff}$ comparison population (§4.4). Solid borders mark validated components. The validation rail summarises the injection-recovery (1200 trials, §4.5), chirp/periodicity, and frequency-occupancy analyses.

the threshold power (EIRP$_{90}$ > 2 EIRP$_{5\sigma}$; §4.5). It is a total-power threshold set by ALMA's coarse native channelisation, not a Hz-resolution narrowband sensitivity. The same reading applies to every subsequent "5σ" in this paper, including the continuum thresholds of §4.4 and Table 21; we state it once here and do not repeat it at each occurrence.

Three terms are defined once here, in strict nesting. A *threshold crossing* (the internal software term is "hit") is a single frequency and drift-rate cell at the star's position whose statistic crosses the nominal 5σ threshold. A *candidate* is a window in which a threshold crossing also passes the spatial-coincidence test against that window's 512-position control ring. A *vetted candidate* is one that survives every subsequent vetting stage (line check, closure phase, cross-target occupancy, and human review) – a status no object of this release attains.

Stellar positions are propagated to each observation's epoch using Gaia DR3 proper motions and parallaxes. The searched drift-rate range is set to the larger of (a) a generic frequency-scaled default (12–13 Hz s^{-1} GHz^{-1}), adopted from the literature fiducial used by Mason et al. (2024) with a threefold safety margin – a chosen design boundary, not a claim that faster drift is physically implausible – and (b) where the target hosts confirmed exoplanets, the maximum line-of-sight orbital acceleration of the innermost known planet, so a transmitter co-located with a short-period planet cannot drift out of the searched window. The scale of the adopted ceiling: $\dot{\nu}$ at frequency ν corresponds to a line-of-sight acceleration $a_{\rm los} = c\dot{\nu}/\nu \approx 3.9\,{\rm m\,s^{-2}}$ (∼0.04 au around a solar-type star); Earth-motion contributions ($\lesssim 0.13\,{\rm Hz\,s^{-1}\,GHz^{-1}}$ combined) are two orders of magnitude below it and subsumed by the grid. These are *apparent* topocentric drift rates: they include any deliberate transmitted chirp as well as acceleration-induced drift, and the survey does not attempt to distinguish the two – the constraints of this paper concern linear apparent drift within the searched range only.

Across the 225 windows the maximum searched drift rates span 1.1–5.9 kHz s^{-1}, the native channel widths 15.3 kHz to 31.25 MHz (median 15.625 MHz; Table 21), and the number of trial drift rates per window 2 to 1944 (median 4). The threefold margin is consequential: at a 1× margin the release would have been blind to two of its eight threshold-crossing windows, including the AU Mic candidate itself; at 5× no disposition changes – so transmitters drifting faster than ∼13 Hz s^{-1} GHz^{-1} remain unconstrained. Intra-integration smearing is negligible almost everywhere: at most 1.10 channels, median 0.001; only three windows have $\eta_{\rm smear} < 0.99$ (marked * in Table 21; Table 12, Appendix C), the worst being the β Pic Band 6 fine window (effective threshold 1.74× nominal). The channel widths are the effective resolutions (verified against the measurement sets' own spectral window tables; Appendix C).

4.1. *Frequency and velocity reference frames*

Every frequency axis in this paper – searched spectra, tabulated ranges, candidate frequencies – is *topocentric*: the frequencies delivered by the ALMA correlator, with no velocity-frame shift applied to any data product at any stage (no LSRK anywhere in the pipeline or tables). Velocities enter at exactly one place: the stellar-frame line-exclusion mask of the vetting stage, through the chain topocentric → barycentric (ephemeris at the execution-block mid-time) → stellar rest frame (catalogued systemic velocity) → offset from each catalogued molecular rest frequency. The chain's residual error is the catalogue radial-velocity uncertainty alone (a few $\mathrm{km\,s^{-1}}$ at 20 pc). The stacking behind the search statistic is coherent in frequency and in the drift-rate trial but preserves no electric-field phase coherence between integrations; no phase-coherent integration of complex visibilities is attempted anywhere in this pipeline.

The native channel width is also the linewidth convention under which every EIRP threshold must be read, stated once here because it is the commonest source of confusion when comparing technosignature limits across surveys. The detection statistic is the flux density in a single channel after drift correction, so the quoted $\mathrm{EIRP}_{5\sigma}$ applies to *any* transmitter narrower than the native channel – from tens of megahertz down to a fraction of a hertz – with the threshold flat in the assumed linewidth (exact for a channel-centred tone; a boundary-straddling tone can cost up to a factor ∼2 in the worst case, and the measured channel-phase dependence is folded into the completeness of §4.5). Dividing the quoted limit by the channel width gives the minimum detectable spectral power for transmitters *wider* than the channel. The worked example – UV Ceti Band 3: rms 0.244 mJy, $S_{\min} = 1.22$ mJy at 2.67 pc, $\mathrm{EIRP}_{5\sigma} = 1.6 \times 10^{13}$ W at 15.625 MHz, or 4×10^9 W had the observation been channelised at 1 Hz ($P_{\min} \propto \sqrt{\Delta\nu}$; the ∼4000× price of the archival channelisation) – together with the smearing table and the channel-width verification, is given in Appendix C (Fig. 7).

The narrowband search covers every distinct spectral window configured for an observation, up to a practical cap of 8 windows per target to bound compute cost. The cap bound exactly one target/band (61 Vir Band 7: four of eight windows searched; Table 21), so the frequency-space selection function it introduces is fully tabulated rather than hidden. Of the 60 target/bands, 53 have every configured window searched; the remaining 3 (τ Cet, Kapteyn's Star, Van Maanen's Star) are represented by their deepest single window, explicitly marked in Table 21, with full multi-window coverage deferred to the next release. Each searched window yields its own independent $\mathrm{EIRP}_{5\sigma}$ limit, reported as a separate row of Table 21; Figure 5 plots only the deepest result per target/band.

Any threshold crossing is cross-checked against the rest frequencies of the eight common mm/submm molecular species (CO, ^{13}CO, C^{18}O, HCN, HCO$^+$, CS, CN, SiO; §5.8). The mask is defined in velocity space: a crossing is rejected as probable line contamination if its recovered frequency, converted to the star's frame, lies within $\pm 50\,\mathrm{km\,s^{-1}}$ of a catalogued transition of any of these species – velocity coincidence, not drift, being the operative discriminant (a technosignature is free to appear at any frequency and drift rate, whereas circumstellar emission sits at rest offset by the gas's own kinematics). The searches as executed used a narrower one-channel-width implementation that agrees with the velocity-space form on every disposition of this release; the tolerance's quantitative justification and the searched bandwidth it removes are documented in §5.4.

Both the narrowband thresholds and the continuum fluxes are corrected for ALMA's primary-beam response at the star's actual offset from the phase centre. For the great majority of targets the correction is negligible (median factor 1.0001, 90th percentile 1.012, none above 2%). Two datasets sit at large offset – Sirius B (corrections 3.6–16% across bands) and ϵ Eri Band 6 (2.9–3.4×) – where the Gaussian beam form is itself an approximation; both are marked provisional in Table 21 and excluded from the headline ensemble statistics. Excluding them changes neither endpoint of the quoted EIRP range nor the median ($2.6 \to 2.9 \times 10^{14}$ W) nor any conclusion; recomputation against the holography beam model is committed for the corrected tagged release.

4.2. *Reference transmitter benchmarks*

For physical context we compare the EIRP thresholds of §5 throughout to two benchmark values for real human transmitters, both from Sheikh et al. (2025a): an Arecibo-like planetary radar (S-band EIRP 2×10^{13} W, the directed on-axis product of peak power and antenna gain – the strongest transmitter built by humans, used here purely as a technological-power benchmark, not as a claim that such a radar radiates at mm/submm frequencies), and unintentional Earth mobile-network leakage (EIRP of order 4×10^9 W). The former represents the most powerful deliberate directed transmitter an Earth-equivalent civilisation could point at us; the latter entirely unintentional background leakage with no signalling intent.

4.3. *Closure-phase point-source vetting*

Threshold-crossing candidates are subjected to a further automated test that single-dish and beamformed surveys lack, since it requires multiple independent baselines: the interferometric closure phase (phase sum around a closed baseline triangle), exactly zero for an unresolved point source at any sky position and immune to per-station calibration errors. Inconsistency with zero at the drift-tracked candidate frequency is evidence against celestial point-source behaviour; the converse does not hold (every unresolved point source passes, as would a coherent far-sidelobe interferer), so this is a strictly one-sided exclusion diagnostic, never an authentication. The statistic is formed in circular form over closed triangles with a null calibrated against the observation's own integrations; Appendix D gives the full specification. The machinery is implemented and simulation-validated, but sensitivity-limited at this survey's flux densities: in the positive-control fields (including the resolved AT Mic binary) per-baseline S/N never exceeds 5.1, so closure phase cannot authenticate or exclude any continuum detection of this release, and for the candidates it retains its intended role as a rejection diagnostic on threshold crossing. The three candidates of this release are all consistent with a point source (z = 0.16 for AU Mic; §5.4).

The spatial-coincidence test it complements is stated here because §5.4 applies it as the definition of a candidate:

a window's narrowband result becomes a candidate when the star's peak signal-to-noise ratio, after drift correction, exceeds the maximum of the same statistic across that window's 512 control positions.

4.4. *Population-level continuum anomaly screen*

Because the sample is a census rather than an interest-selected list, it supports an ancillary screen: whether a star's measured continuum flux is anomalous relative to its own photospheric expectation (Planck function with Gaia DR3 parameters; Pecaut & Mamajek fallback radius where GSP-Phot is unavailable) and to same-$T_{\rm eff}$ comparison stars, in coarse bins with a minimum of five comparisons (Appendix F gives the full specification and error model). This screen is exploratory ancillary analysis in this release, not a headline technosignature constraint: the samples are small (57 measurements: 17 detections, 40 limits; bins of 4–7 stars) and a millimetre excess is at best an indirect, model-dependent observable of waste-heat technology, so we attach no strong technosignature interpretation to it. Its outcome this release: one outlier, χ^1 Ori at Band 3 (excess ratio 2.52, the largest among the 17 detections; resampled tail probability 0.53 in its six-star bin – the ranked outlier of a small sample, not a significant departure), consistent with astrophysical channels (it is a spectroscopic G0V binary; unresolved companion or chromosphere), and no star's technosignature disposition rests on it (§5.7).

4.5. *Injection-recovery sensitivity validation*

The limits quoted in this paper assume a 5σ threshold behaves as an idealised Gaussian-noise calculation predicts. We validate this directly by injecting synthetic narrowband signals of known amplitude into real, calibrated visibility data – injected at the visibility level, as far upstream of the search statistic as the retained products allow, using the same mechanism built into the extraction code – and measuring what fraction the unmodified pipeline recovers. An early 24-tone pilot on Proxima Cen (zero-drift only) located the 50 per cent recovery point near 6σ; the campaign below supersedes it and reproduces that point as threshold marginality.

The campaign injects into six window configurations spanning three bands and both resolution classes of the survey – three coarse Band 7 spws of AT Mic, one coarse Band 3 and one coarse Band 6 spw of AU Mic, and AU Mic's 488-kHz flagband window (49-trial drift grid) – with a complex tone at the star position at amplitudes 4–10× the statistic's own noise, drift fractions $f_{\rm drift} = 0$, 0.25, 0.5, 0.75, 1.0 of each window's searched ceiling, on- and half-grid rates, and channel-centred and boundary placements: 200 trials per configuration, 1200 in total, each run through the unmodified search with the recovery criterion frozen beforehand ($T \geq 5$, peak channel within 3 of the injection channel, peak drift within one grid step; Fig. 3).

The first result is structural and defines the searched class. The pipeline's per-channel time-median subtraction – the step that removes each position's continuum – also absorbs any phase-steady carrier that dwells in one channel for more than roughly 40 per cent of the track. On the coarse windows this is total: within the drift ceiling every drift is sub-channel over the track (the trial grid degenerates to the zero-shift average), so *every* in-grid carrier is static in this sense – the pipeline-identical variant recovers 0 of 500 coarse-window injections at every amplitude from 4σ to 10σ. The coarse-window $\mathrm{EIRP}_{5\sigma}$ limits therefore bound transmitters whose signal is time-variable on track timescales – amplitude-, phase-, or duty-cycle-modulated below the $\sim$50 per cent dwell threshold – and say nothing about phase-steady carriers, which the statistic suppresses exactly as it suppresses static continuum. The fine window measures the same boundary in drift space: $f_{\rm drift} \leq 0.25$ carriers (dwelling $\geq$40 per cent of the track per channel) are fully absorbed, while $f_{\rm drift} \geq 0.5$ carriers survive the subtraction and constitute the searched class.

The second result is the completeness curve of the searched class. For drifting carriers on the fine window ($f_{\rm drift} \geq 0.5$, pooled over grid phase and sub-channel placement), the pipeline recovers 17, 42, 58, 75, and 83 per cent at 4, 5, 6, 8, and 10σ: roughly half at the nominal threshold, saturating by twice the threshold power. The search machinery in isolation (tone injected after the subtraction) is amplitude-faithful to within $-9/+14$ per cent for channel-centred tones at every drift fraction; boundary-straddling placement costs a factor 0.6–0.8 in the statistic – resolved channel-phase-wise, recovery at 5σ is 83 per cent for channel-centred tones and 0 per cent for boundary-straddling ones (100 and 67 per cent at 10σ), with the 50-per-cent crossing near 1.4× threshold amplitude and full recovery at $\sim$1.6× threshold power. This channel-phase dependence is folded into the survey completeness by using the placement-pooled curve above (an average over sub-channel phase), which is conservative for the boundary-straddling case (§6). As named quantities, for the searched class of the fine window: $\mathrm{EIRP}_{50} = 1.10\,\mathrm{EIRP}_{5\sigma}$ (the interpolated half-recovery point, 5.5σ), and $\mathrm{EIRP}_{90} > 2.0\,\mathrm{EIRP}_{5\sigma}$ as a bound – the curve reaches 83 per cent at the largest amplitude injected, so the 90-per-cent point lies beyond the measured range. Coarse windows carry no EIRP_{50} for the searched class by construction. Every table stays in nominal $\mathrm{EIRP}_{5\sigma}$; Table 1 collects the per-class reading, and the occurrence limit of §6 is the one place the measured completeness is folded in numerically.

The campaign's scope is configuration-level, and we state it explicitly. It measures six configurations on three stars – Bands 3, 6, and 7 directly; the Band 8 windows share the coarse-channel construction, over which the static-carrier result holds by the same mechanism, but the drifting-class completeness is transferred by shared construction rather than measured (noted again in §6 and the abstract). Not spanned: other target environments, primary-beam-offset positions, integration durations, and edge-phased drift grids; campaigns on Band 8 data and on the β Pictoris windows themselves are infeasible with the measurement sets retained locally for this release and are declared as such rather than approximated. Configuration-spanning injections are the first priority of the next release.

4.6. *Extended search of retained spectra: chirped drift and periodicity*

TABLE 1
PER-CLASS READING OF THE QUOTED LIMITS (R1 READING RULE). $C_{5\sigma}$ IS THE MEASURED RECOVERY OF THE CLASS AT THE NOMINAL THRESHOLD.

Window class	Searched class	$C_{5\sigma}$	EIRP_{50}	Static-carrier recovery
Fine ($\leq$1 MHz)	drifting	0.42	1.10 $\mathrm{EIRP}_{5\sigma}$	0/20 (absorbed)
Coarse (15.625 MHz)	time-variable	not calibrated[a]	...	0/500 (by construction)

[a]Every in-grid drift on a coarse window is sub-channel over the track, so the injected (constant) tones were all phase-steady; recovery of the time-variable modulation that defines the coarse searched class has not been measured. Static (phase-steady) carriers are suppressed on both classes.

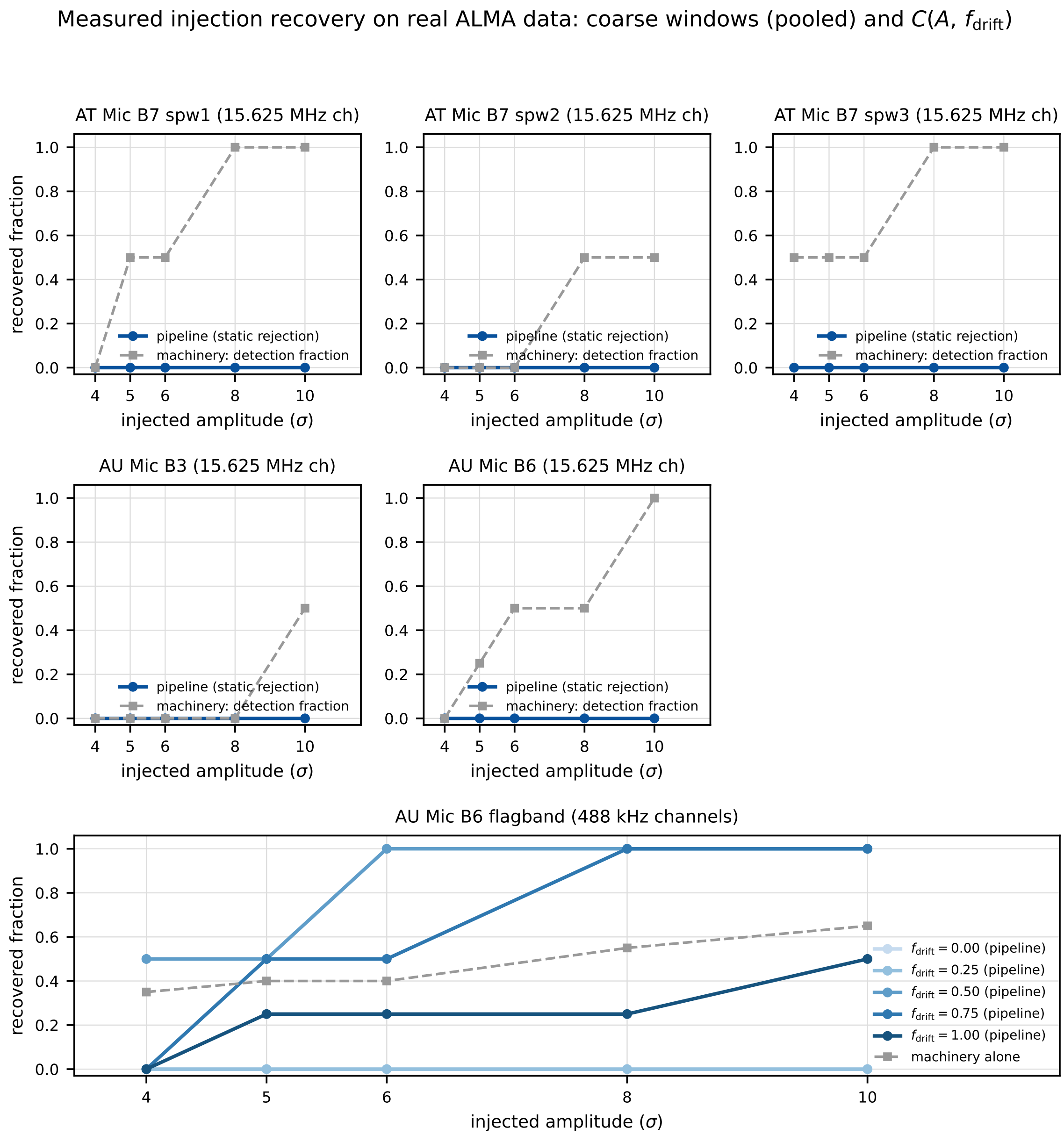

FIG. 3.— Measured injection recovery on real ALMA data (1200 trials, §4.5). Top: the five coarse-channel configurations (15.625 MHz channels; Bands 3, 6, 7), pooled over drift fraction and grid phase – within the searched ceiling every coarse-window drift is sub-channel over the track, so the pipeline-identical variant (solid) recovers zero phase-steady carriers at every amplitude by construction, and the machinery's detection fraction (dashed; $T \geq 5$ and peak channel within 3 of the injection) is the meaningful remaining quantity. Bottom: the 488-kHz-channel configuration, where the drift dimension is real (49-trial grid); the pipeline-identical recovery $C(A, f_{\rm drift})$ rises with amplitude for drifting carriers ($f_{\rm drift} \geq 0.5$) while static and near-static carriers ($f_{\rm drift} \leq 0.25$) are absorbed by the per-channel median subtraction. The frozen recovery criterion includes peak-drift attribution within one grid step; boundary-straddling tones can fail that clause at high drift while remaining detected, which is the origin of the non-monotonic drift dependence near the ceiling.

TABLE 2
Scope of the analyses in this release: what each was applied to, its validation status, and whether it is used in the headline limits.

Analysis	Applied to	Validation status	In headline limits?
Primary narrowband search	225 windows / 46 stars (42 systems)	Injection-calibrated for the searched class (1200 trials; §4.5); false-alarm rate calibrated empirically (§5.4)	Yes
Continuum anomaly screen	57 continuum measurements	Exploratory; small-sample resampling (§4.4, Appendix F)	No
Closure-phase vetting	3 candidates	Simulation-validated; sensitivity-limited on sky (§4.3, Appendix D)	Disposition only
Injection-recovery validation	6 configurations / 3 stars	This release; scope limits stated (§4.5)	Completeness only
Chirp/periodicity screen	122 retained spectra	Demonstration-grade (Appendix G)	No
Frequency-occupancy check	366 overlapping star pairs	Executed (§4.7, Appendix H)	Disposition only

The primary search stacks at constant linear drift rate, so it is by design blind to a transmitter whose drift rate itself changes during the observation and to pulsed or periodically-modulated carriers. We implemented a chirped (quadratic-drift) extension of the same stacking search and a per-channel time-domain periodicity search, each vetting the star's dynamic spectrum against 8 retained control positions. Applied to the 122 retained spectra this screen is a demonstration experiment, not survey-grade (full specification in Appendix G); its null behaviour is as expected (star-exceeds-ensemble rates 10.7 and 13.1 per cent against an 11-per-cent chance rate), no instance is treated as a candidate, and no limit in this paper derives from it. Extending to the full 512-position ring would promote it to survey grade and is planned for the next release.

4.7. *Cross-target frequency-occupancy check*

This single-pointing archival survey has no ON/OFF cadence for rejecting radio-frequency interference; its analogous resource is the many independent, unrelated stars sharing similar correlator setups. A narrowband feature recurring at the same observed *sky frequency* across unrelated targets is essentially certain to be RFI or instrumental. We cross-match threshold crossings towards different targets when they fall within $k = 3$ native channels of the coarser width (kernel and caveats in Appendix H); the null rejects sky-frequency recurrence only – artefacts fixed in the correlator/IF frame defeat it, and a channel-index cross-match (also null) partially covers that case. Applied to the 225 windows and the 107 tabulated crossings of this release: within the 366 of 946 star pairs whose searched-frequency unions overlap by at least 0.01 GHz – 109 involving a crossing-carrying star, so every crossing frequency was tested against genuinely overlapping independent targets – no pair falls within the kernel (or within 0.5 MHz). One bookkeeping caveat, stated in full: the retained products cap each window's list at 50 crossings. Survey-wide the true count is 171 crossings, of which 107 are tabulated here; the difference is one window – the β Pic Band 6 ultra-fine window, with 114 crossings of which 50 are tabulated. The tabulated 50 span 0.748 MHz; the four crossings towards HD 285968 – the only other star with crossings in that frequency region – lie 13.7 MHz from that extent but inside the window's full 54-MHz searched span, so the 64 untabulated crossings, whose individual frequencies the retained products do not carry, could not be tested for coincidence. The null is therefore demonstrated for the tabulated set and declared unresolved for the untabulated remainder; the public release ships uncapped lists (Data Availability). The kernel ($\sim$46 kHz) is narrower than the β Pictoris line offsets of 10.6 and 21.9 MHz, so those candidates were dispositioned by line-catalogue inspection, not by this check.

Table 2 summarises every analysis of this release – the primary narrowband and continuum searches and the five supplementary analyses – with its validation status and whether it enters the headline limits.

5. RESULTS

The counting quantities of this release – systems, stars, star/bands, windows, execution blocks, and projects – are those of Table 7 (§3); levels are not independent statistical trials (unresolved binaries share one measurement set).

The detailed results that follow report the $\leq$20 pc release of version 3.24 (see Scope of this version). Of the 148 target/bands planned for that subsample (spanning 46 unique stars), 60 were searched to completion and yielded a valid result; all values in this section correspond to the frozen pipeline and data snapshot of this paper. The survey is strictly volume-ordered (nearest star first; §3), so the results reported here are weighted towards the very nearest stars in the sample, spanning 1.3–19.6 pc. Full per-target values, including the exact frequency range searched in each band, are given in Table 21 in online-only Appendix I rather than in the main text; this section summarises the results and highlights points of interest.

5.1. *Per-star summary*

The one-line-per-star view of this release – Table 18 in the online-only appendix, complementing the per-window Table 21 – gives the deepest nominal $\mathrm{EIRP}_{5\sigma}$ threshold among all windows of each star, its band, and that band's continuum-lane result. Two stars of the release are continuum-only. All values are read directly from Table 21; the

TABLE 3
THE NARROWBAND DECISION CASCADE OF THIS RELEASE, FROM TRIALS TO DISPOSITION.

Stage	Count
Channel × drift trial cells (all windows)	1.97×10^{7}
Control positions compared per cell-set	512 per window
Threshold-crossing frequencies ($\geq 5\sigma$ at star position)	171 true / 107 tabulated[a]
Windows containing ≥1 threshold crossing	8 (5 stars)
Candidates (crossing + star beats ring)	3 (2 stars)
dispositioned as circumstellar CO	2 (β Pic)
consistent with empirical background	1 (AU Mic)
Vetted candidates (credible technosignatures)	0

[a]The retained products cap each window's crossing list at 50; the deficit is one window (§4.7).

TABLE 4
THE THREE CANDIDATES OF THIS RELEASE AND THE TEST THAT DISPOSITIONED EACH. 'LINE' IS THE STELLAR-FRAME OFFSET FROM THE NEAREST CATALOGUED TRANSITION (TABLE 5); 'SPATIAL / RECURRENCE' THE STAR-VERSUS-CONTROL OUTCOME AND, WHERE ONE EXISTS, THE SAME-STAR SECOND-EPOCH TEST. NO DISPOSITION RESTS ON A SINGLE TEST.

Candidate	Band	Line	Spatial / recurrence	Disposition
β Pic	B3	CO(1→0), $-2.7\,\mathrm{km\,s^{-1}}$	CO-consistent kinematics at 3 epochs, 2 transitions	circumstellar CO
β Pic	B6	CO(2→1), $-3.5\,\mathrm{km\,s^{-1}}$	as above (same gas component)	circumstellar CO
AU Mic	B6	none ($+379\,\mathrm{km\,s^{-1}}$)	$T_{\star} = 4.93$ vs ring max 7.70; no recurrence	empirical background

summary adds no new numbers. Reading rule: EIRP values are nominal native-channel thresholds under the boxed rule of §4, and continuum fluxes are primary-beam corrected. 'Bands' lists the ALMA bands with searched windows.

5.2. *Barnard's Star and Wolf 359*

Barnard's Star and Wolf 359 – the second- and fourth-nearest stellar systems – have not, to our knowledge, previously been searched for technosignatures at ALMA frequencies. Both yield clean non-detections from public Band 6 archival data: $\mathrm{EIRP}_{5\sigma}$ limits of 6.15–6.89 × 10^{13} W (Barnard's Star, 4 windows) and 7.84–9.10 × 10^{13} W (Wolf 359), with continuum upper limits of 0.575 and 0.453 mJy. These are the first reported mm-wave constraints on either star.

5.3. *Data-quality exclusions and open items*

Status of the 148 target/bands planned for the ≤20 pc release: 60 searched to completion and reported here; two attempted and failed at the process level (item (ii) below), excluded from all counts including the headline 60; 86 queued for later releases (60 + 2 + 86 = 148). Within the 60, the entries absent from Table 21 are, with forensic detail in the online-only appendices: (i) *planned but computationally failed* – four target/bands (Barnard's Star [B7], Wolf 359 [B7], SCR J1845−6357, Wolf 358) have no narrowband result after process-level search failures; each retains a valid continuum non-detection and none carries an EIRP threshold. (ii) *excluded for data quality* – the Giclas 9−38 pair [B3]: its only matched member observation unit set points ∼52 arcmin from either star. (iii) *partially withheld* – one spectral window in each of HD 10647 and HD 139664 [B6] shows a reproducible processing defect (12.1 s on-source against 393–520 s for siblings, implausibly small noise); the affected window is excluded, the combined continuum withheld, and their other windows retained. (iv) *relabelled* – the entry previously carried as "γ Lupi" resolves to the F4V debris-disc host HD 139664 at 17.40 pc (Gaia DR3 5989102478619519616); no measured value changes. The supporting audit (Appendix I) is clean across all 225 windows except the four χ^1 Ori Band 3 windows (elevated system temperature; limits weakened, not invalidated), and all 120 census identifiers independently re-resolve through SIMBAD and *Gaia* to 120 distinct sources, none outside 20 pc.

5.4. *Technosignature search*

Fifty-six of the 60 target/bands yielded at least one narrowband result (the four failures are itemised in §5.3); together these span 225 individual spectral-window measurements after removal of 13 duplicate rows identified in the catalogue audit (Appendix I). The resulting $\mathrm{EIRP}_{5\sigma}$ limits span 1.6×10^{13}–7.1×10^{16} W, median 2.7×10^{14} W (distribution in Figure 5, frequency-coded in Figure 6, placed among published searches in Figure 1). As expected for a volume-ordered survey, the deepest limits are for the very nearest systems (1.6×10^{13} W for the UV Ceti pair at 2.7 pc); the shallowest is η Corvi, the first Band 8 target searched, where distance, coarse channels, and receiver performance combine to set the survey's current ceiling. Against the benchmarks of §4.2: only the handful of targets within ∼3 pc reach below the Arecibo-like planetary-radar EIRP, and no target approaches the $\sim 4 \times 10^{9}$ W of unintentional Earth leakage – a sobering calibration of what even a deep, targeted, archival mm-wave search cannot yet see.

Three narrowband features have been promoted to *candidates* – the vocabulary (threshold crossing → candidate → vetted candidate) is that defined after Figure 2. Table 3 follows the decision cascade for the release as a whole; Table 4 gives each candidate's disposition with the test that decided it.

Two questions are kept separate: is a release containing three candidates more than this survey's own calibration predicts (a counting question about the whole release), and is any individual candidate statistically distinguishable from the empirical background (answered per candidate in the dispositions that follow)?

SURVEY-LEVEL FALSE-ALARM CALIBRATION

Each window-level test compares the peak signal-to-noise ratio at the star's position with the same statistic at 512 control positions – an empirical calibration that absorbs the per-window trials factor (the drift-rate grid and the thousands of channels are applied identically to star and controls). From this version the statistic is *symmetric*: the on-source quantity is the single propagated stellar position, exactly what each control contributes. Earlier versions maximised the on-source quantity over a small region centred on that position while each control remained a single position, and that asymmetry was not benign. Under a pure-noise null with exchangeable positions, the maximum of $n_{\rm src}$ source-region positions exceeds the maximum of $n_{\rm ctrl}$ single controls with probability $n_{\rm src}/(n_{\rm src}+n_{\rm ctrl})$; for the $n_{\rm src}=135$ region and 512 controls of a representative window that is 21 per cent, and the measured region-max star-beats-ring rate across this snapshot, 67 of 448 windows (15.0 per cent), is of that order. The symmetric statistic instead produces star-beats-ring in 6 of 448 windows (1.3 per cent), consistent with the exchangeable expectation $1/513 = 0.19$ per cent per window inflated by the real maps' heavy tails, and stable against the 1.3 per cent measured on the $\leq$20 pc subsample alone. Adopting it makes the nominal exchangeability rate the operative one, so the survey-level false-alarm expectation follows directly rather than through the bookkeeping ladder that the asymmetry previously required.

What gates a candidate is the joint criterion, and its threshold leg dominates the rate: only 22 of 448 windows contain any $\geq 5\sigma$ crossing at the stellar position. With a symmetric statistic the survey-level expectation is immediate: each window contributes at most the exchangeable rate $1/(1+n_{\rm ctrl}) = 1/513$, so across 448 windows the expected number of flags under the pure-noise null is $448/513 = 0.87$. Four flagged target/bands against 0.87 expected is not a significant excess, and three of the four carry independent astrophysical attributions (below), leaving one marginal crossing – precisely the order of the expectation. This is the operative survey-level false-alarm estimate of this paper; it absorbs execution-block systematics into the fixed block margins and assumes only that, within a block, crossing status is exchangeable with star-beats-ring status – exactly the no-association null of interest. Secondary bookkeeping models span 1.0–9.2 per cent (Appendix J); the dominant residual uncertainty is the star-beats-ring rate itself (Wilson 95 per cent interval 9.5–18.4 per cent), propagating to a 4–19 per cent bracket on $P(F \geq 3)$. A further limit of the ensemble deserves statement: with 512 controls the smallest resolvable per-window p-value is 1/513, so a Bonferroni threshold over 448 windows (1.1×10^{-4}) lies below what the control ensemble can measure. The controls can therefore establish that a crossing is unremarkable, but cannot on their own certify one as significant survey-wide; that burden falls on localisation, recurrence, and astrophysical attribution, which is how each flag below is decided. The two β Pictoris dispositions do not depend on the bracket at all (they rest on kinematic attribution to a mapped disc), the AU Mic candidate's standing is a rank statement (invariant to the bookkeeping model; conditional probability 41 per cent empirical, 34 per cent idealised), and no model in the bracket promotes any of the three to a vetted candidate: no conclusion of this paper inverts within the bracket.

The expected number of chance crossings is also consistent with the trials count once adjacency is handled. The release evaluates 1.97×10^7 channel$\times$drift trial cells; a Gaussian one-sided 5σ tail (2.87×10^{-7}) over that count predicts 5.6 chance crossings, against 171 observed. The excess is the burst structure the raw count ignores: the drift-tracked statistic is strongly correlated between adjacent channels ($\rho_{\rm lag1} = 0.989$–0.991, i.e. 8.5×10^4–1.1×10^5 effectively independent channel cells rather than 1.97×10^7), so crossings arrive in contiguous runs. Collapsing to contiguous clusters (kernel of 3 native channels) reduces the 171 crossings to 9 – consistent with the Poisson tail of 9 or more given 5.6 expected ($P = 0.12$). It is the cluster count, not the raw crossing count, that must be compared with the trials expectation, and the two agree.

The star-beats-ring rate is not driven by any single stratum of the release. By band: 20 per cent (B3, $n = 40$), 10.9 per cent (B6, $n = 128$), 17.8 per cent (B7, $n = 45$), and zero in the sparsely sampled B4/B5/B8 ($n = 4$ each). By channelisation: 17.0 per cent for fine windows ($n = 47$) against 12.4 per cent for coarse ($n = 178$). By primary-beam offset: 15.7 per cent above the median offset (0.10 arcsec) against 11.0 per cent below. The disc-hosting caveat is empirically inverted in this release: the 132 windows towards disc- or planet-hosting stars contribute all eight crossing windows and all three candidates but only 13 star-beats-ring wins (9.8 per cent), while the 93 towards non-hosts contribute no crossings yet 17 wins (18.3 per cent) – real circumstellar emission shared by star and ring is not what drives the statistic. Excluding the fifteen β Pictoris windows leaves one unattributed candidate among 210 windows, against $4 \times \hat{p}_{\rm sbr} \approx 0.52$ expected ($P(\geq 1) = 41$ per cent) on the empirical calibration.

The control ensemble's geometry carries two properties. The 512 positions form a ring at a radius small compared with the primary beam (targets sit at a median 0.1 arcsec from pointing centres), so all controls share the target's primary-beam gain – no primary-beam-gradient bias enters the comparison. And neighbouring positions are correlated below the synthesised-beam scale, so the effective number of independent spatial trials falls from 512 to $N_{\rm eff} \simeq 30$–43 as the symmetric reprocessing runs measure directly – a floor the empirical 13.3 per cent rate, measured on the actual correlated ensembles, is immune to by construction. A continuum source in the ring does not raise the narrowband statistic unless it also emits a narrow line in-window; the cross-target occupancy screen (§4.7) covers recurring frequencies.

THE SYMMETRIC STAR-VERSUS-CONTROL NULL

The designed principal test of this survey applies the identical source-region, drift-rate, and frequency maximisation at the star's position and at every one of the 512 control locations, giving per-window statistics $T_\star$ and $\{T_i\}$ under a single search rule, with $p = (1+\#\{T_i \geq T_\star\})/(N+1)$. It has been executed on the seven locally reprocessed windows: the six validation windows of the local reprocessing set and, with the same machinery, the AU Mic candidate's first-

TABLE 5
Line-attribution audit of the three candidate windows (first three rows) and the coarse β Pictoris Band 6 window without a candidate that independently shows the same CO component, carried through the frame chain of §4.1. Topocentric offsets are observed minus catalogued rest frequency (CO J=1→0 115.271202 GHz, CO J=2→1 230.538000 GHz, Pickett et al. 1998). The barycentric correction is the Earth-ephemeris term at the execution-block mid-time (block uids in Table 10); the systemic radial velocities are the catalogued values (SIMBAD; *Gaia* DR3 for β Pictoris, Gaia Collaboration et al. 2022; *Gaia* DR2 for AU Mic, Fouqué et al. 2018).

Flag	UTC start	$f_{\rm obs}$ (GHz)	Transition	Topo. offset (MHz; km s^{-1})	Bary. corr. (km s^{-1})	$v_{\rm sys}$ (km s^{-1})	Stellar-frame offset (km s^{-1})
β Pic B3	2022-03-02 23:14	115.260644	CO(1→0)	−10.56; −27.46	−7.90	+16.84	−2.72
β Pic B6	2019-03-12 00:14	230.516128	CO(2→1)	−21.87; −28.44	−8.15	+16.84	−3.45
AU Mic B6	2014-08-18 04:01	230.825230	CO(2→1)	+287.23; +373.51	−10.02	−4.71	+378.82
β Pic B6†	2013-10-06 06:23	230.528134	CO(2→1)	−9.87; −12.83	+7.63	+16.84	−3.62

†Coarse Band 6 window without a candidate (Table 21); shown because it independently exhibits the same CO component as the candidate fine window of the same band.

epoch window itself, reprocessed from the raw execution block with the observatory pipeline's own calibration recipe (§5.5). The identical-treatment requirement cannot be met from the frozen release products, which is what makes the local reprocessing set the test's domain. The result: all seven windows give $T_\star = 3.7$–4.9 against control maxima of 6.7–7.7, $p = 0.81$–1.00 (Figure 9, online-only appendix) – the star position is statistically exchangeable with its control ring wherever the test can be run at full symmetry, and no window shows star-side excess. Extending the test to all 225 windows requires raw-visibility reprocessing per window, which the retained products do not support; it is the designated replacement as the remaining measurement sets are re-run, the machinery ships with the release, and until then the block-permutation model above remains the operative release-wide estimate, with the symmetric statistic a validation subset measured wherever the data allow. The AU Mic candidate also fails the single-position variant of the release statistic (star 5.22 against its ring maximum 5.85, §5.5).

Table 16 (online-only appendix) consolidates the family-wise trials factors: every per-window factor is applied identically at the 512 control positions, which is what makes the control ring, and the empirical star-beats-ring rate built on it, the operative calibration rather than any analytic trials correction.

THE β PICTORIS CANDIDATES: CIRCUMSTELLAR CO

The first two candidates are both towards β Pictoris, a debris-disc system with previously reported CO emission (Matrà et al. 2017): one in Band 3 at 115.2606 GHz, 10.56 MHz below the CO(J=1→0) rest frequency, and one in Band 6 at 230.5161 GHz, 21.87 MHz below CO(J=2→1). Neither coincidence is exact, so the automated line check did not catch them directly. Carried through the full frame chain (Table 5; Fig. 4), the two offsets correspond to the same kinematic component – −2.7 and −3.5 km s^{-1} from stellar systemic, 0.74 km s^{-1} apart – in the two lowest CO rotational transitions of a system in which circumstellar CO has been directly detected; the coarse Band 6 window without a candidate, observed in a third epoch six years earlier, independently shows CO at −3.6 km s^{-1} in the same frame. All three epochs and both transitions land within 0.9 km s^{-1} of each other and 3.6 km s^{-1} of systemic, deep inside the ±50 km s^{-1} mask. The raw topocentric offsets (−27.5 and −28.4 km s^{-1}) become these stellar-frame values only after the barycentric correction and the catalogued systemic velocity are applied – the full frame chain matters. We attribute both candidates to line contamination from this kinematically offset CO emission.

Their empirical significance under the release background is nonetheless extreme, and worth stating so the attribution is not mistaken for dismissal: the Band 3 region peak of 20.3 exceeds all 225 per-window control maxima (add-one $p < 0.0044$), and the Band 6 peak of 12.8 exceeds all but one – that window's own CO-filled control ring, which reaches 15.2 (by far the highest control maximum of the release; $p \approx 0.0088$). The candidates stand out sharply in the empirical background, which is exactly why they demanded astrophysical attribution rather than summary rejection as noise. The attribution came from subsequent inspection of the offsets against known CO kinematics, not from the automated chain (the one-channel line check tolerated their offsets; closure phase, which they pass, does not discriminate line emission from an artificial point source); the alternative interpretation would require a transmitter placed coincidentally at matched offsets from two CO transitions in the same star.

LINE EXCLUSION: DEFINITION, COST, AND LIMITS

The searches of this release ran with an automated line check that matches a crossing against catalogued rest frequencies within one spectral channel at zero drift. The β Pictoris candidates showed that check inadequate: both fell outside its one-channel tolerance (by 10.6 and 21.9 MHz) while corresponding, after the frame chain, to a kinematic component within 3.6 km s^{-1} of stellar systemic of a resolved, previously mapped disc – exactly the regime a channel-width tolerance is blind to, since a native channel is a fraction of a km s^{-1} wide while ordinary circumstellar kinematics displace real emission by many channels. Before publication, every crossing was reclassified with the velocity-space mask – ±50 km s^{-1} in the stellar frame around each catalogued transition – and that criterion is the published pipeline's line-exclusion definition from this release onward. The tolerance is set by the kinematic budget of real circumstellar gas: Keplerian orbital motion where these molecules survive ($\gtrsim$10 au) is $\lesssim$13 km s^{-1}, intrinsic linewidths of cold gas are $\lesssim$1 km s^{-1}, the eight masked species are cold dense-gas tracers (no bulk-wind term for the dwarfs dominating the sample), and the barycentric term left uncorrected is bounded by ±30 km s^{-1}. Empirically it is conservative: the

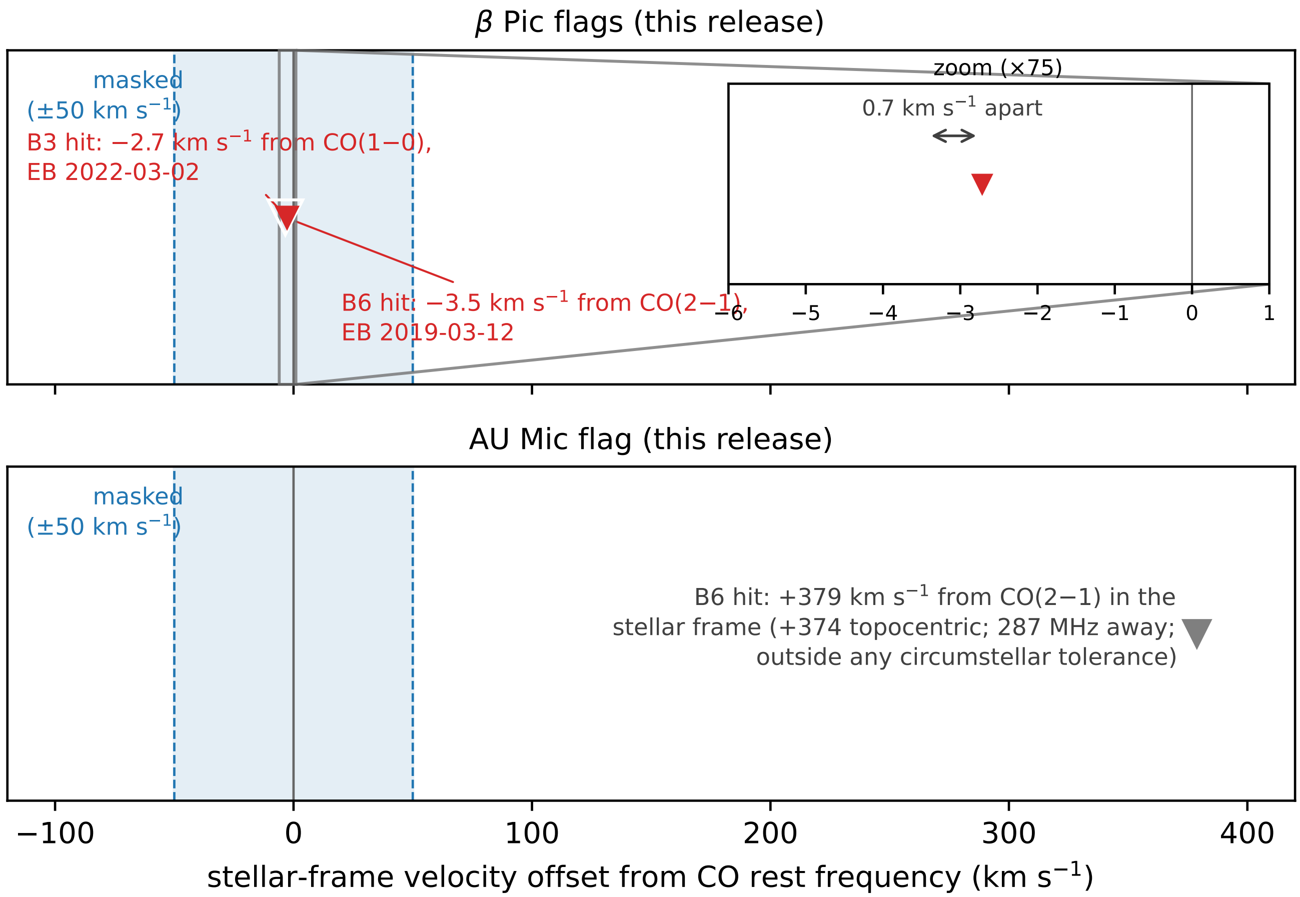

Fig. 4.— Stellar-frame velocity offsets of the candidate windows, carried through the full frame chain (topocentric → barycentric → stellar; Table 5). Shaded band: the $\pm 50\,\mathrm{km\,s^{-1}}$ molecular-line exclusion mask. *Top:* the two β Pictoris candidates fall well inside the mask as a single kinematic component within $3.5\,\mathrm{km\,s^{-1}}$ of stellar systemic, matching the resolved, previously mapped disc (Matrà et al. 2017); the inset resolves the pair ($0.7\,\mathrm{km\,s^{-1}}$ apart). *Bottom:* the AU Mic candidate sits at $+379\,\mathrm{km\,s^{-1}}$ in the stellar frame, far outside any circumstellar tolerance; it is dispositioned by its spatial and recurrence behaviour, not by the mask.

largest attributed circumstellar component (β Pic, -2.7 to $-3.6\,\mathrm{km\,s^{-1}}$) sits at 7 per cent of it, and *zero* candidate frequencies fall between 13 and $50\,\mathrm{km\,s^{-1}}$ from their nearest transition, so a $\pm 13\,\mathrm{km\,s^{-1}}$ Keplerian-only mask would have suppressed exactly the same candidates and released exactly the same five – the wider tolerance costs nothing in candidate yield and buys robustness against radial-velocity catalogue error.

Applied retrospectively, the mask rejects exactly the two β Pictoris candidates and changes no other disposition. Of the remaining six crossing windows, two fall inside the mask (the β Pic coarse window without a candidate at $-3.6\,\mathrm{km\,s^{-1}}$, and HD 285968 at $+8.7\,\mathrm{km\,s^{-1}}$, already rejected by its own control ring, maximum 10.9 against a star peak of 6.5); the rest lie far outside any circumstellar tolerance (AU Mic +379, LHS 1140 $+115\,\mathrm{km\,s^{-1}}$). Both search products are reported explicitly: of the 106 unique tabulated candidate frequencies across the ten crossing windows, 101 are line-coincident within the mask and suppressed by the clean search; the clean product is the remaining five, each dispositioned by the spatial and closure-phase vetting, not by the mask. A counter-argument deserves recording: the same transitions that make astrophysical false positives cheap also make them natural “Schelling points” a deliberate communicator might choose (Cocconi & Morrison 1959). The mask is therefore not a statement that transmitters avoid these frequencies, only that this release cannot distinguish a line-coincident transmitter from gas at its present vetting depth; the released products maintain a known-line-coincident list (every on-source 5σ crossing inside the masked intervals, with its offset from each nearby transition) so that a future search with independent discrimination can revisit exactly the channels set aside.

The masking cost is stated on one definition throughout: every JPL catalogued transition of the eight species (276 transitions), displaced by each system’s radial velocity, $\pm 50\,\mathrm{km\,s^{-1}}$ stellar-frame half-width, intersected with each window and merged. The release effectively searches 82.3 GHz of unique sky frequency – the disjoint union of the searched windows minus the part of the mask inside it (Figure 6) – down from 87.2 GHz before masking (a 5.7 per cent reduction). Summed over the 225 windows, the mask removes 22.2 GHz of the 378.9 GHz of gross window-bandwidth (5.9 per cent; the per-species, per-band breakdown is Table 19, online-only appendix); the per-window usable fraction has median 0.960 (10th percentile 0.790). Union bandwidth is exposure-blind – a window visited for seconds counts

TABLE 6
THE AU MIC CANDIDATE FREQUENCY (230.8252 GHZ) UNDER THE THREE TREATMENTS THAT DECIDE IT: THE RELEASE'S FROZEN PRODUCTS (REGION-MAX STATISTIC), THE FULLY SYMMETRIC REPROCESSING OF THE SAME FIRST-EPOCH EXECUTION BLOCK (2014 AUGUST 18; OBSERVATORY CALIBRATION RECIPE APPLIED VERBATIM), AND THE SECOND-EPOCH BLOCK OF THE SAME PROGRAMME (2014 JUNE 5; IDENTICAL SEARCH TREATMENT).

Statistic	Release (region-max)	Symmetric reprocess.	Second epoch (recurrence)
Star statistic	5.97	4.93	4.77
Control maximum (512)	5.85	7.70	6.78
Star rank p (add-one)	—	0.81	0.97
Single-position star statistic	5.22	—	—
Per-channel S/N (zero drift)	—	2.46	2.22
Peak drift ($\mathrm{Hz\,s^{-1}}$)	1418.5	1593	1281

the same as one integrated for an hour – so we also report the exposure-weighted quantities: the summed per-star frequency union is 371.9 GHz (stars contribute overlapping sky frequency), and the summed product of on-source time and searched bandwidth is 8.3×10^5 s GHz (per-window on-source time spans 12.1–5274 s, median 2570 s). The mask governs dispositions, not the noise-derived thresholds, so the quoted $\mathrm{EIRP}_{5\sigma}$ values are unchanged; no signal was found outside the excluded molecular-line velocity regions in any window of this release. Stated plainly, the mask is a selection effect of the survey's design: no constraint is placed on transmitters within the excluded regions – 5.9 per cent of the searched bandwidth – and a search for deliberately line-adjacent carriers would need a complementary strategy. The exposure is not limited to the candidate windows: 131 of 225 windows (58 per cent) contain at least one catalogued transition of the eight species in-band, and all eight crossing windows do, yet in none does a crossing fall within one native channel of a rest frequency – precisely the regime the velocity-space criterion covers.

5.5. *AU Mic: reclassified, and why the criterion changed*

The AU Mic crossing in ALMA Band 6 at 230.83 GHz was carried as a candidate through version 3.24 and is *not* a candidate under the symmetric criterion adopted here. It is a single-epoch threshold crossing statistically indistinguishable from the empirical control population: 10 of its own 512 controls reach or exceed the stellar statistic, an empirical $p = 0.021$ that is nowhere near the 1/513 floor the criterion requires. It also fails two direct and pre-named tests – it is not localised at the stellar position, and it does not recur at a second epoch (Table 6). It is the worked example that motivated the change: under the old region-max statistic its 135-position source maximum (5.97) cleared the control maximum (5.85) while the star itself (5.22) did not, which is exactly the asymmetry quantified in §5.4.

Localisation. The first-epoch execution block has been reprocessed from the raw science data model with the observatory pipeline's own calibration recipe, and the fully symmetric star-versus-control statistic of §5.4 run on the candidate window itself: $T_\star = 4.93$ against a control maximum of 7.70 ($p = 0.81$) – the star does not beat its control ring, under a reprocessing that retains 310 integrations against the release pipeline's 249 on a finer drift grid. The release's frozen products had indicated the same thing asymmetrically – the source-region peak exceeds the 512-control maximum by only 0.12, a margin smaller than the window-to-window scatter of that statistic, and the candidate fails the single-position variant as well (5.22 against 5.85) – so this window joins the six validation windows with the same outcome: no localised excess at the stellar position.

Recurrence. The archive holds exactly one further public Band 6 block covering 230.83 GHz – the same programme, executed 2014 June 5 – searched with treatment identical to the release: $T_\star = 4.77$ against a control maximum of 6.78 ($p = 0.97$); no channel within ±10 of the first-epoch flagged channel exceeds 3.1; nothing is localised to the stellar position. The second epoch's band rms (1.48 mJy per 488-kHz channel) matches the first epoch's 1.9 mJy, so the test would have recovered a comparable excess: a clean non-recurrence, and the frequency does not recur at any other target in this release. The non-recurrence rejects a continuous or repeating source at this frequency and these epochs – it does not constrain intermittent or precisely-scheduled transmissions outside the sampled windows, a caveat the duty-cycle analysis of §6 quantifies.

Statistical scale. The on-source peak (5.97) lies at the 94.7th percentile of the per-window control-maxima distribution of all 225 windows (median 4.51, 90th percentile 5.81; Figure 10, online-only appendix): twelve windows (5 per cent) have control maxima at or above it, add-one empirical $p = 0.057$, before any correction across 225 windows that would inflate it by more than two orders of magnitude – exactly the kind of marginal excess the rate calculation predicts (one unattributed candidate among 210 non-β Pictoris windows against an expectation of 0.41, $P(\geq 1) = 41$ per cent empirical). The candidate does not coincide with any catalogued molecular transition (287 MHz from CO J=2→1), and closure-phase vetting finds it consistent with a point source – one-sided evidence only (§4.3).

Two pieces of astrophysical context are recorded for future re-examination; neither is the basis of the rejection. AU Mic is an exceptionally active flare star, but the flare hypothesis is tested by the candidate block's own data: its broadband continuum sits at a quiescent 0.15 mJy, the two flares of MacGregor et al. (2020) (16.8 and 6.1 mJy, optically thin, GHz-wide) belong to a later epoch, and an 11 mJy narrowband event over this block's quiescent level would be a factor ∼70 that its own continuum does not show – a flare origin would require a spectrally narrow coherent process leaving the continuum untouched, which the data neither require nor establish, and the same-tuning second-epoch twin shows no burst domination either. And the control positions sit within the same primary-beam footprint, so smooth structure such as AU Mic's debris belt raises star and controls alike and cannot by itself produce a candidate.

The archival epochs also bound what this release can say about duty cycle: each star was observed in one to three execution blocks, searched independently with no epoch stacking, so the non-detection statement concerns the archival epochs observed, not permanent activity. The proper-motion-matched archive census (§6) counts 107 unique public on-star execution blocks over the 44 searched stars against the 57 searched; that surplus is the recurrence pool from which the second-epoch test drew, and §6 turns the epoch structure into a duty-cycle constraint explicitly.

5.6. *Integration-time audit against the ALMA archive*

Because the on-source integrations quoted in this paper were reconstructed from pipeline bookkeeping rather than retained natively, we audited them against the ALMA archive's own independent exposure accounting (`ivoa.obscore` science-intent exposure per unit set; TAP query of 2026 September 8; method, per-block numbers, and caveats in online-only Appendix L). The audit covers all 227 window-level records in 58 star–execution-block groups (the 59 star–EB pairs of the level-1 bookkeeping, with the G 272-61 and AT Mic binaries each collapsed to their shared pointing and AT Mic's narrow flagband window carried separately from its wide windows): 53 of the 58 are conservative, three agree to better than 1%, and two blocks over-count (AT Mic by a factor 1.43 – an internally diagnostic dump-bookkeeping artefact, already corrected to the archive's 877 s in Table 17 – and CD−38 10980 by 7%).

Three statements bound the scientific impact. First, no EIRP threshold in this paper depends on the audited quantity: thresholds derive from the measured per-channel rms, not from integration time, so every limit of Table 21 is unaffected. Second, the audit is conservative for every window of the two candidate hosts of §5.4 – β Pictoris's windows (1824–1845 s) sit inside an archive science exposure of 47 444 s, and AU Mic's inside 109 334 s – so no candidate disposition changes. Third, the exposure-weighted coverage figure of §5.4 is, at worst, optimistic by the same two-block margin: replacing the pipeline values with the archive's lowers the 8.3×10^5 s GHz total by less than 1%. The audit's residual caveats motivate the direct per-EB export committed for the next release. The defect class this audit caught was found by chance in one target.

5.7. *Continuum search (experimental ancillary analysis)*

The continuum lane is an experimental, ancillary analysis rather than a co-equal search: its statistical power is unquantified at this depth (§4.4), and its link to technology is indirect and model-dependent (§4). We report it for completeness and for the population statistics it will accumulate, not as a technosignature result.

Of the 57 target/bands with a usable continuum measurement, seventeen show a $\geq 5\sigma$ continuum detection and forty are non-detections (Figure 5, right panel); a continuum "upper limit" is 5× the map rms at the star's position, primary-beam corrected exactly as the EIRP thresholds are, and the per-target rms values in Table 21 allow rescaling to any other confidence level. The non-detections reach upper limits of 0.036–3.6 mJy (median 0.52 mJy), the deepest towards TRAPPIST-1 in Band 3. Under the combined error model of §4.4, sixteen of the seventeen detections remain above 5σ; the exception (η Corvi Band 7, 4.7σ combined) is treated as marginal. One class of non-detections is worth an explicit sentence: several fields reach bright peaks well away from the target position (ϵ Eri Band 6 at 10.2σ, 18 arcsec from the phase centre; GJ 674 and CD−23 14742 at 10.7 and 9.9σ), none lies within the 3-arcsec positional tolerance of its star, and the pipeline correctly records all as non-detections – whether the ϵ Eri peak is real off-target emission or a primary-beam artefact at ∼0.7 FWHM offset is undecidable from the retained products (holography-beam recomputation committed, §4).

Every detection is astrophysically unsurprising. The G 272-61A/B pair is the unresolved UV Ceti binary (0.85 mJy, attributable to neither component). Of the rest: six are young, active systems with known debris discs or chromospheric activity (τ Cet, β Pic B3/B6, AU Mic, AT Mic A/B); six show the photospheric flux expected for their spectral types (γ Lep and γ Vir in both bands, χ^1 Ori, η Cru; η Corvi's Band 8 limit is likewise photosphere-consistent); and two are late-M dwarfs with documented strong magnetic activity – LP 349−25 (10.5σ, 0.071 mJy) and the auroral emitter LSR J1835+3259 (6.7σ, 0.078 mJy; Hallinan et al. 2007), which merits deeper follow-up. Closure phase cannot discriminate structure for any of the seventeen at these flux densities (§4.3).

The population-level anomaly screen (§4.4) returns one outlier among the seventeen detections: χ^1 Ori Band 3, resampled tail probability 0.53 – the ranked outlier of a small sample, with an astrophysical disposition (unresolved spectroscopic binary plus chromospheric floor; Appendix F).

5.8. *A serendipitous molecular-line catalogue*

The line-exclusion step of the continuum search (§4) identifies every $\geq 5\sigma$ outlier channel in each target's spectrum as a byproduct; no existing mm/submm line survey of the solar neighbourhood is drawn from a distance-limited census of archival coverage, so even a modest catalogue adds value independent of its SETI motivation. Across the subset scanned to date, 18 outlier channel groups were found: twelve are single- or few-channel features with no known transition within 300 MHz, most plausibly statistical. The remaining six are plausibly real – CO(J=1→0) and CO(J=2→1) towards β Pictoris (the stellar-frame component of Table 5 and §5.4); a marginal CO(J=2→1)-consistent feature recurring across two epochs towards HD 285968, reported as tentative only (1–2 channels of significance per epoch); and a single-channel, 195-MHz-offset feature near SiO(J=5→4) towards HD 53143, more likely statistical than real. All are reported openly, including the two we consider probably spurious; the machine-readable catalogue (per-group target, band, channel coordinates, significance, nearest transition) is committed with the tagged public release, and a full-sample re-scan with the completed survey will extend it.

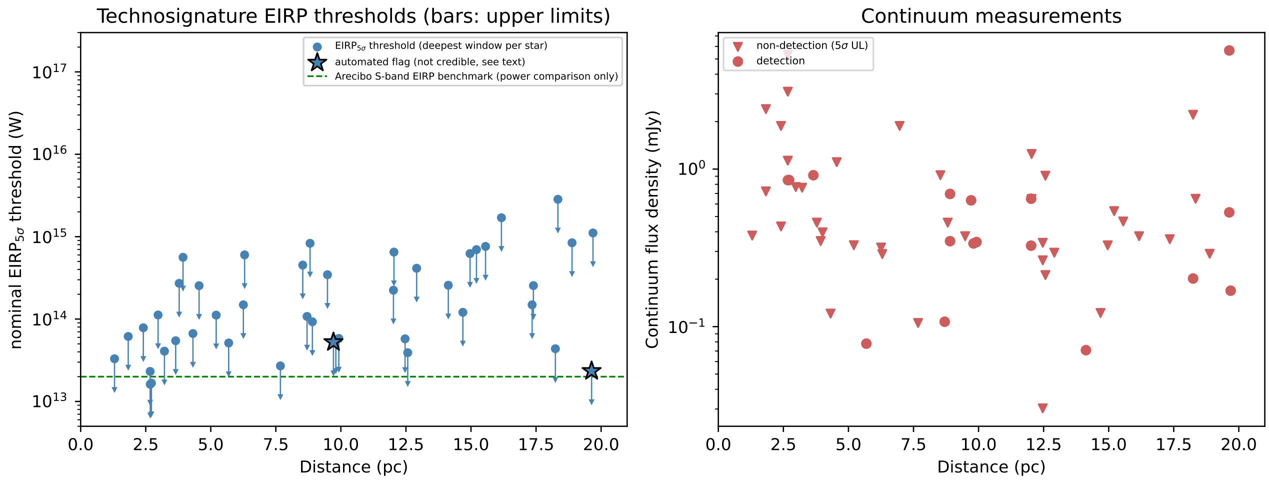

Fig. 5.— Nominal technosignature $\mathrm{EIRP}_{5\sigma}$ thresholds (left; every threshold drawn as a downward bar marking an upper limit) and continuum flux densities (right; downward triangles are 5σ upper limits, i.e. $5\times$ the map rms noise, with rms values given per target in Table 21; filled circles are detections) as a function of distance, for the 60 target/bands (46 unique stars) of this release; the left panel shows one deepest EIRP point per star for the 44 stars with completed narrowband searches (two stars have continuum-only coverage), and the right panel the 57 target/bands with usable continuum maps. The two targets with candidate windows (β Pictoris, Bands 3 and 6, and AU Mic, Band 6) are marked with stars; the β Pictoris candidates are identified in the text as CO disc emission, and the AU Mic candidate as a marginal excess not statistically distinguishable from the empirical background distribution of this release, so neither is a genuine technosignature. The left panel also marks, as a reference line, the EIRP of an Arecibo-like planetary radar (2×10^{13} W, the most powerful deliberate transmitter yet built by humans; Sheikh et al. 2025a, §4.2) – a power comparison only, not a detectability claim at these wavelengths. The ordinate begins at 5×10^{12} W; typical unintentional Earth radio leakage ($\sim 4\times10^{9}$ W of total isotropically radiated power rather than beamed EIRP) lies far below the plotted range and is not shown. Plotted $\mathrm{EIRP}_{5\sigma}$ values are nominal 5σ thresholds, to be read under the boxed reading rule of §4: the injection campaign of §4.5 measures ∼42% recovery at the threshold itself for morphology-matching carriers (∼83% at twice the threshold power), with phase-steady carriers on coarse-channel windows suppressed by construction.

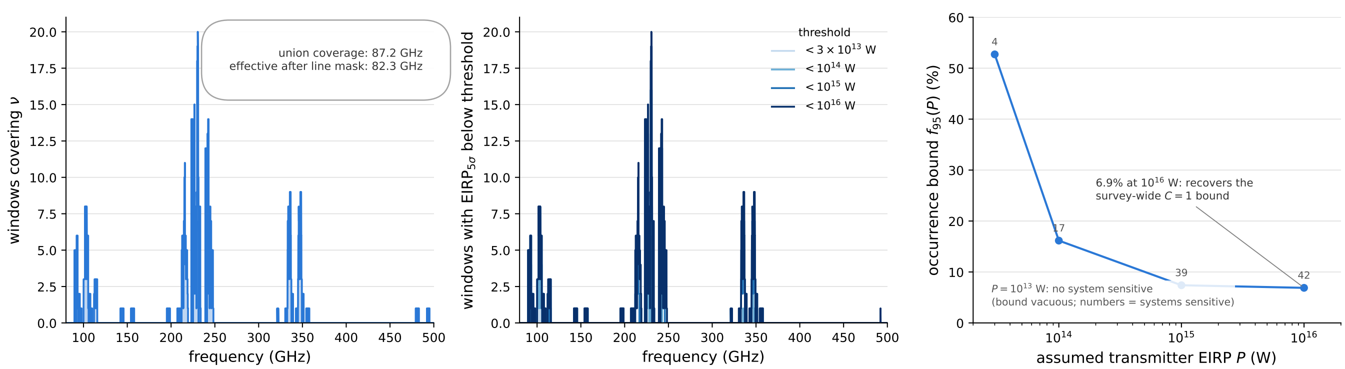

Fig. 6.— Frequency coverage of this release. (a) Number of searched windows covering each frequency (0.01-GHz grid; the union is 87.2 GHz in 36 disjoint intervals from 89.6 to 495.1 GHz). (b) The number of windows whose $\mathrm{EIRP}_{5\sigma}$ threshold lies below the labelled power as a function of frequency – a sensitivity-weighted version of the coverage panel, $N(<\mathrm{EIRP}_{5\sigma};\nu)$: 17, 59, 167, and 220 of the 225 windows lie below 3×10^{13}, 10^{14}, 10^{15}, and 10^{16} W respectively, and none reaches below the survey floor of 1.6×10^{13} W. As throughout this paper, thresholds are nominal 5σ values with the measured recovery carried separately (§4.5).

6. DISCUSSION

This paper is the survey's first stand-alone processed release, and its results are preliminary by construction: 60 of 148 planned target/bands (∼41 per cent) are complete, and the processed subset is deliberately sensitivity-ranked – the survey is ordered to complete the nearest, most sensitively-constrained systems first – so it is not an unbiased draw, and the full sample will extend these results to larger distances and shallower faint-end limits. Readers should treat the survey-level statements (methodology, completeness calibration, benchmark comparisons) as the citable results of this paper, and the per-target values of Table 21 as a snapshot that specific rows will supersede. No credible technosignature has been found among the systems searched so far: no signal matching the searched frequency, drift-rate, temporal and spectral morphology was detected above the nominal instrumental threshold during the analysed archival intervals, and the quoted $\mathrm{EIRP}_{5\sigma}$ thresholds of Table 21 bound any transmitter that was present above them. Two candidates towards β Pic are attributable to CO disc emission – underlining, as Mason et al. (2024) highlight, the importance of checking any mm-band narrowband crossing against known molecular transitions before treating it as a candidate; that check is automated for every crossing (§4) and now uses a velocity-space match applied retroactively to every window of this release (§5.4). The third, marginal candidate (AU Mic, §5.5) is not statistically distinguishable from

TABLE 7

The survey selection function of this release, consolidated. All entries are counts or measurements of this release as frozen, except the measured-recovery row, which is the injection campaign of §4.5. The $\mathrm{EIRP}_{5\sigma}$ and flux-density rows are two forms of the same thresholds: total power delivered within one native channel, and the corresponding flux density (the two are related by $\mathrm{EIRP} = 4\pi d^2 S_{\min}\Delta\nu_{\mathrm{ch}}$).

Quantity	
Census stars (catalogue entries; systems)	
Stars processed / with narrowband windows	
Target/band datasets	
Searched spectral windows (deduplicated)	
Unique frequency union	87.2 (
Effective after line-exclusion mask	82.
Native channel widths	15.3 kHz
Drift-rate ceilings	1.1–5.9
Nominal flux-density thresholds $S_{\min}$	
Nominal $\mathrm{EIRP}_{5\sigma}$ thresholds	1.6×10^{13}
Measured tone recovery (1200 trials, B3/B6/B7)	0/500 static carriers (coarse, by construction); 42% → 83% for driftin
Star/bands with successful narrowband results	
Star/bands with usable continuum measurements	
Public on-star execution blocks available, all bands[c]	
Window-level bookkeeping records (not EBs; ∼4 spw rows per pair)	
Execution blocks; unique projects	
Observing epochs	
Per-star on-source integration	
Threshold crossings / windows with crossings / candidates	

[c]From the proper-motion-matched archive census of 2026 September 9 (§6). Note. Thresholds are nominal 5σ instrumental values; the degree to which a drifting, off-grid, or partially in-channel signal of a given amplitude is actually recovered is a separate measured quantity, carried by the injection campaign of §4.5 rather than assumed from the threshold.

the empirical background distribution and remains recorded for future re-examination rather than dismissed outright.

The scale of the space searched so far can be stated precisely, on one definition throughout. Effectively searched – the unique-frequency union surviving the molecular-line velocity exclusion mask – is 82.3 GHz, in 36 disjoint intervals from 89.6 to 495.1 GHz (87.2 GHz before masking; Figure 6); exposure-weighted, the release accrues 8.3×10^5 s GHz of on-source time-bandwidth product, and the summed per-star frequency union is 371.9 GHz (§5.4). For context on the drift ceiling: of the 37 planets known around the 15 planet-hosting targets of this release, only TRAPPIST-1 b, in its edge-on worst case at $\sim 13\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$, reaches it; GJ 581 e, 61 Vir b, and TRAPPIST-1 c reach roughly half of it, and the remaining 33 planets all sit well inside the searched grid.[5]

Table 7 consolidates the survey selection function of this release – the axes along which a transmitter is or is not detectable – in one place; Table 20 (online-only appendix) decomposes the same release by ALMA band.

Coverage eligibility and search completeness are different quantities and we keep them separate. The census of public usable ALMA data (§3) establishes eligibility: 148 planned target/band combinations within 20 pc, of which the 60 processed here fall within this release's scope; every held execution block was searched – there was no EB-level selection behind the 57. The availability side is measured directly rather than inherited from the release bookkeeping: a proper-motion-matched census of the ALMA archive (2026 September 9; public rows, all bands, pointing within 30 arcsec of the propagated stellar position at the observation epoch) counts 107 unique public on-star EBs for the 44 searched stars, and re-finds 56 of the 57 searched EBs (the one exception, TRAPPIST-1's Band 7 execution A002/Xcccc19/X5564, is carried by `obscore` only through its calibrator-field row – an archive-representation artefact that biases the census low by one for that star). The difference between availability and scope – further epochs of the same target/bands and bands not processed in this release – is the recurrence pool the AU Mic second-epoch test of §5.5 drew on. The full selection funnel, from archive availability to the zero-candidate outcome, is shown in Figure 8 (online-only appendix), and the machine-readable ledger `v323_star_coverage.json` gives per-star distance, availability, scope, and searched frequency intervals.

Figure 14 (online-only appendix) shows the drift ceiling as an explicit selection function over planet period and host mass, and its cost is stated plainly: a transmitter bound to any known planet in the sample other than the innermost TRAPPIST-1 planets would produce drift rates the search covers in full; a transmitter whose drift exceeds the ceiling – a closer-in or more massive companion than any known here, or a rapidly manoeuvring platform – is outside the searched grid and is sacrificed by design (§4.6 recovers some non-linear morphologies). The host-mass dependence is weak ($a \propto M_\star^{1/3}$ at fixed period), so the ceiling is effectively a function of orbital period alone; and the calculation constrains only transmitters co-located with *known* planets – a transmitter on an undiscovered planet, a free-flying platform, or any structure with accelerations of its own choosing is covered only in so far as its drift falls within the searched grid. A coverage fraction along one axis of a multidimensional haystack (frequency, drift rate, sensitivity, duty cycle, sky position) is not a haystack volume, and we quote it as such.

Placing the release against the published haystack of Wright et al. (2018) is likewise best done axis by axis. On

[5] Star–planet relative accelerations $GM_\star/r^2$ from NASA Exoplanet Archive planetary-companion parameters (Mission Star/Planet table, accessed 2026 September), maximised over orbital phase and inclination.

TABLE 8
The 95 per cent upper limit $f_{95}(P)$ on the occurrence of transmitters above EIRP P, in the two forms of the text. n is the number of systems with non-zero completeness at that P. The measured form is defined on the 24 fine-channel systems (18 coarse-only systems have no measured recovery for the drifting class); 'vacuous' means no system has non-zero completeness, so no bound exists.

P (W)	f_{95} thresholded n	limit	f_{95} measured (n)
3×10^{13}	4	52.7%	vacuous (2)
10^{14}	17	16.2%	98.8% (11)
10^{15}	39	7.4%	51.9% (22)
10^{16}	42	6.9%	44.5% (24)

their frequency axis (10 MHz–115 GHz), only the Band 3 portion of our union lies inside their boundaries (20.6 GHz of 115 GHz, or 0.18 of the axis, and a part their Hz-resolution surveys reach far more finely); the remaining 66.6 GHz (Bands 4–8, 125–495 GHz) lies entirely outside their published haystack – the genuinely new territory of this release. On their transmitted-bandwidth axis (0–20 MHz), a single-channel search of native channels probes a fraction set by the channel width (window-mean 0.63, median 0.78). On their sensitivity axis, the median threshold of this release (7.3 mJy across a 15.625-MHz channel, i.e. $\phi \approx 1.1 \times 10^{-21}\,\mathrm{W\,m^{-2}}$) detects a 10^{13} W transmitter only within $\sim$0.9 pc. Against the only prior ALMA-archival technosignature search, Mason et al. (2024) reach 28 bycatch targets at $\geq$1.01 kpc with a deepest quoted $\mathrm{EIRP_{min}}$ of 6.91×10^{17} W, whereas this release's 42 systems all lie inside 20 pc at median per-target thresholds $\sim$2500$\times$ deeper: the two searches occupy nearly disjoint corners of the same haystack, near versus deep. We deliberately do not multiply these fractions into a combined haystack coverage – the axes are strongly coupled (distance and sensitivity above all), and a product of marginal fractions would misstate the true overlap by orders of magnitude.

6.1. *Occurrence limits*

What a zero-detection result bounds is a fully conditioned quantity. Writing the per-system detection completeness as

$$C_i(P) = F_i\, R_i(P)\, D_i\, C_{i,\mathrm{drift}}, \tag{1}$$

where F_i is the probability that the transmitter's sky frequency falls inside the spectral intervals actually searched for system i, $R_i(P)$ the probability that a transmitter of EIRP P would have crossed system i's detection threshold had it been emitting at that frequency, D_i the probability that it was emitting during the archival interval analysed, and $C_{i,\mathrm{drift}}$ the recovery probability for a carrier of the searched drift and temporal morphology (§4.5), the probability of zero credible detections when a fraction f of systems hosts a matching transmitter is

$$P(N_{\mathrm{det}} = 0 \,|\, f) = \prod_{i=1}^{N_{\mathrm{sys}}} (1 - f\, C_i)\,, \tag{2}$$

and the one-sided 95% upper limit f_{95} solves $P(N_{\mathrm{det}} = 0 \,|\, f_{95}) = 0.05$.

Reading rule. Among the ALMA-observed systems, fewer than $f_{95}(P)$ per cent host a transmitter of EIRP $\geq P$ that was emitting during the archival observations, at a sky frequency within the particular spectral intervals searched for that system, with drift and temporal morphology lying within the searched class.

We evaluate Eq. 2 on the 42 independently observed systems (44 narrowband-searched stars; the UV Ceti and AT Mic binaries share observations and are counted once each) in two forms (Table 8). The *thresholded* form is the step function the retained products support directly: $R_i(P) = 1$ when system i's deepest window threshold lies below P and 0 otherwise, $F_i = D_i = 1$ by construction. The *measured* form integrates the injection-recovery curve of §4.5 over each system's searched bandwidth, $R_i(P) = \sum_w \Delta\nu_w \,\mathrm{rec}(P/\mathrm{EIRP}_{5\sigma,w})/\sum_w \Delta\nu_w$, and so folds in the 42 $\to$ 83 per cent threshold-to-2$\times$-threshold recovery of the searched class (including the channel-phase dependence of R1-25's placement-pooled curve). The measured calibration covers the 24 systems with at least one fine-channel window; the 18 coarse-channel-only systems have no measured recovery for the drifting class (static carriers there are suppressed by construction), so the measured ladder is a 24-system bound, not a 42-system one – the thresholded ladder remains the only survey-wide form at this depth. The Band 8 windows share the coarse-channel construction and their completeness is likewise transferred by shared construction, not measured (§4.5).

Three caveats bound how far these numbers can be pushed. First, even the measured form fixes $F_i = D_i = 1$ and the per-system recovery at the configuration level: it is a frequency-integrated completeness for the searched class, not a per-system calibration, and a single uniform completeness $C = 0.42$ (the measured threshold-level recovery) would give $f_{95} = 16.4$ per cent survey-wide at $P \gtrsim 10^{16}$ W – between the two forms, and quoted only to show the size of the completeness correction. Second, the population is heterogeneous in threshold, so each point applies to transmitters above that P, not to a single well-defined class. Third, systems are treated as independent trials in a randomly chosen archival interval. The frequency-integrated duty-cycle dependence is quantified separately below. These are first ALMA-side numbers of their kind, not competitive prevalence constraints; removing the remaining

TABLE 9
Duty-cycle dependence of the survey-wide 95 per cent upper limit at $P \geq 10^{16}$ W, from the product form of Eq. 2 with $D_i = D$: thresholded form (42 systems) and measured-completeness form (24 systems). At $D \leq 0.1$ the thresholded bound is quoted for completeness only – the single-interval assumption behind it has failed by then, which is the honest content of the table.

Duty cycle D	f_{95} thresholded	f_{95} measured
1 (continuous)	6.9%	44.5%
0.5	13.8%	89.0%
0.1	68.8%	vacuous
0.01	vacuous	vacuous

idealisations requires per-configuration completeness folded into a per-system $C_i(P)$, which the retained injection campaign supports.

The archival epoch structure converts directly into a duty-cycle constraint. Writing D_i for the probability that a transmitter with duty cycle D was emitting during system i's analysed intervals ($D_i = D$ to the accuracy the epoch structure supports; Table 17 gives the per-star epochs and integrations), the same product form gives the $f_{95}(D)$ ladder of Table 9: for continuous emitters the limits are those of Table 8; at $D = 0.5$ the measured-form limit degrades to 89 per cent (it is already weak at $D = 1$); and at $D \leq 0.1$ no bound of this form exists – transmitters active in a small fraction of randomly chosen intervals are not constrained by single-epoch archival coverage, and only the recurrence-pool reprocessing of future releases can touch them. The non-recurrence logic of §5.5 points the same way: it rejects continuous or repeating sources at the tested frequency and epochs, not intermittent ones.

A cumulative look-elsewhere consideration accompanies multi-release use: as releases accumulate, the probability that at least one statistically unremarkable crossing is reported somewhere in the programme grows toward unity even if every release is individually clean. The guard is procedural – dispositions stay anchored to per-release control rings, and any candidate promoted across releases must clear a recurrence test at its exact sky frequency (pre-registered for the AU Mic frequency, §5.5) rather than inherit aggregated significance.

The data-quality exclusions of §5.3 carry two open engineering items, recorded here rather than buried: the continuum-imaging step still lacks the pointing-offset guard the narrowband step has, and the window-level noise-handling defect affecting HD 10647 and HD 139664 is excluded from results but not yet fixed at source. The numerical values of this paper correspond to the frozen pipeline release named in the Data Availability statement; both defects are corrected, and the affected datasets reprocessed, in the single corrected public release. The general lessons these cases teach about archival mining are drawn together in online-only Appendix K.

The four supplementary analyses – closure-phase vetting, the population-level anomaly check, the chirp screen, and the cross-target occupancy screen – are infrastructure investments rather than sources of results at this stage. Their role in the Type-I error budget should be stated exactly: the survey's false-alarm probability is carried by the primary lane alone (the candidate definition, its empirical calibration against the 512-position control rings, and the permutation null of §5.4); the supplementary screens gate nothing in that budget and are correlated with it by construction, and the three candidates are dispositioned on astrophysical grounds that depend on none of them. A reader should not multiply the screen-level non-detections into a stronger constraint than the primary lane's own.

Because our selection is on ALMA archival coverage alone, useful subsamples fall out without compromising the census framing, and one general caveat attaches: close multiples unresolved at ALMA's angular resolution (the UV Ceti and AT Mic pairs) share a single measurement, so completeness analyses count such pairs once. First, 15 of the 46 processed stars (33 per cent; 36 planets) are confirmed exoplanet hosts – including Proxima Centauri, τ Ceti, ϵ Eridani, β Pictoris, and the seven-planet TRAPPIST-1 system – a sub-survey a reader interested in known planetary systems can extract directly from Table 21; HN Lib and LHS 1140, each host to a habitable-zone planet, now carry a genuine ALMA technosignature non-detection alongside their planet-search literature. Second, four of the 46 are white dwarfs (Sirius B, Van Maanen's Star, Wolf 219, CD−38 10980), all continuum non-detections with $\mathrm{EIRP}_{5\sigma}$ limits of 2.3×10^{13}–8.4×10^{14} W – among the very few mm/submm SETI constraints reported for white dwarfs (Huang et al. 2026), a channel motivated by post-main-sequence survival scenarios and the compact debris discs of metal-polluted white dwarfs.

7. CONCLUSIONS

We have described the design, methodology, and processed release of a technosignature search of the 120 stars within 20 pc with public ALMA archival coverage, irrespective of disc-hosting or planet-host status: a complete census of ALMA-covered stars within 20 pc – every entry passing an independent positional and identity audit against SIMBAD and *Gaia* (§5.3) – with no additional astrophysical selection, but covering only ∼5% of the catalogued stars within 20 pc and inheriting the archive's own non-uniform targeting, so the contrast with interest-selected samples (§3) is one of degree rather than of kind. This release processes 60 of the 148 planned target/bands – 46 of the 120 census stars, 42 of the 113 systems – and is deliberately sensitivity-ranked, not an unbiased draw.

The release yields $\mathrm{EIRP}_{5\sigma}$ limits of 1.6×10^{13}–7.1×10^{16} W (unchanged if the two primary-beam-affected datasets, Sirius B and ϵ Eri, are excluded) and forty non-detections in the parallel, ancillary continuum screen (median 5σ limit 0.52 mJy) – a screen of unquantified statistical power whose zero outcome is not symmetric in strength with the narrowband non-detection. No credible technosignature was identified: no signal matching the searched frequency, drift-rate, temporal and spectral morphology was detected above the nominal instrumental threshold during the analysed archival intervals. Of the three candidates, two are attributed to CO disc emission towards β Pictoris by targeted astrophysical inspection beyond the automated line check, and one, towards AU Mic, is statistically consistent with the survey's quantified false-candidate rate (one unattributed candidate among 210 non-β Pictoris windows against an expectation of 0.41). The chance probability of the observed candidate count is 6.1 per cent under the execution-block-respecting permutation null, inside a 4–19 per cent bracket across bookkeeping models and rate uncertainty (§5.4; Appendix J), with the designed principal null – the fully symmetric star-versus-control statistic at all 512 control positions – measured wherever the retained measurement sets allow (seven windows, all consistent with exchangeability; Figure 9) and shipping for the remainder.

The result is a constraint rather than a bare non-detection: the 225 searched spectral windows cover 82.3 GHz of unique effective frequency space after molecular-line masking, at nominal thresholds that constrain transmitters matching the searched spectral and drift morphology during those same intervals – with the measured recovery carried alongside them wherever they are used (boxed reading rules of §4 and §6; §4.5) – and no signal was found outside the explicitly excluded molecular-line velocity regions. These limits should not be read as narrowband-transmitter parity with Hz-resolution centimetre-wave surveys: the sensitivity comes from the proximity of the targets overcoming ALMA's coarse native channelisation, not from spectral resolution (Figure 7); the contribution of this release is to bound total transmitter power in a virtually unexplored regime, for the nearest stars, at archive-quality resolution. The non-detection is likewise bounded by epoch coverage: with one to three execution blocks per star (median 46 minutes on source), transmitters active only outside the sampled epochs, or with duty cycles below these baselines, are not constrained (§6). Turned into a population statement under the stated idealisations (§6, Eq. 2), zero credible detections among the 42 independently observed systems bounds the occurrence of transmitters of the searched class that were emitting during the analysed intervals at EIRP $\gtrsim 10^{16}$ W to fewer than 6.9 per cent (95 per cent confidence, thresholded completeness, 42 systems), or 44.5 per cent with the injection-measured, frequency-integrated completeness folded in (24 calibrated systems); a uniform threshold-level completeness of 0.42 would place the survey-wide limit near 16 per cent.

The release includes the first reported ALMA technosignature searches of Barnard's Star and Wolf 359 (§5.2) and, to our knowledge, the first application of interferometric closure-phase vetting, a population-level spectral-type-normalised continuum anomaly check, a chirped-drift and periodicity extension of the narrowband search re-using already-retained spectra, and a cross-target frequency-occupancy check for sky-frequency-recurrent interference in a technosignature survey. All four ship with the pipeline at two grades – closure-phase vetting and the occupancy check survey-grade; the anomalous-continuum screen and the chirped-drift/periodicity extension demonstration-grade, with one astrophysically attributed continuum outlier (§4.4) and no limit deriving from either – and accumulate statistical power as the survey proceeds. A modest, serendipitous molecular-line catalogue and the exoplanet-host and white-dwarf subsamples are by-products of the census. The survey continues, and we anticipate extending it to 30, 40, and 50 pc once the 20 pc sample is complete. Numerical values throughout correspond to the frozen pipeline and data snapshot named in the Data Availability statement.

ACKNOWLEDGEMENTS

This paper makes use of ALMA data from the public ALMA Science Archive. The individual execution blocks searched in this release (57 unique execution-block identifiers, listed with their project codes and epochs in Appendix A) should be cited from that list in the standard JAO.ALMA#XXXX.X.NNNNN.S form; the machine-readable per-window-to-execution-block linkage accompanies the released products. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), MOST and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database, operated at CDS, Strasbourg, France.

COMPETING INTERESTS

G.J.W. declares no competing interests. R.D. is an employee of VBRL Holdings Inc.

AUTHOR CONTRIBUTIONS (CREDIT)

Glenn J. White: Conceptualisation, Methodology, Supervision, Writing – review & editing.
Robin Dey: Software, Investigation, Formal analysis, Data curation, Visualisation, Writing – original draft.

DATA AVAILABILITY

The reduced data products underlying this paper, together with the full target-selection and analysis pipeline, are available at `https://github.com/Tilanthi/SETI`, and the repository is live at submission. Two snapshots must be distinguished. All numerical values in this paper were produced by the *submitted-analysis snapshot*: the repository commit carrying the tag `submitted-v3.25`, created when this manuscript was submitted, which freezes the pipeline code and input data snapshot exactly as run. That commit is retained unchanged and reproduces every number, every table, and every figure in this paper exactly, through the regeneration scripts shipped with it. The *public release* is a single corrected release built from that snapshot: the two pipeline defects recorded in §5.3 (the missing pointing-offset guard in the continuum lane and the window-level noise-handling defect) are corrected, and the affected datasets reprocessed, *before* the public release is frozen – so the public release carries one set of products, not a

TABLE 10
EXECUTION BLOCKS SEARCHED IN THIS RELEASE, ORDERED BY EPOCH. EPOCHS (UTC START, TO THE MINUTE) AND PROPOSAL CODES ARE FROM THE ALMA SCIENCE ARCHIVE **IVOA.OBSCORE** SERVICE AT **ALMASCIENCE.ESO.ORG**, QUERIED BY EXECUTION BLOCK IDENTIFIER ON 2026 SEPTEMBER 8; MJD IS THE ARCHIVE **T_MIN**. THESE METADATA WERE RECOVERED AFTER THE SEARCH PRODUCTS WERE FROZEN AND ARE NOT PART OF THE FROZEN PRODUCTS.

Execution block	Proposal	UTC date	MJD	Execution block	Proposal	UTC date	MJD
uid://A002/X6f1341/X1484	2012.1.00142.S	2013-10-06 06:23	56571.26616	uid://A002/Xd0fb35/X90dc	2017.1.00698.S	2018-08-22 15:32	58352.64755
uid://A002/X74deb6/Xff1	2012.1.00268.S	2013-12-02 07:15	56628.30252	uid://A002/Xd16a1e/X8d4	2017.1.01644.S	2018-08-31 22:26	58361.93490
uid://A002/X877e41/X14c1	2012.1.00993.S	2014-05-22 11:24	56799.47525	uid://A002/Xd248b5/X5186	2017.1.00698.S	2018-09-22 11:12	58383.46702
uid://A002/X899169/X1838	2012.1.00198.S	2014-08-18 04:01	56887.16777	uid://A002/Xd28a9e/Xbcdf	2017.1.00698.S	2018-09-28 10:57	58389.45629
uid://A002/X99c183/X4f1e	2013.1.00645.S	2015-01-17 00:35	57039.02475	uid://A002/Xd6ebab/X41a3	2018.1.01149.S	2018-12-20 09:49	58472.40910
uid://A002/X9f2ff8/X25b0	2013.1.01214.S	2015-04-26 12:19	57138.51354	uid://A002/Xd86f11/X584b	2018.1.01833.S	2019-01-20 19:05	58503.79517
uid://A002/Xb0be8b/X785f	2015.1.00307.S	2016-03-22 23:25	57469.97599	uid://A002/Xd95042/X2d99	2018.1.01149.S	2019-03-09 06:13	58551.25945
uid://A002/Xb25e1a/Xed74	2015.1.00783.S	2016-05-01 17:17	57509.72060	uid://A002/Xd9668b/X3a90	2018.1.00072.S	2019-03-12 00:14	58554.01019
uid://A002/Xb2e94e/X361a	2015.1.00307.S	2016-05-13 04:19	57521.18017	uid://A002/Xd98580/X3bf6	2018.1.01833.S	2019-03-15 20:39	58557.86052
uid://A002/Xb39557/X3789	2015.1.00307.S	2016-05-26 11:30	57534.47920	uid://A002/Xd9d7c7/X6fd2	2018.1.01833.S	2019-03-21 15:03	58563.62709
uid://A002/Xb9a2bb/X1899	2016.1.00817.S	2016-10-22 02:07	57683.08829	uid://A002/Xda2c82/X765d	2018.1.01833.S	2019-03-28 21:57	58570.91480
uid://A002/Xb9cc97/X150a	2016.1.00803.S	2016-10-25 02:14	57686.09356	uid://A002/Xdc621b/X2bb0	2018.1.01833.S	2019-05-20 04:28	58623.18665
uid://A002/Xba3ea7/X2ab4	2016.1.00817.S	2016-11-06 20:01	57698.83471	uid://A002/Xe5808e/X80a1	2019.1.01681.S	2019-12-20 17:03	58837.71056
uid://A002/Xbe7502/X2047	2016.1.00637.S	2017-03-28 05:22	57840.22390	uid://A002/Xf5d76d/X32f1	2021.1.00485.S	2022-03-02 23:14	59640.96830
uid://A002/Xc067f7/X822	2016.2.00015.S	2017-05-17 01:51	57890.07740	uid://A002/Xfdb69b/X667a	2021.1.01209.S	2022-08-30 16:22	59821.68235
uid://A002/Xc19d6f/X19ce	2016.2.00015.S	2017-07-02 08:43	57936.36365	uid://A002/Xfdf45f/X478e	2021.1.01209.S	2022-09-05 02:32	59827.10618
uid://A002/Xc6141c/X3af0	2017.1.00786.S	2017-10-27 03:04	58053.12832	uid://A002/Xfeb5e6/X13ee	2021.1.01209.S	2022-09-22 01:39	59844.06896
uid://A002/Xc66ce1/Xaa9	2017.1.00786.S	2017-11-04 03:17	58061.13698	uid://A002/Xff294f/X2db6	2022.1.00338.L	2022-10-04 04:49	59856.20118
uid://A002/Xc74b5b/X7c16	2017.1.01644.S	2017-11-29 07:29	58086.31237	uid://A002/X106a1f1/X6a0c	2022.1.01163.S	2023-05-06 03:35	60070.14972
uid://A002/Xc8b2b0/X9c35	2017.1.01644.S	2018-01-05 00:25	58123.01783	uid://A002/X107fa96/Xce38	2022.1.01163.S	2023-06-02 02:51	60097.11926
uid://A002/Xc9831a/X443	2017.1.00986.S	2018-01-22 20:39	58140.86104	uid://A002/X10a341d/X82d2	2022.1.00134.S	2023-07-17 18:28	60142.76964
uid://A002/Xc99ad7/X348	2017.1.00698.S	2018-01-24 23:51	58142.99391	uid://A002/X10b0d7d/X323f	2022.1.00134.S	2023-08-06 09:20	60162.38930
uid://A002/Xcbb583/Xb78	2017.1.00704.S	2018-04-06 23:22	58214.97419	uid://A002/X11e10bc/X12662	2023.1.00390.S	2024-09-28 16:03	60581.66907
uid://A002/Xcc626d/X579	2017.1.00698.S	2018-04-21 21:13	58229.88426	uid://A002/X1207c39/X9417	2024.1.00722.S	2024-11-15 08:17	60629.34515
uid://A002/Xcccc19/X5564	2017.1.00215.S	2018-05-03 11:42	58241.48813	uid://A002/X12277ed/X22e53	2024.1.00165.S	2024-12-22 10:09	60666.42301
uid://A002/Xcd97a1/X3f1e	2017.1.01583.S	2018-05-19 07:56	58257.33086	uid://A002/X1237e23/Xdb0	2024.1.01095.S	2025-01-06 18:51	60681.78602
uid://A002/Xcda49e/Xdebb	2017.1.00561.S	2018-05-21 09:50	58259.41000	uid://A002/X12925a3/Xa4b1	2024.1.01095.S	2025-05-17 08:00	60812.33402
uid://A002/Xcf0c6b/X6513	2017.1.00167.S	2018-06-27 10:14	58296.42698	uid://A002/X12a8fc8/X10b8	2024.A.00040.S	2025-06-11 08:49	60837.36781
uid://A002/Xcf196f/Xabc5	2017.1.00200.S	2018-06-29 09:32	58298.39773				

submitted/corrected pair – and no value printed in this paper is silently altered: wherever a corrected product differs from the corresponding submitted-snapshot product, the change-log entry names the table and row affected.

The datasets reprocessed under these corrections are precisely those excluded from this paper's results by §5.3 (the Giclas 9−38 pair and the two defective Band 6 windows); the Sirius B and ε Eri holography-beam recomputation committed in §4 is folded into the same freeze, with its outcome bounded by the systematic already quoted there. A change-log in the release maps every affected product and the reason for each change. The machine-readable per-window table carries archive-consistent execution times: where the printed integration-time audit differs (the CD−38 10980 row is optimistic by 7%, §5.6), the machine-readable value is the archive's. Per-window hit lists in the public release are uncapped (the interim retained product capped them at 50 frequencies; §4.7).

The complete release will also be deposited on Zenodo, where a DOI will be minted upon final acceptance and inserted here; the deposit comprises: (i) the per-window search products for all 225 windows (spectral extracts, per-drift-trial peak statistics, and the machine-readable line-exclusion mask of §5.4); (ii) the control-ring calibration products at all 512 control positions; (iii) the closure-phase and population-anomaly test outputs; (iv) the full injection-recovery campaign products (all 1200 trials of §4.5, including the 24-tone pilot); (v) the frozen pipeline code and input data snapshot; (vi) the audit tables of online-only Appendix L in machine-readable form; (vii) the machine-readable molecular-line catalogue of §5.8; and (viii) the proper-motion-matched execution-block availability census of §6 with its per-star ledger. The public release is distributed under the Creative Commons Attribution 4.0 International licence (CC BY 4.0) for data products and the MIT licence for pipeline code. Raw ALMA visibility data are publicly available from the ALMA Science Archive at `https://almascience.org`.

SOFTWARE

The analysis pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; calibration, imaging, and the local reprocessing of §§5.4–5.5), Python 3 with astropy, numpy and scipy (spectral extraction and statistics), and the ALMA Science Archive and SIMBAD TAP interfaces for data and catalogue access; all dependencies are free and open-source, and exact version numbers are recorded with the frozen release.

APPENDIX

EXECUTION-BLOCK METADATA

Table 10 lists the 57 unique ALMA execution blocks searched in this release, with their proposal codes and observing epochs. These metadata were recovered from the ALMA Science Archive by execution-block identifier after the search products were frozen; they are not part of the frozen products themselves, and we note that provenance explicitly. The per-window linkage between execution block and searched spectral window, together with per-observation antenna configurations, synthesised beam sizes, and polarisation setup, will accompany the machine-readable table of the corrected tagged release committed in the Data Availability statement. The observing epochs span 2013 October 6 to 2025 June 11 (MJD 56571–60837) across 36 unique proposal codes; the AU Mic candidate window of §5.4 derives from the 2014 August 18 execution block of project 2012.1.00198.S. One further public execution block of that project (uid://A002/X7d80c7/X15c6, executed 2014 June 5, 1633 s of science exposure, public since 2015 April 8) also covers the candidate 230.83 GHz window but was not processed in this release; it is the second epoch of the recurrence test committed in §5.4.

TABLE 11
Survey vocabulary.

Term	Meaning in this paper
Census	every star within 20 pc with public ALMA archival coverage (120 entries; §3)
Bycatch	census stars observed by ALMA for unrelated science goals
Target/band dataset	one star in one ALMA band; the unit of the 148-dataset plan (60 complete)
Spectral window	one correlator spw of one execution block, after deduplication; the unit of search and bookkeeping (225)
Execution block (EB)	one contiguous ALMA observing session (57 searched; Appendix A)
MOUS	member observation unit set: the ALMA data hierarchy level at which pipeline products are delivered (one calibration recipe per MOUS)
QA2	the ALMA observatory pipeline's second-stage quality-assurance calibration and scoring, whose products this survey consumes
`obscore`	the IVOA table protocol of the ALMA Science Archive metadata service from which coverage and EB metadata are queried
Native channel	the correlator channel width of the window, 15.3 kHz–31.25 MHz here
Threshold crossing ("hit")	a $\geq 5\sigma$ spectral feature at the star's position in a searched window
Candidate	a threshold crossing promoted by the joint threshold-and-localisation criterion for manual vetting (three)
Control ring	512 off-source test positions per window, used for empirical false-alarm calibration
Star-beats-ring	a window whose star statistic exceeds all 512 control statistics (30/225)
Drift trial	one trial Doppler drift rate of the stack grid; 2–1944 per window
Nominal threshold	noise-derived 5σ detection threshold, *not* a calibrated completeness limit (boxed rule, §4)
$\mathrm{EIRP}_{5\sigma}$	equivalent isotropic radiated power at the nominal threshold; $4\pi d^2 S_{\min}\Delta\nu_{\rm ch}$
$t_{\rm on}$ / $T_{\rm span}$	summed on-source integration / elapsed block span per star (Table 17)
Molecular-line exclusion mask	the $\pm 50\,\mathrm{km\,s^{-1}}$ stellar-frame region around catalogued transitions where crossings are excluded (§4.1)
Closure phase	interferometric phase triplet test discriminating point-source structure (§4.3)
Stellar frame	topocentric $\rightarrow$ barycentric $\rightarrow$ systemic-RV frame chain (§4.1)
Funnel	the candidate-narrowing cascade from channels searched to vetted candidates (Table 3)

GLOSSARY OF SURVEY VOCABULARY

Terms as used in this paper; quantitative definitions live in the sections cited.

FREQUENCY CONVENTIONS, LINEWIDTH READING, AND SMEARING

This appendix collects the conventions behind every quoted limit.

REFERENCE FRAMES

Every frequency axis in this paper is topocentric, as stated in §4.1. The conversion chain used by the line-exclusion mask – topocentric $\rightarrow$ barycentric (Earth ephemeris at the execution-block mid-time) $\rightarrow$ stellar rest frame (catalogued systemic radial velocity; JPL catalogue rest frequencies, §5.8) – has as its residual error the catalogue radial-velocity uncertainty alone (at most a few $\mathrm{km\,s^{-1}}$ within 20 pc); second-order relativistic terms at the $50\,\mathrm{km\,s^{-1}}$ scale are $\lesssim 5$ kHz at 350 GHz, below the finest native channel searched (15.3 kHz). Kinematic attributions of the β Pictoris and AU Mic windows (§5.4, Fig. 4) use this same chain, so their quoted offsets are directly comparable to the exclusion tolerance. Enlarging the trial drift-rate grid needs no additional frame machinery, because the searched drift rates are topocentric quantities read from the same untransformed frequency axis; and the frame transformations that do enter are applied only at the vetting stage, never to the searched spectra. The trial grid itself depends only on elapsed time and channel width, not on sky position, and is applied identically to the star and to every control position.

CHANNEL WIDTHS AND THEIR VERIFICATION

Channel widths are read from the measurement sets' own spectral window tables (CHAN_WIDTH). For the local reprocessing set we verified they are the *effective* resolution rather than a pre-averaging bookkeeping value: the tables carry no EFFECTIVE_BANDWIDTH column (no online spectral averaging was recorded) and the full-resolution spectral windows are explicitly named FULL_RES in their metadata – the 15.625-MHz coarse and 488.28125-kHz fine channelisations are therefore the noise bandwidths the thresholds refer to.

LINEWIDTH CONVENTION

The detection statistic is the flux density in a single channel after drift correction, so the quoted $\mathrm{EIRP}_{5\sigma}$ applies to any transmitter narrower than the native channel with the threshold flat in the assumed linewidth. Two sub-channel qualifications: the flatness is exact for a tone centred in a channel (a boundary-straddling tone divides its power between two channels, rising toward a factor of two in the worst case; the measured dependence is given in §4.5); and the transition between adjacent channels follows the correlator's channel passband shape (including any Hanning weighting or spectral averaging), which varies with correlator generation and is not retained in the frozen products, so the flat-in-linewidth statement is a channel-centred idealisation, accurate at the tens-of-per-cent level.

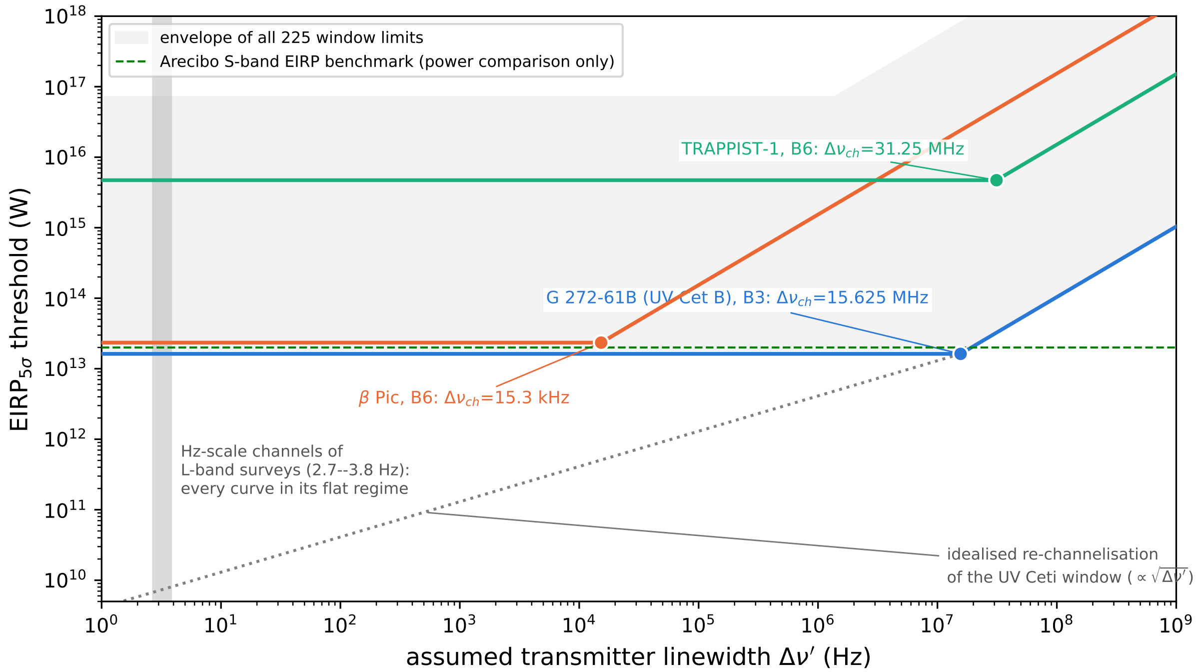

Fig. 7.— $\mathrm{EIRP}_{5\sigma}$ threshold as a function of the assumed transmitter linewidth $\Delta\nu'$, for three representative windows (coloured curves) and the envelope of all 225 window limits of this survey (grey band): the UV Ceti Band 3 window has 15.625-MHz channels (the worked example above), the β Pictoris Band 6 fine window has 15.3-kHz channels, and the TRAPPIST-1 Band 6 window has 31.25-MHz channels. Each curve is flat at the quoted limit for any carrier narrower than the window's native channel width $\Delta\nu_{\rm ch}$ (dot at each knee); above $\Delta\nu_{\rm ch}$ the required power rises linearly with linewidth. The grey dotted curve shows the deepest window under an idealised re-channelisation, for which the threshold would fall as $\sqrt{\Delta\nu'}$. The shaded vertical band marks the Hz-scale channel widths of L-band technosignature backends, within which every curve of this survey is in its flat regime. The green dashed line is an Arecibo-like planetary radar (2×10^{13} W).

A robustness test with injected tones at randomised sub-channel positions is committed with the validation items of §6. For the deepest window (UV Ceti Band 3) the quoted limit is 1.6×10^{13} W at a 15.625-MHz channel and the minimum detectable spectral power is $1.0 \times 10^{6}\,\mathrm{W\,Hz^{-1}}$: a 1-Hz carrier there needs the full 1.6×10^{13} W. Had the same observation been channelised at 1 Hz from the start, the threshold would fall as $P_{\rm min} \propto \sqrt{\Delta\nu}$ to 4×10^{9} W; archival data cannot be re-channelised below native resolution, so the ratio $\sqrt{\Delta\nu_{\rm ch}/\Delta\nu'}$ (~4000 for that window) is the unavoidable price of the archival channelisation (Fig. 7).

Worked example: from flux to EIRP. Take UV Ceti (G 272-61B) Band 3. The per-channel rms of the searched window is 0.244 mJy, so $S_{\rm min} = 5 \times 0.244 = 1.22$ mJy in a single 15.625-MHz native channel. At 2.67 pc this corresponds to $1.22 \times 10^{-29}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$, and $\mathrm{EIRP}_{5\sigma} = 4\pi d^2 S_{\rm min}\Delta\nu = 4\pi(2.67\,\mathrm{pc})^2 \times 1.22 \times 10^{-29}\,\mathrm{W\,m^{-2}\,Hz^{-1}} \times 15.625\,\mathrm{MHz} = 1.6 \times 10^{13}$ W. Any carrier narrower than 15.625 MHz must deliver that *total* power to be flagged in this window; a transmitter wider than the channel needs $S_{\rm min}$ per channel.

INTRA-INTEGRATION SMEARING

The frequency excursion of a signal at the maximum searched drift rate across one integration, $\dot\nu_{\rm max} t_{\rm int}$, is evaluated for every window: across the 225 windows it is at most 1.10 channels (median 0.001, 90th percentile 0.06). A linearly drifting signal smeared by Δ channels within an integration retains a fraction $\eta_{\rm smear} = \mathrm{sinc}(\pi\Delta/2)$ of its peak-channel amplitude, so the effective threshold for a worst-drift transmitter is $\mathrm{EIRP}_{5\sigma}/\eta_{\rm smear}$. Only three of the 225 windows have $\eta_{\rm smear} < 0.99$ (Table 12, marked * in Table 21); the median window is essentially unsmeared. A signal drifting midway between two adjacent trial rates accumulates at most 0.03 channels of residual drift across the grid as realised, so no significant sensitivity is lost to signals falling between trial rates.

CLOSURE-PHASE VETTING: FULL SPECIFICATION

The closure phase is the phase sum $\arg(V_{ij}) + \arg(V_{jk}) - \arg(V_{ik})$ around a closed triangle of baselines i-j-k: theoretically exactly zero for an unresolved point source at any sky position (a point source's visibility phase is a pure linear function of (u, v) that cancels around any closed loop) and immune to per-station calibration errors. The test is independent of the spatial-coincidence test: closure phase does not depend on where the source is, so it asks whether the signal behaves like a coherent point source at all – a coherent far-sidelobe interferer (satellite, aircraft) would itself produce near-zero closure phase and pass.

Consistency with zero is assessed by a z-score – the mean drift-tracked closure phase divided by its standard error across the closed triangles – with three properties handled explicitly. First, the triangles of one observation are not

TABLE 12
THE ONLY WINDOWS OF THIS RELEASE WITH NON-NEGLIGIBLE INTRA-INTEGRATION SMEARING AT THE MAXIMUM SEARCHED DRIFT RATE ($\eta_{\rm smear} < 0.99$; ALL 222 OTHER WINDOWS HAVE $\eta_{\rm smear} \geq 0.99$ AND THEIR NOMINAL AND EFFECTIVE THRESHOLDS AGREE TO BETTER THAN 1%). Δ IS THE FREQUENCY EXCURSION IN CHANNELS ACROSS ONE INTEGRATION, $t_{\rm int}$ THE PER-INTEGRATION ON-SOURCE TIME, AND $\mathrm{EIRP}_{5\sigma,\rm eff}$ THE SMEARED EFFECTIVE THRESHOLD $\mathrm{EIRP}_{5\sigma}/\eta_{\rm smear}$.

Window	$\Delta\nu_{\rm ch}$ (kHz)	$t_{\rm int}$ (s)	Δ (ch)	$\eta_{\rm smear}$	$\mathrm{EIRP}_{5\sigma,\rm eff}$ (10^{13} W)
β Pic B6	15.3	6.05	1.096	0.574	4.08
η Crv B8	61.0	6.05	0.585	0.865	365.8
η Crv B7	61.0	6.05	0.411	0.932	4.68

independent (any two sharing a baseline share its thermal noise), so the effective number of independent closure quantities is smaller than the triangle count (of order 10^4 for a 40-antenna configuration) and the aggregate score must not treat triangles as independent trials. Second, closure phases wrap at $\pm\pi$, so the statistic is formed in circular form (unit-phasor averaging). Third, the z-score's null distribution is calibrated empirically against the observation's own integrations rather than assumed Gaussian, which also absorbs the shared-baseline covariance.

The calculation was validated against synthetic visibility data before any ALMA application: zero closure phase for a noiseless simulated point source, statistical consistency with zero for a noisy one, clearly inconsistent scatter for a simulated RFI-like non-point-source signal. The positive-control run – continuum detections driven through the closure machinery at their detection frequencies – was executed on four fields of the local reprocessing set (AT Mic Band 7, the known ∼2-arcsec binary, as the resolving control; AU Mic Band 6 and Band 3 continuum; and the AU Mic Band 6 candidate frequency 230.8252 GHz), with triangle closure phases from time- and channel-averaged unflagged cross visibilities and $\sigma_{\rm CP}$ propagated from per-baseline integration-to-integration scatter. The result is a sensitivity verdict: per-baseline S/N is 0.5–0.7 at the median and never exceeds 5.1 (0.02 per cent of triangles have all three baselines above S/N = 5), so every field's closure phases are statistically indistinguishable from pure noise – including the resolved binary, whose ∼0.4 mJy total flux sits below the per-baseline noise of these short integrations. Closure phase at this survey's continuum flux densities cannot authenticate or exclude any of the seventeen detections; it retains its intended role as a rejection diagnostic for brighter narrowband candidates, triggering only on threshold crossing at negligible computational cost.

PRIOR LOW-FREQUENCY TECHNOSIGNATURE SURVEYS

Table 13 summarises the surveys cited in §2 and §2, replacing a footnote of the original submission. Content is limited to what the cited abstracts state; a dash marks quantities the cited work does not quote. The table is context for the present survey's place in the field, not a sensitivity ranking: band, backend, and threshold conventions differ across rows by construction (§2, Figure 1).

THE CONTINUUM ANOMALY SCREEN: FULL SPECIFICATION AND FIRST EXECUTION

This appendix gives the full specification of the population-level continuum anomaly screen summarised in §4.4, and states exactly how far it is exercised in this release.

Radius sourcing. Inputs are Gaia DR3 photometric temperatures and, where available, Gaia GSP-Phot radii; where a direct Gaia radius estimate is unavailable – the case for close to half of the sample by construction – the pipeline substitutes a coarse Pecaut & Mamajek main-sequence $T_{\rm eff}$–radius relation. Any systematic offset or excess scatter introduced by that fallback propagates directly into the excess-ratio distribution, and individual stars can also depart from a mean relation through age, metallicity, or unresolved multiplicity. A sensitivity check – rerunning the entire screen with Pecaut & Mamajek parameters imposed on every star, so that the fallback's scatter can be judged against the outlier threshold – is executed with this release: the sole flag survives, the M and G/F bin scales are unchanged (those bins are already fallback-dominated), and the A/F-hot median moves from 4.3 to 3.9 (§4.4). Every excess ratio derived through the fallback relation nonetheless carries a systematic of the order of that relation's own scatter.

Error model. The error model applied before any flag is evaluated is $\sigma^2_{\rm tot} = \sigma^2_{\rm map} + (f_{\rm cal}\, S_\nu)^2 + \sigma^2_{\rm model}$, with $\sigma_{\rm map}$ the image rms, $f_{\rm cal}$ the band-dependent absolute-flux systematic (5 per cent in Bands 3–6, 10 per cent in Bands 7–8), and $\sigma_{\rm model}$ the photospheric-model uncertainty propagated from the Gaia temperature and radius errors (and from the main-sequence-relation scatter for the fallback stars). The $\sigma_{\rm tot}$ column of Table 21 carries the map and calibration terms, $\sigma_{\rm tot} = \sqrt{\sigma^2_{\rm map} + (f_{\rm cal} S_\nu)^2}$; the model term enters the population-level excess-ratio screen, where it matters, rather than the tabulated per-star errors. Excess ratios near unity should be read with the model term (dominant for fallback stars) and the calibration term (dominant for the brightest detections) both in mind.

Binning and the outlier statistic. Excess ratios are compared against other stars of similar effective temperature in coarse, equal-width $T_{\rm eff}$ bins whose width is a fixed pipeline configuration constant, published with the machine-readable data release and chosen so that populated bins meet the five-comparison-star minimum, using a robust median/MAD outlier statistic. That minimum makes the screen explicitly exploratory at this depth: with five comparison stars the bin median rests on two or three values and the MAD-based scale is itself a small-sample statistic, so outlier flags at the current depth motivate individual inspection and are not calibrated detections. Raising the minimum to ten comparison stars, which the completed survey will reach in the occupied bins, is deferred to the completed survey. Because the comparison is repeated for every star and bin, a small expected number of chance flags

TABLE 13
Selected prior technosignature programmes below 10 GHz and the sole prior ALMA search, as cited in the text. All entries are quoted from the cited works' own abstracts and summaries.

Programme / study	Facility	Scale	Frequency	Note as stated by the cited work
Project Ozma (Drake 1960)	—	2 stars	1.42 GHz	τ Cet and ϵ Eri
Project Phoenix (Backus & Project Phoenix Team 2002)	multi-site	—	—	pioneered multi-site RFI verification
Allen Telescope Array (Tarter 2001)	ATA	—	—	first purpose-built commensal SETI instrument
ATA TRAPPIST-1 (Tusay et al. 2024)	ATA	—	—	—
ATA ISO survey (Sheikh et al. 2025b)	ATA	—	—	—
BL first data release (Enriquez et al. 2017)	GBT	692 stars	1.1–1.9 GHz	EIRP $> 10^{13}$ W
BL 1327-star search (Price et al. 2020)	—	1327 stars	1.10–3.45 GHz	—
BL Earth Transit Zone (Gajjar et al. 2019)	GBT	—	3.95–8.00 GHz	—
Margot et al. (Margot et al. 2023)	GBT	11,680 stars	1.15–1.73 GHz	—
ML vetting (Ma et al. 2023)	GBT	820 stars	—	β-CVAE cuts human-vetting load by $\sim$100$\times$
turboSETI correction (Choza et al. 2024)	—	—	—	S/N mischaracterisation; retrospective corrections
FAST commensal ISOs (Zhang et al. 2020; Li et al. 2026)	FAST	—	—	interstellar-object targets
Occultation timing (Tusay et al. 2025)	—	—	—	eclipsing/occultation strategy
COSMIC (Tremblay et al. 2024)	VLA	—	—	commensal system
MeerKAT dragnet (Czech et al. 2021)	MeerKAT	—	—	—
MWA back-ends (Morrison et al. 2023)	MWA	—	—	—
ALMA stellar bycatch (Mason et al. 2024)	ALMA B3	28 stars	90.642/93.151 GHz	first ALMA technosignature search; $\mathrm{EIRP_{min}} = 6.9 \times 10^{17}$ W (nearest star)

follows from the multiple comparisons; the test is a screening tool that motivates individual inspection of flagged stars, not a per-star hypothesis test at a fixed false-alarm rate.

Exact status in this release. The screen is live: stellar parameters were ingested for every continuum-measured target (Gaia GSP-Phot radii and temperatures where the catalogue provides them – 16 of the 44 narrowband-searched stars – and the Pecaut & Mamajek relation otherwise), all 57 usable continuum measurements were evaluated, and the 17 detections carry measured excess ratios binned M/G/F/A-hot (7/6/4 members of the three populated bins) with the per-bin median ratios and log-scatters published in §4.4 and with the machine-readable release. One measurement crosses the outlier threshold (χ^1 Ori, Band 3, $z = 4.1$); its disposition and the two committed robustness reruns (fallback-only parameters; leave-one-out bin scatter) are recorded in §4.4. The ten-comparison-star bin minimum remains a next-release item: at the current depth the populated bins hold 4–7 detections, so every bin statistic here is a small-sample statistic, and the screen remains a discovery filter that motivates individual inspection rather than a calibrated per-star hypothesis test.

CHIRPED-DRIFT AND PERIODICITY SCREEN: FULL SPECIFICATION

This appendix expands the demonstration experiment summarised in §4.6. Both extensions operate on the retained products kept for every target processed to date – the star's own raw dynamic spectrum together with 8 sampled control positions, a strict subset of the primary search's 512-position control ring for the same window, not an independent ensemble (§4) – specifically so that analyses like these do not require re-downloading or re-extracting anything.

The chirped extension is a quadratic-drift generalisation of the same drift-corrected stacking search as the primary pipeline, on a coarser trial grid of 21 linear-drift $\times$ 5 chirp-rate trials. The periodicity search is an FFT along the time axis of each channel, per position, with the peak power compared against a robust per-channel noise estimate. Both vet the star's spectrum against the 8 retained control positions exactly as the primary search vets the star against its control ring. Because the control ensemble is 8 positions rather than 512 and the trial grids are coarser, the screen's role is to establish that the extension runs, behaves correctly under the null, and is worth extending – nothing rests on it as a detection analysis. Its exceedance rates (10.7% chirp, 13.1% periodicity, against the $\sim$11% chance rate of a star being the maximum of a 9-member ensemble) behave exactly as expected under the null hypothesis, a clean if modest validation. Extending both to the full 512-position control ring – at negligible additional cost, since the spectra are

TABLE 14
Target pairs with the deepest searched-frequency overlap (computed from the merged per-window frequency ranges of this release). These pairs are the exposure behind the cross-target frequency-occupancy check: a shared tuning is the circumstance in which a recurring sky-frequency feature – RFI or an instrumental spur – would be caught.

Target pair	Overlap (GHz)
γ Lep – γ Vir	14.25
γ Vir – TRAPPIST-1	10.78
γ Lep – TRAPPIST-1	10.77
HD 10647 – HR 1010	10.13
AU Mic – β Pic	8.18
GL 3379 – HD 285968	7.32
GJ 849 – LHS 1140	7.32
TRAPPIST-1 – Wolf 359	7.31
Proxima Cen – Wolf 359	7.31
Proxima Cen – TRAPPIST-1	7.31

already retained – is what promotes the screen to survey grade, and is deferred to the completed survey.

CROSS-TARGET FREQUENCY-OCCUPANCY CHECK: KERNEL, COORDINATES, AND SCOPE

This appendix expands the cross-target occupancy check summarised in §4.7.

Matching kernel. The check cross-matches, across every target/spw result in this release, both the best-fit-channel frequency (regardless of whether it formally crossed the detection threshold) and the list of formal $\geq 5\sigma$ threshold crossings, pairing two crossings at frequencies ν_1, ν_2 towards different targets when $|\nu_1 - \nu_2| < \max(k\,\Delta\nu_1,\ k\,\Delta\nu_2)$: the kernel scales with the coarser of the two native channel widths rather than being a fixed frequency interval. We adopt $k = 3$ native channels; the 5 MHz tolerance of the original formulation corresponds to k between $\sim$0.2 and $\sim$330 across this release's native channel widths (15.3 kHz to 31.25 MHz).

The three frequency coordinates. Three coordinates are at play and deserve separate names: the *sky frequency* of a hit as reconstructed from the archive spectral-window metadata in the observation's recorded frame; the *topocentric frequency*, which differs from a barycentric convention by up to the $\pm 30\,\mathrm{km\,s^{-1}}$ barycentric term ($\sim$23 MHz at 230 GHz – far wider than the matching kernel); and the *correlator channel index* of the native spectral window. The sky-frequency cross-match probes the first and the channel-index cross-match the third; a frame-convention inconsistency between two observations, or an artefact fixed in the second, would each defeat sky-frequency pairing, which is why the frame-convention disclosure queued with the symmetric-null recomputation (§5.4) feeds this check as well, and why the null result should be read as the rejection of sky-frequency recurrence only – not as a strong general RFI rejection. Instrumental artefacts fixed in the correlator frame – correlator birdies, local-oscillator spurs, quantization spurs, and bandpass-edge ripple – recur at the same intermediate frequency or channel number rather than the same sky frequency; a channel-index cross-match (same channel number of an identical correlator setup) is possible with the retained products and also returns zero pairs, while a full intermediate-frequency reconstruction, which would pair artefacts across different tunings, is not recoverable from the retained metadata and remains with the completed survey.

Exposure and exclusions. The power of the null is the number of target pairs that share frequency coverage, quantified directly from the per-window frequency ranges of the 44 narrowband-covered stars: 366 of the 946 star pairs (39%) have searched-frequency unions overlapping by at least 0.01 GHz – 359 pairs (38%) by at least 0.5 GHz and 345 (36%) by at least 1 GHz – and 109 of the overlapping pairs involve one of the five stars carrying formal crossings. Table 14 lists the deepest overlaps. Two execution blocks are shared between catalogue entries – that of the UV Ceti binary (observed as one field but carried as the G 272-61A/B entries) and that of the AT Mic pair (carried as AT Mic B and AT Mic AB; §3) – and their intra-pair "coincidences" are identities by construction, excluded from the cross-target test. The per-window hit lists of the interim retained product are capped at 50 frequencies; one window (β Pictoris Band 6, 114 formal crossings) exceeds that cap, so its occupancy match covers the 50 strongest crossings – a limitation of the retained product, not of the criterion, and one the public release removes by shipping uncapped hit lists (Data Availability).

TABLE 15
SELECTION FUNCTION OF THE 120-STAR ALMA-COVERED CENSUS AGAINST THE 20 PC REFERENCE CENSUS. CENSUS SIDE: SIMBAD TAP QUERY OF 2026 SEPTEMBER 8 ($\pi \geq 50$ MAS, $\sigma_\pi/\pi \leq 0.1$; 3526 OBJECTS, OF WHICH 389 ARE SOLAR-SYSTEM PLANETS AND 415 BROWN DWARFS, LEAVING 2722 STELLAR OBJECTS OF WHICH 2392 CARRY MK CLASSIFICATIONS, COUNTING OLD-STYLE DWARF PREFIXES AT FACE VALUE). SAMPLE SIDE: THE 120 CENSUS ENTRIES, BINNED BY SIMBAD SPECTRAL TYPE (114 CLASSIFIED; THE SIX UNCLASSIFIED ENTRIES ARE FAINT HIGH-PROPER-MOTION OBJECTS AND COMPANIONS). PERCENTAGES ARE OF CLASSIFIED OBJECTS ON BOTH SIDES.

Property	Census	ALMA 120	Ratio
Spectral class (fraction of classified)			
A	0.6%	0.9%	1.5
F	3.8%	7.0%	1.8
G	5.0%	9.6%	1.9
K	13.8%	10.5%	0.8
M	70.8%	66.7%	0.9
WD	6.0%	5.3%	0.9
Distance			
$d \leq 5$ pc	59 (2.5%)	16 (13.3%)	5.4
5–10 pc	267 (11.2%)	22 (18.3%)	1.6
10–20 pc	2066 (86.4%)	82 (68.3%)	0.8
Other axes			
Planet hosts	n/a[a]	24 (20%)	–
Systems / entries	–	113 / 120	–

[a] Planetary-hosting status is not derivable from the SIMBAD census query (solar-system planet otypes only); the processed-subset rate is 15/46 (33%), against the sample-wide 24/120, already showing the planet-host enrichment among ALMA targets.

TABLE 16
CONSOLIDATED FAMILY-WISE TRIALS ACCOUNTING FOR THIS RELEASE. "ABSORBED BY" NAMES THE STEP THAT CARRIES EACH FACTOR'S MULTIPLICITY IN THE OPERATIVE FALSE-ALARM MODEL; THE BOOKKEEPING LADDER IS DETAILED IN APPENDIX J.

Trials factor	Unit	Count
Channels per window (median; maximum)	window	118; 3770
Drift trials per window (median; maximum)	window	4; 1944
Channel × drift trials	release	1.97×10^7
Effectively independent channel cells[b]	release	8.5×10^4–1.1×10^5
Control positions / $N_{\rm eff}$	window	512 / ∼30–43
Windows (star-level tests)	release	225
Threshold-crossing frequencies (true / tabulated)	release	171 / 107 (8 windows, 5 stars)
Contiguous crossing clusters	release	9 (vs. 5.6 Gaussian-expected)
Cross-target frequency pairs tested	release	946
Star-beats-ring rate $\hat{p}_{\rm sbr}$	window/release	30/225 (13.3%)
Execution blocks spanned	release	57 (sizes 1–8)
Principal false alarm: EB-block permutation	release	$P(F{\geq}3) = 6.1\%$ ($E[F]{=}1.04$)
Expected flags under the symmetric null ($N/513$)	snapshot	0.87
Wilson-rate bracket on $P(F{\geq}3)$	release	4–19%

[b] From the measured lag-one channel autocorrelation of the drift-tracked statistic, $\rho_{\rm lag1} = 0.989$–0.991.

SUPPLEMENTARY MATERIAL (ONLINE ONLY)

The appendices below are published as online-only supplementary material and are not part of the printed article; they are included in the preprint source for completeness and for the reproducibility of every number quoted in the main text. Each is cross-referenced from the main text at the point where its summary is used, as are the figures and tables relocated here from the main text.

FULL PER-TARGET RESULTS TABLE

Table 21 lists the full per-target/band results underlying the summary given in Section 5: spectral type, the narrowband technosignature frequency range actually searched, the number of spectral windows configured simultaneously by the correlator for that observation ("$N_{\rm spw}$"), the resulting $\mathrm{EIRP}_{5\sigma}$ limit, the native channel width at which it applies ($\Delta\nu_{\rm ch}$), and the continuum flux density (or, for non-detections, its 5σ upper limit) together with the underlying map rms noise. $\mathrm{EIRP}_{5\sigma}$ is the nominal 5σ threshold – the minimum EIRP that would have produced a 5σ detection in that spectral window – i.e. $\mathrm{EIRP}_{5\sigma} = 4\pi d^2(5\sigma_{\rm comb})\Delta\nu$, using the same 5σ threshold, and the same primary-beam correction, as the continuum column, so the two limits in each row are directly comparable and neither is a bare, unnormalised measured value. The $\Delta\nu_{\rm ch}$ column gives the native channel width at which each row's $\mathrm{EIRP}_{5\sigma}$ applies, so that the limit is independently reproducible from the tabulated quantities: the per-channel flux-density threshold it corresponds to is $S_{\rm min} = \mathrm{EIRP}_{5\sigma}/(4\pi d^2\,\Delta\nu_{\rm ch})$, which is the narrowband per-channel noise of that window and *not* the tabulated RMS column – that column is the line-excluded, full-bandwidth continuum noise of the same target.

The $\mathrm{EIRP}_{5\sigma}$ column is the nominal 5σ threshold; the measured recovery behind it (42% at the threshold, 83% at

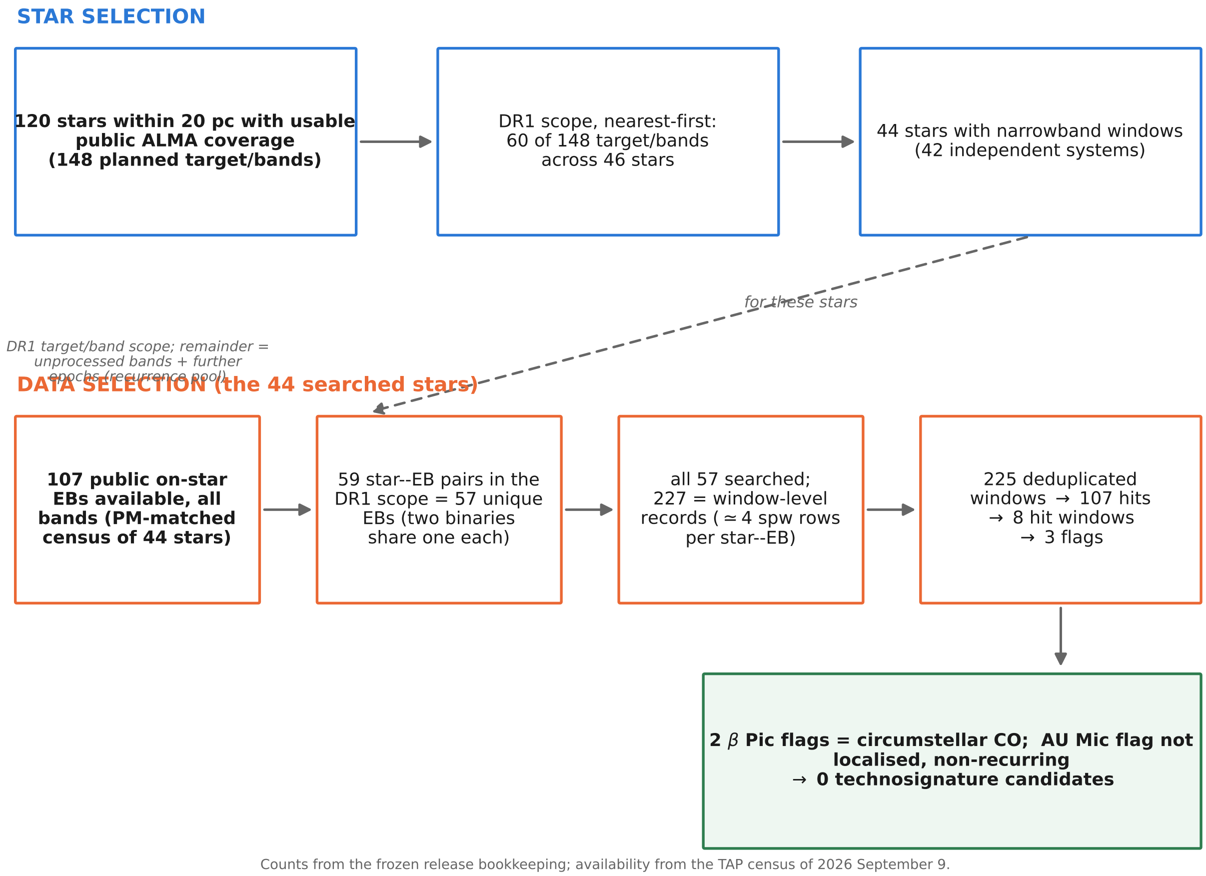

Fig. 8.— The survey's selection funnel, from archive availability to the zero-candidate outcome. Top: star selection. Bottom, for the 44 searched stars: public on-star execution blocks available (all bands; proper-motion-matched archive census of 2026 September 9), the star–EB pairs within the release scope (59 pairs = 57 unique EBs; every held EB was searched – there was no EB-level selection), the window-level records with their correct unit (227 records, ∼4 spectral-window rows per pair), and the search outcome. The difference between availability and scope – further epochs and unprocessed bands – is the recurrence pool drawn on by the AU Mic second-epoch test (§5.4).

twice the threshold power, for morphology-matching carriers) is carried by the injection campaign of §4.5, and every value in the machine-readable table of the released products should be read under the boxed reading rule of §4. As described in §4, the pipeline searches every distinct configured spectral window, and each searched window appears as its own row; the handful of target/bands still represented by a single window are noted per row, with full multi-window coverage deferred to subsequent processing.

A catalogue audit of this release removed 13 duplicate rows: window results enumerated twice from the same execution block, identified by identical name, band, frequency span, and noise statistic, with the more conservative (larger) EIRP retained where the two copies differed. Across the 13 pairs the copies' EIRP values agree to relative differences of 1.4×10^{-6} (minimum), 6.1×10^{-4} (median), and 0.138 (maximum); no pair differs in crossing count or flag status disposition, so the deduplication changes no result of this paper. The single largest difference is fully accounted for rather than merely tolerated: the pair is Sirius B Band 5, whose two copies enumerate the same execution block with identical noise, star, and control statistics (the rms, star peak, and control maximum agree exactly) and differ *only* in the primary-beam correction – one copy carries the uncorrected $S_{\rm min}$, the other the primary-beam-corrected value, and the ratio of the two EIRP entries is exactly the Band 5 primary-beam correction factor at Sirius B's offset (1.161; §4). The deduplication retained the corrected copy, in line with the conservative-retention rule; the remaining twelve pairs agree at the 10^{-3} level and trace to double enumeration in the per-execution-block join, not to any difference in the underlying measurement.

For the audit trail behind the exclusions of §5.3, we record the extent of the quality checks that produced them. The window-level noise-consistency diagnostic – the measured per-integration rms against the radiometer-equation expectation for the same integration – is computed by the pipeline for every window, and we have swept it across all 225 window measurements of this release: 221 of 225 lie within 0.58–1.24 of the expectation (median 1.07), and the four exceptions are the four χ^1 Ori Band 3 windows, which sit ∼20 times *above* the expectation – consistent with elevated system temperature in that execution block, which weakens that target's limits but does not invalidate them, since every limit is computed from the measured noise rather than the nominal one. The anomalously *small* combined noise of item (iii) in §5.3 lives at the integration level (12.1 s of on-source time against 393–520 s for sibling windows)

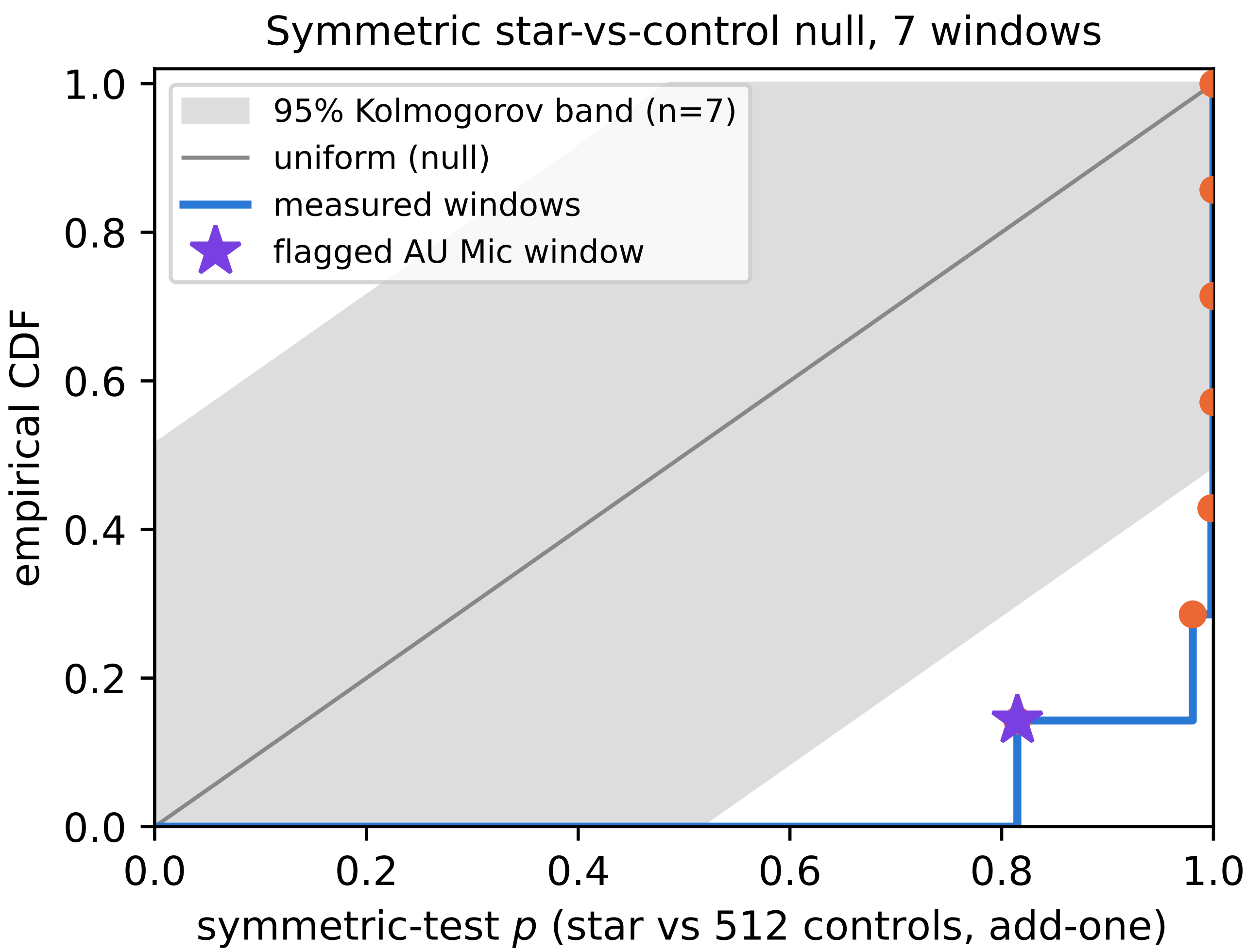

Fig. 9.— The symmetric star-versus-control null, executed on the seven locally reprocessed windows (six validation windows plus the AU Mic candidate window). Each window's p is the add-one rank of the star statistic $T_\star$ against the 512 control statistics under identical treatment; under the null, p is Uniform(0,1). All seven measured windows lie at $p \geq 0.81$; with $n = 7$ the 95 per cent Kolmogorov band is wide, and the figure's content is the absence of star-side excess, not a precise distribution. The control ring's effective independence is $N_{\rm eff} \simeq 30$–43.

and is caught by the cross-target comparison, not by this per-integration diagnostic.

The positional side of the audit is complete. The earlier name-based audit covered the 11 targets carrying traditional designations and caught item (iv); for this release, every one of the 120 census identifiers was independently re-resolved through SIMBAD[6] and checked for uniqueness, type consistency, and distance. All 120 resolve, and to 120 distinct sources: no two census names resolve to the same object. The one identifier that fails a direct name lookup, the AT Mic AB pair, resolves through its Gaia DR3 source identifier to the A component, a 9.9 pc M4.5Ve dwarf. Every census parallax lies at or above 50 mas – no star falls outside the 20 pc boundary – with the deepest case, LP 638-50 at 50.04 ± 0.04 mas (19.99 ± 0.02 pc), sitting within the boundary at about the 1σ level. The γ Lupi name artefact is confirmed independently: the identifier resolves to the 17.4 pc F4V dwarf (HD 139664), exactly as discussed in §5.3. The table therefore contains 225 window measurements.

In a separate name audit, every target carrying a traditional or Bayer-style designation (11 of the 120) was cross-checked against SIMBAD parallaxes and classifications; ten agree, and the audit reclassified Wolf 219 as a DQ8 white dwarf (previously carried as an unclassified late-type dwarf; §6); this classification is independently confirmed by the star's entry as WDJ034434.85+182609.82 (Gaia EDR3 44901791432527232) with a 99.35 per cent white-dwarf probability in the Gaia EDR3 white-dwarf catalogue (Gentile Fusillo et al. 2021), whose astrometry (parallax 52.96 mas, distance 18.87 pc) is consistent with the pipeline's values to within their respective errors. The remaining entry, formerly carried as "γ Lupi", has been resolved as HD 139664 (Gaia DR3 5989102478619519616) by the parallax and position match of §5.3, and is tabulated under that name. The error entered through the name-resolution path used for the 11 traditional/Bayer-style designations (HD 139664 itself carries the SIMBAD identifier "* g Lup"); the remaining targets,

[6] TAP query of the identifier table joined to the basic source data (coordinates, object and spectral type, parallax), `simbad.cds.unistra.fr`, accessed 2026 September 7.

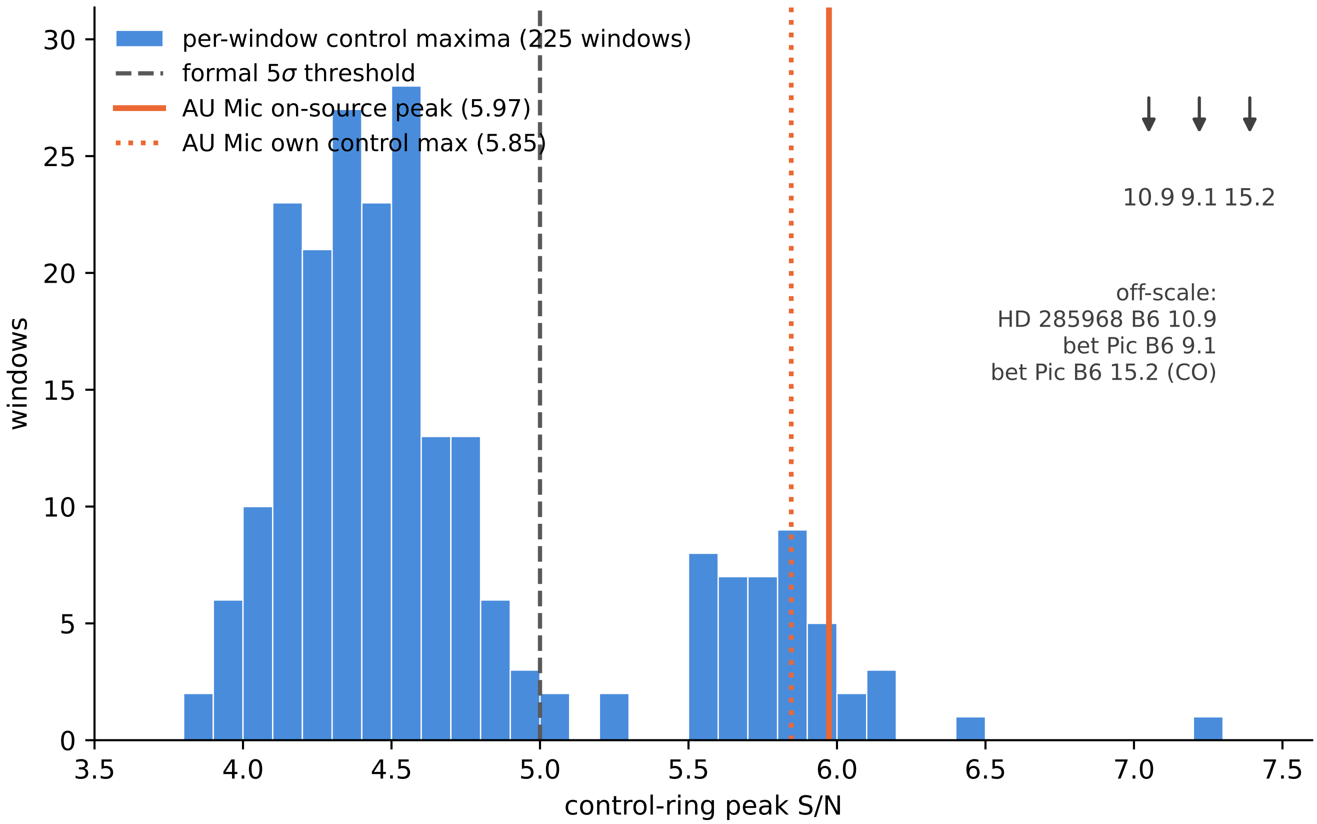

Fig. 10.— Diagnostic distribution for the marginal AU Mic candidate: the per-window control-ring peak signal-to-noise ratios of all 225 windows of this release (histogram; median 4.51, 90th percentile 5.81), i.e. the distribution of maxima expected from noise plus background structure alone. The dashed line marks the formal 5σ crossing threshold; the solid line marks the on-source peak of the AU Mic candidate window (5.97, the 94.7th percentile of this distribution) and the dotted line its own control maximum (5.85). Twelve windows (5%) have control maxima at or above the AU Mic on-source value. Three windows lie beyond the plotted range, marked at upper right: HD 285968 Band 6 at 10.9, and β Pictoris Band 6 at 9.1 and at 15.2, the latter the CO-filled control ring discussed in the text.

which carry Gaia DR3 parallaxes and identifiers throughout, do not pass through that path. A full positional re-audit of the input list against Gaia DR3 is nevertheless planned with the completed survey.

Two target/bands, Giclas 9−38A and Giclas 9−38B (Band 3), are excluded from the table entirely, and two further rows, HD 10647 and HD 139664 (both Band 6), have one narrowband window each excluded and their continuum measurement withheld, all for the reasons set out in §5.3. Excluded and withheld entries are being re-investigated. This table will continue to grow, and existing rows will be superseded by per-window results, as the survey proceeds; the snapshot reproduced here is the frozen release underlying the summary statistics quoted in the main text, so that those statistics are fully traceable. The machine-readable version of this table, accompanying the released products, carries the per-window fields behind every row in addition to the printed columns: execution-block identifier, per-window on-source time and time span, the window's drift ceiling and trial count, the smearing factor $\eta_{\rm smear}$, the applied primary-beam attenuation and offset, and the nearest-catalogued-line attribution with its offset.

FALSE-ALARM ACCOUNTING UNDER THE SYMMETRIC CRITERION

Versions up to 3.24 carried a four-model bookkeeping ladder (execution-block permutation, hypergeometric, block bootstrap, and a Poisson rate heuristic, spanning $P(F{\geq}3) = 6.1$–9.2 per cent) whose sole purpose was to calibrate a false-alarm rate for an *asymmetric* statistic, whose per-window null rate could not be written down. The symmetric statistic adopted in §5.4 removes that need: star and controls are exchangeable single positions under the null, so the per-window rate is $1/(1{+}n_{\rm ctrl}) = 1/513$ by construction and the survey-level expectation is $N/513$, here $448/513 = 0.87$ flags. The ladder is therefore retired rather than updated, and this appendix records only what replaces it and what the replacement cannot do.

Two limits are worth stating plainly. First, $1/513$ is a *floor*, not a measurement: with 512 controls no window can be assigned a smaller empirical p-value, so a Bonferroni threshold over 448 windows (1.1×10^{-4}) lies an order of magnitude below the ensemble's resolution. Enlarging the control ring is the only way to buy resolution here, and it is bounded by the primary beam. Second, windows are not fully independent: they are grouped into 101 execution blocks that share weather, calibration, and correlator setup, so $N/513$ should be read as an expectation over a mildly correlated sample rather than N independent trials. Both limits push in the conservative direction for the conclusion actually drawn – that the flagged windows are consistent with chance – because they widen, not narrow, the plausible

TABLE 17
PER-STAR OBSERVING DUTY CYCLE OF THE 44 STARS WITH NARROWBAND COVERAGE, ORDERED BY DISTANCE. THE TWO SPATIALLY UNRESOLVED BINARIES (UV CETI = G 272-61A/B; AT MIC = THE B AND AB ENTRIES) SHARE EVERY OBSERVATION AND APPEAR AS SEPARATE CATALOGUE ROWS FOR COMPLETENESS; THEY CONSTITUTE SINGLE TRIALS IN ANY STATISTICAL ACCOUNTING, GIVING 42 INDEPENDENT SYSTEMS. $N_{\rm win}$: SEARCHED SPECTRAL-WINDOW MEASUREMENTS CONTRIBUTING TO THIS RELEASE (225 IN TOTAL, AFTER DEDUPLICATION); $N_{\rm EB}$: DISTINCT EXECUTION BLOCKS SEARCHED; $t_{\rm on}$: SUMMED ON-SOURCE INTEGRATION IN MINUTES, TAKING EACH EXECUTION BLOCK'S MEDIAN ACROSS ITS WINDOWS (PER-SPECTRAL-WINDOW FLAGGING MAKES THE WINDOWS OF ONE BLOCK DIFFER SLIGHTLY); $T_{\rm span}$: SUMMED ELAPSED SPAN OF THE EXECUTION BLOCKS IN HOURS, WHICH INCLUDES CALIBRATION AND OVERHEADS AND THEREFORE EXCEEDS $t_{\rm on}$. THESE QUANTITIES WERE NOT RETAINED BY THE FROZEN EXTRACTION PRODUCTS; THEY HAVE SUBSEQUENTLY BEEN RECONSTRUCTED FROM THE ALMA SCIENCE ARCHIVE (QUERIED 2026 SEPTEMBER 8) AND AUDITED AGAINST THE ARCHIVE'S INDEPENDENT EXPOSURE ACCOUNTING (ONLINE-ONLY APPENDIX L); FOR THE TWO EXCEPTION GROUPS OF THAT AUDIT (AT MIC AND CD−38 10980, WHOSE PIPELINE ON-SOURCE TIMES ARE OVER-COUNTED BY DUMP-COUNTING DEFECTS) THE ARCHIVE-CONSISTENT EXPOSURES ARE USED HERE. CALENDAR DATES AND MJDS OF THE EXECUTION BLOCKS ARE GIVEN IN APPENDIX A, AND PER-WINDOW ON-SOURCE AND TIME-SPAN VALUES ACCOMPANY THE MACHINE-READABLE TABLE.

Target	$N_{\rm win}$	$N_{\rm EB}$	$t_{\rm on}$ (min)	$T_{\rm span}$ (h)
Proxima Cen	4	1	50.0	1.05
Barnard's Star	4	1	50.0	1.01
Wolf 359	4	1	50.0	1.10
Sirius B	12	3	17.2	0.27
G 272-61B	4	1	31.0	0.62
G 272-61A	4	1	31.0	0.62
CD-23 14742	4	1	50.0	1.10
eps Eri	4	1	42.8	0.89
tau Cet	1	1	23.2	0.47
BD+05 1668	4	1	23.7	0.46
HD 33793	1	1	14.1	0.26
Wolf 28	1	1	48.4	0.98
GJ 674	4	1	50.0	1.04
GL 3379	4	1	45.9	0.91
LSR J1835+3259	4	1	43.1	0.80
HN Lib	4	1	24.2	0.49
GJ 581	4	1	50.0	1.03
lp 876-10	4	1	46.5	0.74
61 Vir	4	1	37.8	0.95
chi01 Ori	4	1	46.4	0.92
GJ 849	4	1	44.4	0.88
gam Lep	8	2	63.7	1.23
HD 285968	4	1	46.9	0.93
AU Mic	12	3	136.1	2.93
AT Mic B	4	1	14.6	0.32
AT Mic AB	4	1	14.6	0.32
gam Vir	8	2	67.7	1.28
HR 1010	8	2	70.1	2.02
TRAPPIST-1	12	3	168.4	2.91
HD 69830	8	2	92.4	2.04
CD-38 10980	4	1	9.6	0.18
LP 349-25	4	1	44.2	0.82
GJ14	4	1	51.1	0.90
LHS 1140	4	1	43.3	0.95
HD 38858	4	1	0.3	0.03
HD 207129	4	1	47.9	1.04
HD 23484	4	1	45.9	0.91
HD 10647	8	3	107.2	1.98
g Lup	3	1	8.7	0.12
eta Crv	8	2	98.6	2.04
HD 53143	4	1	41.8	0.84
Wolf 219	4	1	1.0	0.02
bet Pic	15	3	93.7	1.81
eta Cru	4	1	30.2	0.59

null range.

LESSONS FOR ARCHIVAL PIPELINES

Three of the four audit items of §5.3 are instances of failure modes generic to any survey that mines a heterogeneous archive with automated crossmatch and processing pipelines, and we record them as design lessons rather than isolated incidents. First, every processing lane needs its own geometric guard. The narrowband step aborts on the 52-arcmin pointing mismatch of item (ii) of §5.3; the continuum-imaging step had no equivalent offset guard and would have reported a spurious, incorrectly-offset "detection" for that field – the guard is asymmetric by omission, not by necessity, and the corrected release carries it on both lanes. Second, window-level bookkeeping anomalies are best caught by

TABLE 18

One-row-per-star summary of this release, ordered by distance: deepest nominal $\mathrm{EIRP}_{5\sigma}$ threshold of any searched window (W; band in the next column) and that band's continuum result (mJy; '<' a 5σ upper limit; a dash marks a withheld or not-applicable entry). Compiled from Table 21; see the reading rules there.

Star	d (pc)	Bands	$\mathrm{EIRP}_{5\sigma}$	B	Cont. (mJy)
Proxima Cen	1.30	6	3.30e+13	6	<0.666
Barnard's Star	1.83	6,7	6.15e+13	6	<0.575
Wolf 359	2.41	6,7	7.84e+13	6	<0.453
Sirius B	2.67	3,4,5	2.31e+13	4	<0.270
G 272-61B	2.67	3	1.62e+13	3	0.851
G 272-61A	2.72	3	1.67e+13	3	0.852
CD-23 14742	2.98	6	1.12e+14	6	<0.388
eps Eri	3.22	6	4.08e+13	6	<0.373
tau Cet	3.65	6	5.46e+13	6	0.914
BD+05 1668	3.79	6	2.71e+14	6	<0.665
HD 33793	3.93	6	5.62e+14	6	<0.578
SCR J1845-6357	4.01	6	—	—	<0.653
Wolf 28	4.31	6	6.67e+13	6	<0.161
GJ 674	4.55	6	2.54e+14	6	<0.516
GL 3379	5.21	6	1.12e+14	6	<0.532
LSR J1835+3259	5.69	3	5.12e+13	3	0.078
HN Lib	6.25	6	1.49e+14	6	<0.521
GJ 581	6.30	6	6.01e+14	6	<0.451
Wolf 358	6.97	7	—	—	<3.380
lp 876-10	7.68	7	2.70e+13	7	<0.097
61 Vir	8.53	7	4.52e+14	7	<1.047
chi01 Ori	8.70	3	1.08e+14	3	0.107
GJ 849	8.81	6	8.30e+14	6	<0.712
gam Lep	8.91	6,7	9.27e+13	6	0.349
HD 285968	9.49	6	3.45e+14	6	<0.700
AU Mic	9.71	3,6	5.26e+13	6	0.633
AT Mic B	9.81	7	5.60e+13	7	0.337
AT Mic AB	9.92	7	5.82e+13	7	0.344
gam Vir	12.02	6,7	2.24e+14	6	0.326
HR 1010	12.04	6,7	6.50e+14	6	<0.498
TRAPPIST-1	12.47	3,6,7	5.76e+13	7	<0.106
HD 69830	12.57	6,7	3.91e+13	6	<0.101
CD-38 10980	12.91	6	4.14e+14	6	<0.116
GJ14	14.69	6	1.21e+14	6	<0.175
LHS 1140	14.96	6	6.24e+14	6	<0.586
HD 38858	15.21	6	6.96e+14	6	<0.686
HD 207129	15.56	6	7.62e+14	6	<0.563
HD 23484	16.17	6	1.70e+15	6	<0.497
HD 10647	17.35	6,7	1.49e+14	7	<0.220
HD 139664	17.40	6	2.55e+14	6	—
eta Crv	18.24	7,8	4.36e+13	7	0.202
HD 53143	18.34	6	2.84e+15	6	<0.877
Wolf 219	18.88	6	8.43e+14	6	<0.463
bet Pic	19.63	3,6	2.34e+13	6	5.659
eta Cru	19.69	6	1.11e+15	6	0.169

TABLE 19

Masked bandwidth (GHz) by species and ALMA band on the operative mask (definition in the text). Columns B3–B8 and 'Total' are window-summed (denominator: the 378.9 GHz gross searched bandwidth); 'Union' is the same intervals merged across windows and stars (denominator: the 87.2 GHz unique-frequency union). Species rows overlap where two species' masks share a channel, so rows exceed the all-species totals (22.2 GHz window-summed; 4.96 GHz of the union). Per-star intervals are in the released veto-mask export.

Species	B3	B4	B5	B6	B7	B8	Total	Union
CO	0.12	0.00	0.00	1.51	0.86	0.00	2.49	0.33
^{13}CO	0.35	0.05	0.07	1.71	0.51	0.00	2.69	0.74
C^{18}O	0.04	0.00	0.00	0.00	0.00	0.00	0.04	0.04
HCN	0.00	0.00	0.00	0.00	0.12	0.00	0.12	0.12
HCO^{+}	0.10	0.00	0.07	0.00	0.36	0.00	0.53	0.50
CS	0.00	0.00	0.07	0.57	0.11	0.00	0.75	0.29
CN	0.87	0.00	0.00	11.04	2.59	0.00	14.50	2.69
SiO	0.00	0.00	0.00	0.27	0.81	0.00	1.08	0.26
All species (merged)	1.48	0.05	0.21	15.10	5.36	0.00	22.20	4.96

cross-target pattern recognition, not per-target sanity checks: the defect of item (iii) – 12.1 s of recorded on-source time against 393–520 s for sibling windows, together with a noise estimate ∼1000× deeper than radiometer scaling allows – recurred with matching correlator parameters across two otherwise unrelated targets, and it was the recurrence, not either instance alone, that established it as a pipeline artifact rather than a target idiosyncrasy. Third, name resolution is not position resolution. The γ Lupi entry was misidentified not through any positional error but through the SIMBAD identifier list: HD 139664 itself carries the historical main identifier "* g Lup", so a name-based lookup of "γ Lupi" returns HD 139664 rather than the B2IV star seven times farther away (parallax 7.75 mas, 3.7° distant on the

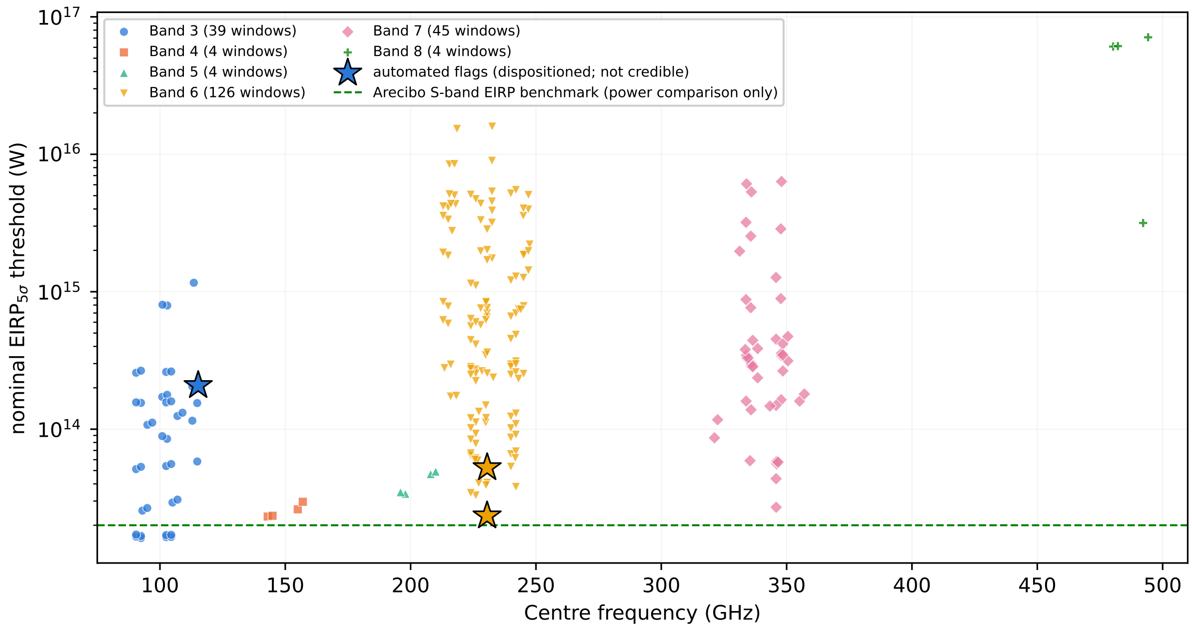

Fig. 11.— $\mathrm{EIRP}_{5\sigma}$ limits of this release as a function of the centre frequency of each searched spectral window, colour- and marker-coded by ALMA band (225 deduplicated windows; the three windows of the automated flags towards β Pictoris and AU Mic carry a distinct star marker, retaining their band colour, so they are identifiable without a size cue – all three are dispositioned in §5.4 and none is credible). The horizontal line marks the Arecibo-like planetary-radar benchmark of 2×10^{13} W (§4.2). The frequency dependence within and between bands reflects the combination of ALMA receiver sensitivity, native channel width, and the distance mix of the targets observed in each band. As in Figure 5, plotted values are nominal 5σ thresholds under the boxed reading rule of §4 (measured recovery $\sim$42% at the threshold, $\sim$83% at twice the threshold power; §4.5).

TABLE 20
Per-band survey characterisation. "Systems" counts unique stellar systems searched in the band (a system observed in two bands counts in both rows; 42 systems, 44 stars, in total); "EBs" the execution blocks searched, attributed by the provenance of the deduplicated windows; "Win." the deduplicated searched windows; $\Delta\nu_{\rm ch}$ the native channel width in MHz (median; range); "union" the unique searched frequency in GHz; $\mathrm{EIRP}_{5\sigma}$ the median nominal threshold across the band's windows; $\dot{\nu}_{\rm ceil}$ the drift-rate ceiling range in Hz s^{-1} GHz^{-1}.

Band	Sys.	EBs	Win.	$\Delta\nu_{\rm ch}$ (MHz)	Union (GHz)	Med. $\mathrm{EIRP}_{5\sigma}$ (W)	$\dot{\nu}_{\rm ceil}$
3	8	9	40	15.63 (0.244–15.63)	20.6	1.1×10^{14}	12.0–13.3
4	1	1	4	15.63 (0.016–15.63)	6.9	2.5×10^{13}	12.0
5	1	1	4	15.63 (0.016–15.63)	6.9	4.1×10^{13}	12.0
6	33	34	128	15.63 (0.015–31.25)	26.9	5.7×10^{14}	12.0–13.3
7	10	11	45	15.63 (0.061–15.63)	20.2	3.3×10^{14}	12.0–13.3
8	1	1	4	15.63 (0.061–15.63)	5.5	6.1×10^{16}	12.0
All	42	57	225	15.63 (0.015–31.25)	87.2	2.7×10^{14}	12.0–13.3

sky). What caught it was cheap and should be standing practice: the observed star's Gaia DR3 parallax, 57.476 mas, matched digit-for-digit by inverting the pipeline distance, with $T_{\rm eff} \approx 6600$ K and cross-identifiers HR 5825, HIP 76829 and GJ 594 consistent with the F4V classification. An archival survey inherits every historical artefact in the catalogues it resolves against; position-anchored verification is the inexpensive antidote.

INTEGRATION-TIME AUDIT: METHOD AND EXCEPTIONS

This appendix documents the archive cross-check summarised in §5.6. The comparison uses the ALMA Science Archive's `ivoa.obscore` service (queried 2026 September 8): each row reports, per observing unit set and pointed science source, the science-intent exposure time `t_exptime`, aggregated over all executions of that unit set. For each of the 57 searched execution blocks, the pointed science source was identified from the archive's field list (designation aliases resolved by hand: Sirius B enters through the Sirius A pointings, the UV Ceti and G 272-61 pairs share one field each, AT Mic's block points at the pair), and the pipeline's on-source time was compared with the archive's exposure for that source and unit set. Two properties of `obscore` bound the comparison's sharpness and are stated as caveats: it counts science-intent scans only, so time spent on the target during calibrator-intent scans is invisible to it (the archive figure is a lower bound on true target time, and a pipeline over-count against it is therefore conservative evidence); and per-execution durations are not exposed individually, so a unit set observed in several executions is compared through its aggregate.

The AT Mic block carries the diagnostic signature of a dump-counting defect: three of its windows report 208

TABLE 21
Per-target/band results, ordered by distance. Band numbers are given without the customary "B" prefix to avoid confusion with B-type spectral classifications elsewhere in the table.

Target	Band	Sp. type	Dist. (pc)	Freq. range (GHz)	$N_{\rm spw}$	$EIRP_{5\sigma}$ (W)	$\Delta\nu_{\rm ch}$	Flux (mJy)	RMS (mJy)	$\sigma_{\rm tot}$ (mJy)	Notes
Proxima Cen	6	M6V	1.30	225.09–226.91	5	3.30e+13	15.62 MHz	<0.666	0.1333	0.1374	
Proxima Cen	6	M6V	1.30	223.09–224.91	5	3.43e+13	15.62 MHz	—	—	—	
Proxima Cen	6	M6V	1.30	241.09–242.91	5	3.80e+13	15.62 MHz	—	—	—	
Proxima Cen	6	M6V	1.30	239.09–240.91	5	5.33e+13	15.62 MHz	—	—	—	
Barnard's Star	7	M5V	1.83	...	1	...	...	<3.622	0.7243	0.8098	narrowband search failed; partial-crossmatch epochs excluded from continuum (§5.3)
Barnard's Star	6	M5V	1.83	225.09–226.91	4	6.15e+13	15.62 MHz	<0.575	0.1150	0.1185	first ALMA technosignature search of this star, see §5.2
Barnard's Star	6	M5V	1.83	223.09–224.91	4	6.40e+13	15.62 MHz	—	—	—	
Barnard's Star	6	M5V	1.83	239.09–240.91	4	6.58e+13	15.62 MHz	—	—	—	
Barnard's Star	6	M5V	1.83	241.09–242.91	4	6.89e+13	15.62 MHz	—	—	—	
Wolf 359	7	M6V	2.41	...	1	...	...	<3.380	0.6761	0.7559	narrowband search failed; partial-crossmatch epochs excluded from continuum
Wolf 359	6	M6V	2.41	225.09–226.91	4	7.84e+13	15.62 MHz	<0.453	0.0907	0.0935	first ALMA technosignature search of this star, see §5.2
Wolf 359	6	M6V	2.41	223.09–224.91	4	8.41e+13	15.62 MHz	—	—	—	
Wolf 359	6	M6V	2.41	239.09–240.91	4	8.67e+13	15.62 MHz	—	—	—	
Wolf 359	6	M6V	2.41	241.09–242.91	4	9.10e+13	15.62 MHz	—	—	—	
Sirius B	4	DA2	2.67	142.13–143.86	5	2.31e+13	15.62 MHz	<0.270	0.0540	0.0557	primary-beam corrected (§4)¶; $\sigma_{\rm PB}$ = 0.024 mJy
Sirius B	4	DA2	2.67	144.06–145.80	5	2.34e+13	15.62 MHz	—	—	—	
Sirius B	4	DA2	2.67	154.13–155.86	5	2.61e+13	15.62 MHz	—	—	—	
Sirius B	4	DA2	2.67	156.13–157.86	5	2.96e+13	15.62 MHz	—	—	—	
Sirius B	3	DA2	2.67	92.13–93.86	5	2.56e+13	15.62 MHz	<0.196	0.0392	0.0404	primary-beam corrected (§4)¶; $\sigma_{\rm PB}$ = 0.007 mJy
Sirius B	3	DA2	2.67	94.06–95.80	5	2.67e+13	15.62 MHz	—	—	—	
Sirius B	3	DA2	2.67	104.13–105.86	5	2.93e+13	15.62 MHz	—	—	—	
Sirius B	3	DA2	2.67	106.13–107.86	5	3.07e+13	15.62 MHz	—	—	—	
Sirius B	5	DA2	2.67	197.06–198.80	5	3.39e+13	15.62 MHz	<0.529	0.1057	0.109	primary-beam corrected (§4)¶; $\sigma_{\rm PB}$ = 0.075 mJy
Sirius B	5	DA2	2.67	195.13–196.86	5	4.04e+13	15.62 MHz	—	—	—	
Sirius B	5	DA2	2.67	207.13–208.86	5	4.71e+13	15.62 MHz	—	—	—	
Sirius B	5	DA2	2.67	209.13–210.86	5	4.92e+13	15.62 MHz	—	—	—	
G 272-61B	3	M6Ve	2.67	91.56–93.30	5	1.62e+13	15.62 MHz	0.851	0.0260	0.0499	unresolved UV Cet AB pair, blended flux
G 272-61B	3	M6Ve	2.67	101.63–103.36	5	1.64e+13	15.62 MHz	—	—	—	
G 272-61B	3	M6Ve	2.67	103.63–105.36	5	1.65e+13	15.62 MHz	—	—	—	

Column definitions: frequency range in GHz; $N_{\rm spw}$ the number of spectral windows configured simultaneously (or the number searched, a lower bound); $EIRP_{5\sigma}$ the nominal, primary-beam-corrected 5σ threshold. **These are nominal thresholds, not calibrated completeness limits; see the boxed reading rule of §4 (and the injection campaign of §4.5).** EIRP assumes an isotropic radiator. Flux is the measured peak continuum flux density for a detection, or the resulting 5σ upper limit ("<"), also primary-beam corrected; RMS is the underlying (post-correction) continuum image rms; both in mJy; $\sigma_{\rm tot} = \sqrt{{\rm RMS}^2 + (f_{\rm cal}\,{\rm Flux})^2}$ is the combined map-plus-calibration error ($f_{\rm cal}$: 5 per cent in Bands 3–6, 10 per cent in Bands 7–8; §4.4), evaluated at the quoted flux or limit. A target's single continuum measurement per band is given once, in the first row.
All three automated flags are noted in the Notes column (§5.4); ellipses mark target/bands excluded by failed validation (§5.3, items (i) and (iii)), with withheld continuum measurements indicated in the Notes. Spectral types marked † are approximate (Gaia DR3 $T_{\rm eff}$ via the Pecaut & Mamajek 2013 scale); those marked ‡ carry only a coarse class. Rows marked ¶ carry the large-offset primary-beam systematic of §4, quantified as $\sigma_{\rm PB} = |A-1|\,S_\nu$ (A the applied Gaussian-response correction at the star's offset): its value in mJy is given in the Notes of the continuum-bearing row of each marked target/band, to be added in quadrature to $\sigma_{\rm tot}$, and the same relative systematic applies to every quoted value of the marked rows, including the $EIRP_{5\sigma}$ thresholds (Sirius B: 3.6–4.6% in Band 3, 9.0–10.4% in Band 4, 14.1–15.7% in Band 5; ϵ Eri Band 6: 65–71%, i.e. the stated 2.9–3.4× correction range), pending the holography-beam recomputation. Rows marked * are the three windows with intra-integration smearing $\eta_{\rm smear} < 0.99$ at the maximum searched drift rate (Table 12); their effective threshold for a maximally drifting signal rises by $1/\eta_{\rm smear}$, at most ×1.74. This table spans 8 parts (Tables 21–28) for typesetting reasons only; it is a single table.

TABLE 22
Table 21 continued.

Target	Band	Sp. type	Dist. (pc)	Freq. range (GHz)	N_{spw}	$EIRP_{5\sigma}$ (W)	$\Delta\nu_{ch}$	Flux (mJy)	RMS (mJy)	σ_{tot} (mJy)	Notes
G 272-61B	3	M6Ve	2.67	89.63–91.36	5	1.66e+13	15.62 MHz	—	—	—	
G 272-61A	3	M5.5Ve	2.72	91.56–93.30	5	1.67e+13	15.62 MHz	0.852	0.0260	0.0499	unresolved UV Cet AB pair, blended flux
G 272-61A	3	M5.5Ve	2.72	101.63–103.36	5	1.70e+13	15.62 MHz	—	—	—	
G 272-61A	3	M5.5Ve	2.72	103.63–105.36	5	1.70e+13	15.62 MHz	—	—	—	
G 272-61A	3	M5.5Ve	2.72	89.63–91.36	5	1.71e+13	15.62 MHz	—	—	—	
CD-23 14742	6	M5V	2.98	225.09–226.91	5	1.12e+14	15.62 MHz	<0.388	0.0776	0.08	
CD-23 14742	6	M5V	2.98	223.09–224.91	5	1.20e+14	15.62 MHz	—	—	—	
CD-23 14742	6	M5V	2.98	239.09–240.91	5	1.23e+14	15.62 MHz	—	—	—	
CD-23 14742	6	M5V	2.98	241.09–242.91	5	1.30e+14	15.62 MHz	—	—	—	
eps Eri	6	K2V†	3.22	229.65–231.38	4	4.08e+13	488.3 kHz	<0.373	0.0745	0.0768	Teff-derived sptype; large-offset beam systematic¶ (§4); σ_{PB} = 0.262 mJy
eps Eri	6	K2V†	3.22	215.10–216.84	4	1.73e+14	15.62 MHz	—	—	—	
eps Eri	6	K2V†	3.22	217.35–219.09	4	1.75e+14	15.62 MHz	—	—	—	
eps Eri	6	K2V†	3.22	232.10–233.84	4	2.37e+14	15.62 MHz	—	—	—	
tau Cet	6	K0V	3.65	229.63–231.47	1	5.46e+13	488.3 kHz	0.914	0.1323	0.14	reprocessing queued (other spws)
BD+05 1668	6	M5V†	3.79	225.13–226.87	4	2.71e+14	15.62 MHz	<0.665	0.1330	0.1371	Teff-derived sptype
BD+05 1668	6	M5V†	3.79	223.13–224.87	4	2.83e+14	15.62 MHz	—	—	—	
BD+05 1668	6	M5V†	3.79	239.13–240.87	4	2.96e+14	15.62 MHz	—	—	—	
BD+05 1668	6	M5V†	3.79	241.13–242.87	4	3.11e+14	15.62 MHz	—	—	—	
HD 33793	6	sdM1	3.93	223.09–224.91	1	5.62e+14	15.62 MHz	<0.578	0.1157	0.1193	Kapteyn's Star; reprocessing queued (other spws)
SCR J1845-6357	6	M8.5V+T6‡	4.01	...	1	...	...	<0.653	0.1306	0.1346	narrowband search failed; M8.5V+T6 binary (Teff unavailable); partial-crossmatch epochs excluded from continuum
Wolf 28	6	DZ7	4.31	223.13–224.87	1	6.67e+13	15.62 MHz	<0.161	0.0321	0.0331	Van Maanen's Star (white dwarf); reprocessing queued (other spws)
GJ 674	6	M3V	4.55	225.09–226.91	5	2.54e+14	15.62 MHz	<0.516	0.1032	0.1064	
GJ 674	6	M3V	4.55	223.09–224.91	5	2.74e+14	15.62 MHz	—	—	—	
GJ 674	6	M3V	4.55	239.09–240.91	5	2.84e+14	15.62 MHz	—	—	—	
GJ 674	6	M3V	4.55	241.09–242.91	5	2.97e+14	15.62 MHz	—	—	—	
GL 3379	6	M4V†	5.21	229.04–230.88	4	1.12e+14	488.3 kHz	<0.532	0.1064	0.1097	Teff-derived sptype
GL 3379	6	M4V†	5.21	227.04–228.87	4	5.72e+14	15.62 MHz	—	—	—	
GL 3379	6	M4V†	5.21	214.05–215.87	4	5.86e+14	15.62 MHz	—	—	—	
GL 3379	6	M4V†	5.21	212.05–213.87	4	6.21e+14	15.62 MHz	—	—	—	
LSR J1835+3259	3	M8.5V‡	5.69	89.63–91.36	4	5.12e+13	15.62 MHz	0.078	0.0117	0.0123	known strongly-magnetically-active UCD (Hallinan et al. 2007)

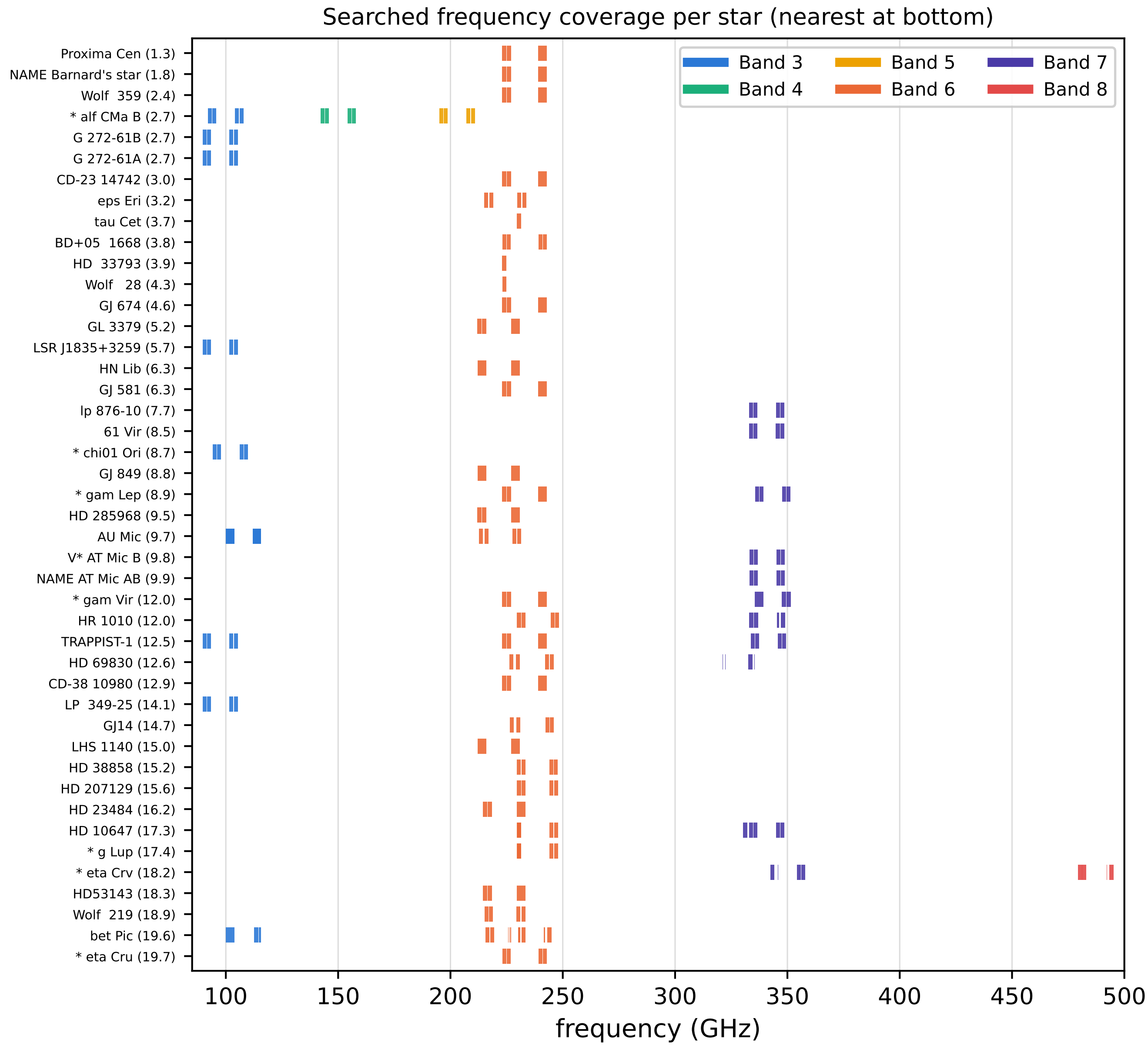

Fig. 12.— Searched frequency coverage per star (nearest at the bottom; the parenthetical is the heliocentric distance in pc). Horizontal bars are the deduplicated searched windows, coloured by ALMA band. The per-star merged intervals, EB and spectral-window counts, and distances are tabulated in the machine-readable ledger `v323_star_coverage.json` accompanying this release; summing the per-star interval unions gives 371.9 star×GHz, and summing the window bandwidths 382.6 star×GHz (the difference is band-overlap within a star). Neither number duplicates a headline: the 87.2 GHz union of Fig. 6 counts unique frequency once across the whole survey, and the 378.9 GHz star-by-bandwidth headline of the abstract is the frozen release product's own sum, which differs from the 382.6 GHz ledger figure by window-listing conventions (the ledger counts every deduplicated window row of the retained products, including near-duplicate re-listings of the same execution block that the frozen sum collapses).

integrations where 145 exist, and 1258 s of on-source time against the block's own 1140 s elapsed span. The duty table of this paper uses the archive-consistent 877 s for both AT Mic catalogue entries; the over-counted values survive only in per-window machine-readable fields, to be corrected in the next release's export. The CD−38 10980 excess (7%) is smaller than the intent-filter's conservative margin and its limit is unaffected (thresholds derive from measured rms, not integration time). A Wolf 359 discrepancy that appears in the raw comparison is an audit-side artefact – an over-aggressive calibrator-source exclusion in the query itself – and disappears when the star's science-source exposure (9463 s, ample against the 2999 s searched) is included; we record it here because it is exactly the kind of error the audit is designed to catch, and it was caught in the audit rather than the pipeline.

EXCLUSION AND DEFECT FORENSICS

The itemised record behind the data-quality exclusions summarised in §5.3, published here in full so that the absence of the corresponding entries in Table 21 is auditable.

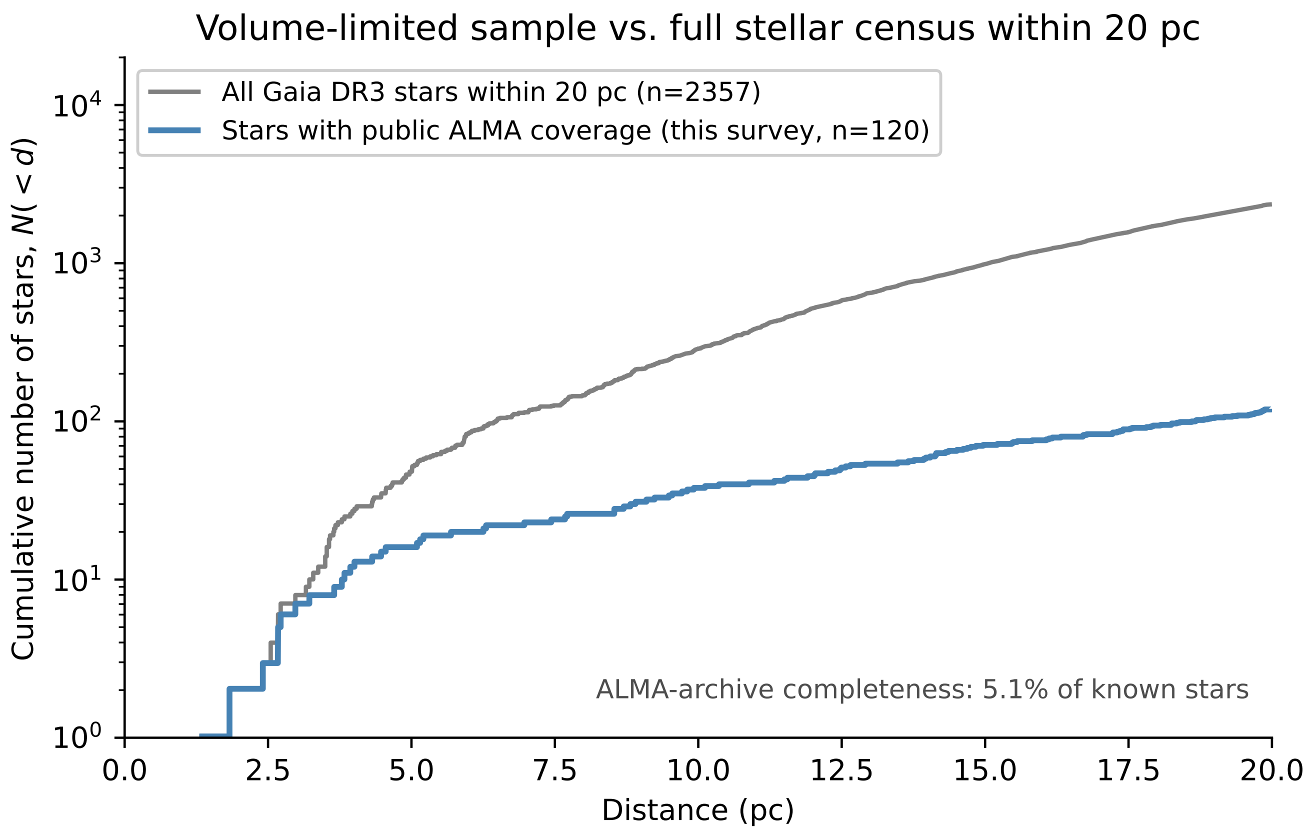

Fig. 13.— The ALMA-covered subset versus the stellar population within 20 pc: cumulative $N(< d)$. Grey: the reference Gaia DR3 catalogue (2357 stars); blue: the census sample of this paper, only ∼5 per cent of the total. The curves are not expected to track one another; the census is complete conditional on ALMA coverage, not an unbiased sample of the local population (§3).

(i) Narrowband process failures. Four target/bands (Barnard's Star [B7], Wolf 359 [B7], SCR J1845−6357, and Wolf 358) have no narrowband result, because the narrowband search failed at the process level for these datasets; each retains a valid continuum non-detection measurement, and no EIRP threshold is quoted for them at this time.

(ii) Giclas 9−38 pair. Giclas 9−38A and Giclas 9−38B [B3] are excluded entirely. The only matched member observation unit set (MOUS) for this pair points ∼52 arcmin from either star, 35× its own primary-beam FWHM. The narrowband step correctly aborts on this mismatch; the continuum-imaging step lacks the equivalent offset guard, and for such a field would report a spurious, incorrectly-offset "detection", so the entire target/band is excluded pending a guard on the continuum step (§6).

(iii) Window-level noise defect. One spectral window in each of two Band 6 targets sharing a similar correlator setup (HD 10647 and HD 139664) reports only 12.1 s of on-source integration against 393–520 s for its sibling windows from the same execution block, yet an implausibly *small* combined noise estimate, ∼1000× deeper than its siblings and deeper than the survey's best continuum limit – the opposite of what radiometer-noise scaling predicts for a ∼40× shorter integration; the defect recurs with matching parameters across two independent targets, so we treat it as a reproducible window-level processing defect. The 12.1 s value is not a transcription of an observing log into this paper: it is recorded, together with the implausibly small noise estimate, in the same machine-written per-window product that carries every other window's integration time and noise, and it is the recorded on-source time itself that is defective alongside the noise. The affected window's narrowband result is excluded, and the combined continuum measurement is withheld for both target/bands, since it is built from all configured windows together; their other, unaffected narrowband windows are retained.

(iv) The "γ Lupi" resolution. The catalogue entry previously carried as "γ Lupi" is resolved: the object actually observed is the F4V nearby debris-disc host HD 139664 at 17.40 pc (Gaia DR3 5989102478619519616), a natural ALMA Band 6 target – *not* the naked-eye γ Lupi. All entries formerly tabulated under "γ Lupi" are relabelled HD 139664; none of their measured values changes, and the target/band's defect status (item (iii)) is unaffected.

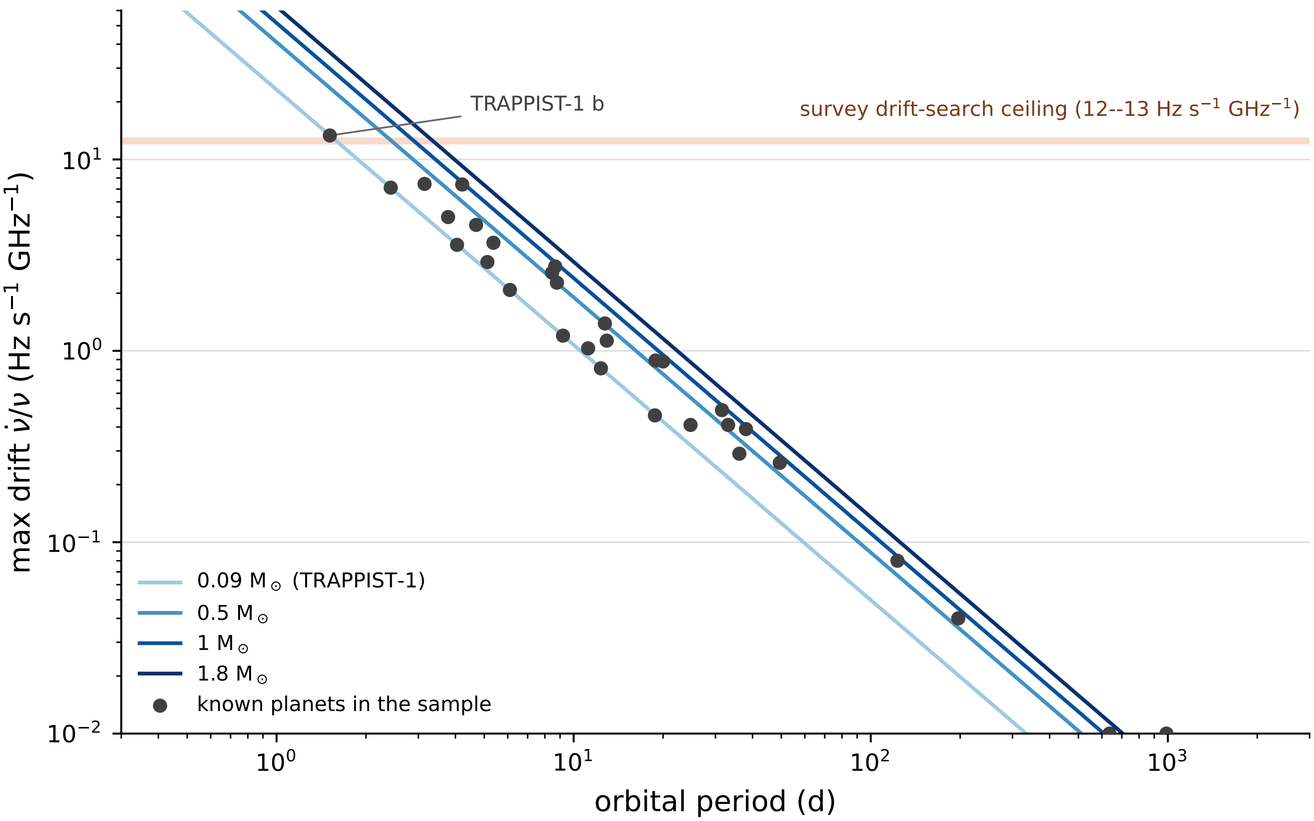

FIG. 14.— The drift-rate ceiling as a physical selection function. Maximum star–planet relative acceleration, edge-on and maximised over orbital phase, expressed as the carrier drift rate it induces (Hz s^{-1} per GHz of carrier frequency), against orbital period, for host-star masses 0.09–1.8 $M_\odot$ (lines) and for the 37 known planets of the 15 planet-hosting targets in this release (points). The shaded band is the survey's drift-search ceiling (12–13 Hz s^{-1} GHz^{-1}; §5.4): a transmitter whose drift falls above the band is outside the searched grid and is sacrificed by design. Only TRAPPIST-1 b ($P = 1.51$ d) reaches the band; the next-highest planets reach roughly half of it (§6).

TABLE 23
Table 21 continued.

Target	Band	Sp. type	Dist. (pc)	Freq. range (GHz)	N_{spw}	$EIRP_{5\sigma}$ (W)	$\Delta\nu_{ch}$	Flux (mJy)	RMS (mJy)	σ_{tot} (mJy)	Notes
LSR J1835+3259	3	M8.5V‡	5.69	101.63–103.36	4	5.39e+13	15.62 MHz	—	—	—	
LSR J1835+3259	3	M8.5V‡	5.69	103.63–105.36	4	5.58e+13	15.62 MHz	—	—	—	
HN Lib	6	M4V	6.25	229.07–230.91	4	1.49e+14	488.3 kHz	<0.521	0.1042	0.1074	known exoplanet host (HN Lib b, habitable zone)
HN Lib	6	M4V	6.25	227.08–228.91	4	7.57e+14	15.62 MHz	—	—	—	
HN Lib	6	M4V	6.25	214.08–215.91	4	7.82e+14	15.62 MHz	—	—	—	
HN Lib	6	M4V	6.25	212.08–213.91	4	8.43e+14	15.62 MHz	—	—	—	
GJ 581	6	M4V	6.30	225.09–226.91	4	6.01e+14	15.62 MHz	<0.451	0.0903	0.0931	
GJ 581	6	M4V	6.30	223.09–224.91	4	6.35e+14	15.62 MHz	—	—	—	
GJ 581	6	M4V	6.30	239.09–240.91	4	6.61e+14	15.62 MHz	—	—	—	
GJ 581	6	M4V	6.30	241.09–242.91	4	6.97e+14	15.62 MHz	—	—	—	
Wolf 358	7	M6V†	6.97	...	1	...	...	<3.380	0.6761	0.7559	narrowband search failed; partial-crossmatch epochs excluded from continuum
lp 876-10	7	M6V†	7.68	344.97–346.70	5	2.70e+13	488.3 kHz	<0.097	0.0194	0.0217	
lp 876-10	7	M6V†	7.68	334.92–336.66	5	1.38e+14	15.62 MHz	—	—	—	
lp 876-10	7	M6V†	7.68	332.97–334.70	5	1.60e+14	15.62 MHz	—	—	—	
lp 876-10	7	M6V†	7.68	346.97–348.70	5	1.64e+14	15.62 MHz	—	—	—	
61 Vir	7	G5V	8.53	344.87–346.71	4	4.52e+14	488.3 kHz	<1.047	0.2094	0.2341	
61 Vir	7	G5V	8.53	334.82–336.56	4	2.54e+15	15.62 MHz	—	—	—	
61 Vir	7	G5V	8.53	346.82–348.56	4	2.87e+15	15.62 MHz	—	—	—	
61 Vir	7	G5V	8.53	332.93–334.66	4	3.20e+15	15.62 MHz	—	—	—	
chi01 Ori	3	G0V	8.70	94.13–95.86	4	1.08e+14	15.62 MHz	0.107	0.0084	0.01	
chi01 Ori	3	G0V	8.70	96.06–97.80	4	1.12e+14	15.62 MHz	—	—	—	
chi01 Ori	3	G0V	8.70	106.13–107.86	4	1.25e+14	15.62 MHz	—	—	—	
chi01 Ori	3	G0V	8.70	108.13–109.86	4	1.32e+14	15.62 MHz	—	—	—	
GJ 849	6	M4V	8.81	229.08–230.92	4	8.30e+14	488.3 kHz	<0.712	0.1424	0.1468	
GJ 849	6	M4V	8.81	214.09–215.91	4	4.11e+15	15.62 MHz	—	—	—	
GJ 849	6	M4V	8.81	212.09–213.91	4	4.18e+15	15.62 MHz	—	—	—	
GJ 849	6	M4V	8.81	227.09–228.91	4	4.37e+15	15.62 MHz	—	—	—	
gam Lep	6	F7V	8.91	225.09–226.91	4	9.27e+13	15.62 MHz	0.349	0.0165	0.024	
gam Lep	6	F7V	8.91	239.09–240.91	4	1.02e+14	15.62 MHz	—	—	—	
gam Lep	6	F7V	8.91	223.09–224.91	4	1.02e+14	15.62 MHz	—	—	—	
gam Lep	6	F7V	8.91	241.09–242.91	4	1.09e+14	15.62 MHz	—	—	—	
gam Lep	7	F7V	8.91	337.56–339.30	4	2.36e+14	15.62 MHz	0.695	0.0623	0.0933	
gam Lep	7	F7V	8.91	347.63–349.36	4	2.65e+14	15.62 MHz	—	—	—	

TABLE 24
Table 21 continued.

Target	Band	Sp. type	Dist. (pc)	Freq. range (GHz)	N_{spw}	$EIRP_{5\sigma}$ (W)	$\Delta\nu_{ch}$	Flux (mJy)	RMS (mJy)	σ_{tot} (mJy)	Notes
gam Lep	7	F7V	8.91	335.63–337.36	4	2.86e+14	15.62 MHz	—	—	—	
gam Lep	7	F7V	8.91	349.63–351.36	4	3.14e+14	15.62 MHz	—	—	—	
HD 285968	6	M1V	9.49	229.04–230.88	4	3.45e+14	488.3 kHz	<0.700	0.1400	0.1443	
HD 285968	6	M1V	9.49	214.05–215.87	4	1.83e+15	15.62 MHz	—	—	—	
HD 285968	6	M1V	9.49	212.05–213.87	4	1.93e+15	15.62 MHz	—	—	—	
HD 285968	6	M1V	9.49	227.04–228.87	4	1.97e+15	15.62 MHz	—	—	—	
AU Mic	6	M1Ve	9.71	229.68–231.40	4	5.26e+13	488.3 kHz	0.633	0.0374	0.049	automated flag, not credible, see text
AU Mic	6	M1Ve	9.71	227.63–229.36	4	2.64e+14	15.62 MHz	—	—	—	
AU Mic	6	M1Ve	9.71	212.63–214.36	4	2.78e+14	15.62 MHz	—	—	—	
AU Mic	6	M1Ve	9.71	215.13–216.86	4	2.95e+14	15.62 MHz	—	—	—	
AU Mic	3	M1Ve	9.71	114.03–115.75	8	5.82e+13	976.6 kHz	—	—	—	narrowband complete; continuum step incomplete
AU Mic	3	M1Ve	9.71	101.91–103.74	8	8.51e+13	15.62 MHz	—	—	—	
AU Mic	3	M1Ve	9.71	100.02–101.85	8	8.90e+13	15.62 MHz	—	—	—	
AU Mic	3	M1Ve	9.71	112.02–113.85	8	1.15e+14	15.62 MHz	—	—	—	
AU Mic	3	M1Ve	9.71	114.01–115.74	8	1.55e+14	976.6 kHz	—	—	—	
AU Mic	3	M1Ve	9.71	100.00–101.83	8	1.72e+14	15.62 MHz	—	—	—	
AU Mic	3	M1Ve	9.71	101.96–103.79	8	1.78e+14	15.62 MHz	—	—	—	
AU Mic	3	M1Ve	9.71	112.00–113.83	8	2.05e+14	15.62 MHz	—	—	—	
AT Mic B	7	M4V	9.81	345.18–346.90	4	5.60e+13	488.3 kHz	0.337	0.0292	0.0446	
AT Mic B	7	M4V	9.81	335.07–336.80	4	2.88e+14	15.62 MHz	—	—	—	
AT Mic B	7	M4V	9.81	333.17–334.91	4	3.30e+14	15.62 MHz	—	—	—	
AT Mic B	7	M4V	9.81	347.07–348.80	4	3.43e+14	15.62 MHz	—	—	—	
AT Mic AB	7	M5V	9.92	345.18–346.90	4	5.82e+13	488.3 kHz	0.344	0.0300	0.0456	
AT Mic AB	7	M5V	9.92	335.07–336.80	4	3.00e+14	15.62 MHz	—	—	—	
AT Mic AB	7	M5V	9.92	333.17–334.91	4	3.44e+14	15.62 MHz	—	—	—	
AT Mic AB	7	M5V	9.92	347.07–348.80	4	3.56e+14	15.62 MHz	—	—	—	
gam Vir	6	F0V	12.02	225.09–226.91	4	2.24e+14	15.62 MHz	0.326	0.0135	0.0212	
gam Vir	6	F0V	12.02	223.09–224.91	4	2.49e+14	15.62 MHz	—	—	—	
gam Vir	6	F0V	12.02	239.09–240.91	4	2.49e+14	15.62 MHz	—	—	—	
gam Vir	6	F0V	12.02	241.09–242.91	4	2.59e+14	15.62 MHz	—	—	—	
gam Vir	7	F0V	12.02	337.52–339.35	4	3.86e+14	15.62 MHz	0.648	0.0244	0.0692	
gam Vir	7	F0V	12.02	347.58–349.41	4	4.18e+14	15.62 MHz	—	—	—	
gam Vir	7	F0V	12.02	335.58–337.41	4	4.43e+14	15.62 MHz	—	—	—	
gam Vir	7	F0V	12.02	349.58–351.41	4	4.73e+14	15.62 MHz	—	—	—	

TABLE 25
Table 21 continued.

Target	Band	Sp. type	Dist. (pc)	Freq. range (GHz)	N_{spw}	$EIRP_{5\sigma}$ (W)	$\Delta\nu_{ch}$	Flux (mJy)	RMS (mJy)	σ_{tot} (mJy)	Notes
HR 1010	6	G2V	12.04	229.61–231.45	4	6.50e+14	976.6 kHz	<0.498	0.0997	0.1028	
HR 1010	6	G2V	12.04	231.62–233.35	4	1.75e+15	15.62 MHz	—	—	—	
HR 1010	6	G2V	12.04	244.62–246.35	4	1.87e+15	15.62 MHz	—	—	—	
HR 1010	6	G2V	12.04	246.62–248.35	4	2.21e+15	15.62 MHz	—	—	—	
HR 1010	7	G2V	12.04	345.32–346.24	4	1.27e+15	488.3 kHz	<1.353	0.2706	0.3025	
HR 1010	7	G2V	12.04	335.05–336.87	4	5.33e+15	15.62 MHz	—	—	—	
HR 1010	7	G2V	12.04	333.09–334.92	4	6.09e+15	15.62 MHz	—	—	—	
HR 1010	7	G2V	12.04	347.05–348.87	4	6.34e+15	15.62 MHz	—	—	—	
TRAPPIST-1	3	M8V	12.47	91.56–93.30	4	1.55e+14	15.62 MHz	<0.036	0.0071	0.0073	
TRAPPIST-1	3	M8V	12.47	101.63–103.36	4	1.57e+14	15.62 MHz	—	—	—	
TRAPPIST-1	3	M8V	12.47	89.63–91.36	4	1.57e+14	15.62 MHz	—	—	—	
TRAPPIST-1	3	M8V	12.47	103.63–105.36	4	1.59e+14	15.62 MHz	—	—	—	
TRAPPIST-1	7	M8V	12.47	345.73–347.46	4	5.76e+13	488.3 kHz	<0.106	0.0212	0.0237	
TRAPPIST-1	7	M8V	12.47	335.62–337.36	4	2.86e+14	15.62 MHz	—	—	—	
TRAPPIST-1	7	M8V	12.47	333.73–335.46	4	3.31e+14	15.62 MHz	—	—	—	
TRAPPIST-1	7	M8V	12.47	347.62–349.36	4	3.47e+14	15.62 MHz	—	—	—	
TRAPPIST-1	6	M8V	12.47	225.08–226.92	4	4.72e+15	31.25 MHz	<0.531	0.1061	0.1094	
TRAPPIST-1	6	M8V	12.47	223.08–224.92	4	5.08e+15	31.25 MHz	—	—	—	
TRAPPIST-1	6	M8V	12.47	239.08–240.92	4	5.19e+15	31.25 MHz	—	—	—	
TRAPPIST-1	6	M8V	12.47	241.08–242.92	4	5.49e+15	31.25 MHz	—	—	—	
HD 69830	7	K0V†	12.57	335.34–335.44	5	5.89e+13	244.1 kHz	<0.116	0.0231	0.0258	
HD 69830	7	K0V†	12.57	321.17–321.27	5	8.65e+13	244.1 kHz	—	—	—	
HD 69830	7	K0V†	12.57	322.41–322.51	5	1.17e+14	244.1 kHz	—	—	—	
HD 69830	7	K0V†	12.57	332.58–334.41	5	3.80e+14	15.62 MHz	—	—	—	
HD 69830	6	K0V†	12.57	229.20–230.92	4	3.91e+13	488.3 kHz	<0.101	0.0202	0.0208	
HD 69830	6	K0V†	12.57	226.30–228.03	4	4.04e+13	488.3 kHz	—	—	—	
HD 69830	6	K0V†	12.57	242.19–243.92	4	2.33e+14	15.62 MHz	—	—	—	
HD 69830	6	K0V†	12.57	244.19–245.92	4	2.53e+14	15.62 MHz	—	—	—	
CD-38 10980	6	DA2	12.91	225.09–226.91	4	4.14e+14	15.62 MHz	<0.116	0.0232	0.0239	white dwarf; see §6
CD-38 10980	6	DA2	12.91	223.09–224.91	4	4.44e+14	15.62 MHz	—	—	—	
CD-38 10980	6	DA2	12.91	239.09–240.91	4	4.54e+14	15.62 MHz	—	—	—	
CD-38 10980	6	DA2	12.91	241.09–242.91	4	4.86e+14	15.62 MHz	—	—	—	
LP 349-25	3	M dwarf‡	14.13	89.63–91.36	4	2.57e+14	15.62 MHz	0.071	0.0067	0.0076	

TABLE 26
Table 21 continued.

Target	Band	Sp. type	Dist. (pc)	Freq. range (GHz)	N_{spw}	$EIRP_{5\sigma}$ (W)	$\Delta\nu_{ch}$	Flux (mJy)	RMS (mJy)	σ_{tot} (mJy)	Notes
LP 349-25	3	M dwarf‡	14.13	101.63–103.36	4	2.61e+14	15.62 MHz	—	—	—	
LP 349-25	3	M dwarf‡	14.13	103.63–105.36	4	2.63e+14	15.62 MHz	—	—	—	
LP 349-25	3	M dwarf‡	14.13	91.56–93.30	4	2.66e+14	15.62 MHz	—	—	—	
GJ14	6	M0V†	14.69	229.25–230.98	4	1.21e+14	488.3 kHz	<0.175	0.0350	0.0361	
GJ14	6	M0V†	14.69	226.36–228.09	4	1.34e+14	488.3 kHz	—	—	—	
GJ14	6	M0V†	14.69	242.25–243.98	4	7.50e+14	15.62 MHz	—	—	—	
GJ14	6	M0V†	14.69	244.25–245.98	4	7.85e+14	15.62 MHz	—	—	—	
LHS 1140	6	M4.5V	14.96	229.08–230.92	4	6.24e+14	488.3 kHz	<0.586	0.1172	0.1208	known exoplanet host (LHS 1140 b/c, b in habitable zone)
LHS 1140	6	M4.5V	14.96	227.08–228.91	4	3.31e+15	15.62 MHz	—	—	—	
LHS 1140	6	M4.5V	14.96	214.08–215.91	4	3.35e+15	15.62 MHz	—	—	—	
LHS 1140	6	M4.5V	14.96	212.08–213.91	4	3.56e+15	15.62 MHz	—	—	—	
HD 38858	6	G5V	15.21	229.64–231.36	4	6.96e+14	488.3 kHz	<0.686	0.1373	0.1415	
HD 38858	6	G5V	15.21	231.59–233.32	4	3.18e+15	15.62 MHz	—	—	—	
HD 38858	6	G5V	15.21	244.09–245.82	4	3.56e+15	15.62 MHz	—	—	—	
HD 38858	6	G5V	15.21	246.08–247.82	4	3.97e+15	15.62 MHz	—	—	—	
HD 207129	6	G2V	15.56	229.64–231.48	4	7.62e+14	488.3 kHz	<0.563	0.1126	0.1161	
HD 207129	6	G2V	15.56	231.65–233.38	4	3.89e+15	15.62 MHz	—	—	—	
HD 207129	6	G2V	15.56	244.15–245.88	4	4.03e+15	15.62 MHz	—	—	—	
HD 207129	6	G2V	15.56	246.15–247.88	4	5.06e+15	15.62 MHz	—	—	—	
HD 23484	6	K5V	16.17	229.61–231.45	5	1.70e+15	976.6 kHz	<0.497	0.0995	0.1026	
HD 23484	6	K5V	16.17	216.57–218.40	5	5.02e+15	15.62 MHz	—	—	—	
HD 23484	6	K5V	16.17	214.57–216.40	5	5.12e+15	15.62 MHz	—	—	—	
HD 23484	6	K5V	16.17	231.57–233.40	5	5.36e+15	15.62 MHz	—	—	—	
HD 10647	7	G0V	17.35	344.93–346.65	5	1.49e+14	488.3 kHz	<0.220	0.0440	0.0492	
HD 10647	7	G0V	17.35	334.82–336.55	5	7.66e+14	15.62 MHz	—	—	—	
HD 10647	7	G0V	17.35	332.92–334.65	5	8.77e+14	15.62 MHz	—	—	—	
HD 10647	7	G0V	17.35	346.82–348.55	5	8.92e+14	15.62 MHz	—	—	—	
HD 10647	7	G0V	17.35	330.35–332.19	5	1.97e+15	488.3 kHz	—	—	—	
HD 10647	6	G0V	17.35	229.67–231.40	3	3.61e+14	488.3 kHz	—	—	—	corrupted spectral window excluded (see §5); continuum withheld (1 corrupted window excluded)
HD 10647	6	G0V	17.35	244.12–245.86	3	1.85e+15	15.62 MHz	—	—	—	
HD 10647	6	G0V	17.35	246.12–247.86	3	1.98e+15	15.62 MHz	—	—	—	
HD 139664	6	F4V	17.40	229.69–231.42	3	2.55e+14	488.3 kHz	—	—	—	formerly tabulated as “g Lup”; resolved to HD 139664 (§5.3); corrupted spectral window excluded; continuum withheld (1 corrupted window excluded)
HD 139664	6	F4V	17.40	244.14–245.88	3	1.26e+15	15.62 MHz	—	—	—	

TABLE 27
Table 21 continued.

Target	Band	Sp. type	Dist. (pc)	Freq. range (GHz)	N_{spw}	$EIRP_{5\sigma}$ (W)	$\Delta\nu_{ch}$	Flux (mJy)	RMS (mJy)	σ_{tot} (mJy)	Notes
HD 139664	6	F4V	17.40	246.14–247.88	3	1.43e+15	15.62 MHz	—	—	—	
eta Crv	7	F2V	18.24	345.77–345.82	4	4.36e+13	61.04 kHz	0.202	0.0380	0.043	second independent band for this target (Band 8 already reported)*
eta Crv	7	F2V	18.24	342.48–344.21	4	1.47e+14	488.3 kHz	—	—	—	
eta Crv	7	F2V	18.24	354.32–356.04	4	1.60e+14	488.3 kHz	—	—	—	
eta Crv	7	F2V	18.24	356.19–357.92	4	1.80e+14	488.3 kHz	—	—	—	
eta Crv	8	F2V	18.24	492.19–492.24	4	3.16e+15	61.04 kHz	<3.609	0.7218	0.807	*
eta Crv	8	F2V	18.24	479.34–481.17	4	6.08e+16	15.62 MHz	—	—	—	
eta Crv	8	F2V	18.24	481.23–483.06	4	6.13e+16	15.62 MHz	—	—	—	
eta Crv	8	F2V	18.24	493.30–495.13	4	7.10e+16	15.62 MHz	—	—	—	
HD 53143	6	K0V	18.34	229.60–231.44	4	2.84e+15	976.6 kHz	<0.877	0.1754	0.1808	
HD 53143	6	K0V	18.34	214.57–216.39	4	8.45e+15	15.62 MHz	—	—	—	
HD 53143	6	K0V	18.34	216.57–218.39	4	8.49e+15	15.62 MHz	—	—	—	
HD 53143	6	K0V	18.34	231.57–233.39	4	8.97e+15	15.62 MHz	—	—	—	
Wolf 219	6	DQ8	18.88	229.26–230.99	4	8.43e+14	488.3 kHz	<0.463	0.0926	0.0954	
Wolf 219	6	DQ8	18.88	217.09–218.83	4	4.34e+15	15.62 MHz	—	—	—	
Wolf 219	6	DQ8	18.88	215.20–216.93	4	4.35e+15	15.62 MHz	—	—	—	
Wolf 219	6	DQ8	18.88	231.59–233.33	4	4.53e+15	15.62 MHz	—	—	—	
bet Pic	6	A6V	19.63	230.49–230.54	12	2.34e+13	15.26 kHz	5.659	0.9626	1.003	CO(2-1) line, automated flag, see text*
bet Pic	6	A6V	19.63	226.80–226.91	12	5.87e+13	122.1 kHz	—	—	—	
bet Pic	6	A6V	19.63	226.24–226.35	12	5.90e+13	122.1 kHz	—	—	—	
bet Pic	6	A6V	19.63	226.58–226.69	12	5.92e+13	122.1 kHz	—	—	—	
bet Pic	6	A6V	19.63	225.62–225.73	12	6.00e+13	122.1 kHz	—	—	—	
bet Pic	6	A6V	19.63	241.55–241.98	12	6.17e+13	122.1 kHz	—	—	—	
bet Pic	6	A6V	19.63	243.05–244.88	12	7.41e+14	15.62 MHz	—	—	—	
bet Pic	6	A6V	19.63	230.08–231.00	12	2.01e+15	244.1 kHz	—	—	—	
bet Pic	6	A6V	19.63	215.68–217.41	12	2.77e+15	488.3 kHz	—	—	—	
bet Pic	6	A6V	19.63	217.65–219.35	12	1.53e+16	15.62 MHz	—	—	—	
bet Pic	6	A6V	19.63	231.68–233.41	12	1.59e+16	15.62 MHz	—	—	—	
bet Pic	3	A6V	19.63	114.83–115.69	5	2.08e+14	244.1 kHz	0.531	0.0242	0.0359	CO(1-0) line, automated flag, see text
bet Pic	3	A6V	19.63	101.99–103.82	5	7.94e+14	15.62 MHz	—	—	—	
bet Pic	3	A6V	19.63	100.10–101.93	5	8.04e+14	15.62 MHz	—	—	—	
bet Pic	3	A6V	19.63	112.60–114.43	5	1.16e+15	15.62 MHz	—	—	—	

TABLE 28
Table 21 continued.

Target	Band	Sp. type	Dist. (pc)	Freq. range (GHz)	N_{spw}	$EIRP_{5\sigma}$ (W)	$\Delta\nu_{ch}$	Flux (mJy)	RMS (mJy)	σ_{tot} (mJy)	Notes
eta Cru	6	F2V	19.69	225.13–226.87	4	1.11e+15	15.62 MHz	0.169	0.0210	0.0226	
eta Cru	6	F2V	19.69	223.13–224.87	4	1.15e+15	15.62 MHz	—	—	—	
eta Cru	6	F2V	19.69	239.13–240.87	4	1.21e+15	15.62 MHz	—	—	—	
eta Cru	6	F2V	19.69	241.13–242.87	4	1.29e+15	15.62 MHz	—	—	—	

TABLE 29
Exceptions of the integration-time audit (§5.6). All 53 remaining star–execution-block groups have pipeline on-source time less than or equal to the archive science exposure, i.e. the comparison is conservative for them; three further groups (Wolf 28, the narrow AT Mic window, and G 272-61) agree to better than 1%.

Star / block	Pipeline (s)	Archive (s)	Ratio
AT Mic (A002_Xcd97a1_X3f1e, wide windows)	1258	877	1.43
CD−38 10980 (A002_Xe5808e_X80a1)	617	575	1.07

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